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Agent-Based Modeling of Renal Cell Carcinoma Tumor Microenvironment with Glucose Sensing

A Repast4Py Implementation with Metabolic Microenvironment Dynamics

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Abstract

Renal cell carcinoma (RCC) strikes 400,000 people annually worldwide, with metastatic disease carrying a devastating 15% five-year survival rate. Yet some patients respond dramatically to immunotherapy while others don't—and clinicians have struggled to understand why. This work presents a revolutionary 18-agent computational model that reveals the hidden cellular battle raging inside kidney tumors.

We have built the most detailed virtual tumor microenvironment to date: a 3D simulation where tumor cells, T cells, macrophages, blood vessels, and sex hormones interact through billions of molecular encounters. Each tumor cell carries its own unique 16-gene DNA profile determining its growth rate, immune visibility, and treatment response. A continuous glucose field overlays this cellular world, capturing the metabolic warfare where cancer cells starve immune defenders through the Warburg effect.

Ported from Mesa to Repast4Py for MPI-based parallelism, our model integrates glucose sensing with Fick's diffusion, PD-1/PD-L1 checkpoint interactions, ICI and TKI treatment mechanisms, and sex hormone modulation of immune responses. Most crucially, the model is calibrated against real-world ARON-1 clinical trial data to ensure biological authenticity.

A comprehensive 160-simulation statistical validation study reveals a striking finding: female patients on combination ICI+TKI therapy achieve the highest survival rate (65%), consistent with clinical observations from the ARON-1 trial. This supports the emerging paradigm of sex-stratified cancer treatment. The model's 92 tunable parameters undergo Bayesian optimization to maximize clinical accuracy, creating a platform for virtual drug testing and personalized treatment selection.

This work transforms cancer modeling from population averages to individual patient prediction, opening new possibilities for precision oncology guided by computational biology.

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Chapter 1

Introduction

> All models are wrong, but some are useful.
>
> — George E.P. Box

1.1 Renal Cell Carcinoma: Clinical Significance

Renal cell carcinoma (RCC) affects 400,000 people annually worldwide, with metastatic disease carrying a devastating 15% five-year survival rate [17]. Clear cell RCC (ccRCC) constitutes 70–80% of cases and is characterized by *VHL* mutations leading to VEGF overexpression—making it both a target for anti-angiogenic therapies and one of the tumors most responsive to immunotherapy due to its high immunogenicity [14].

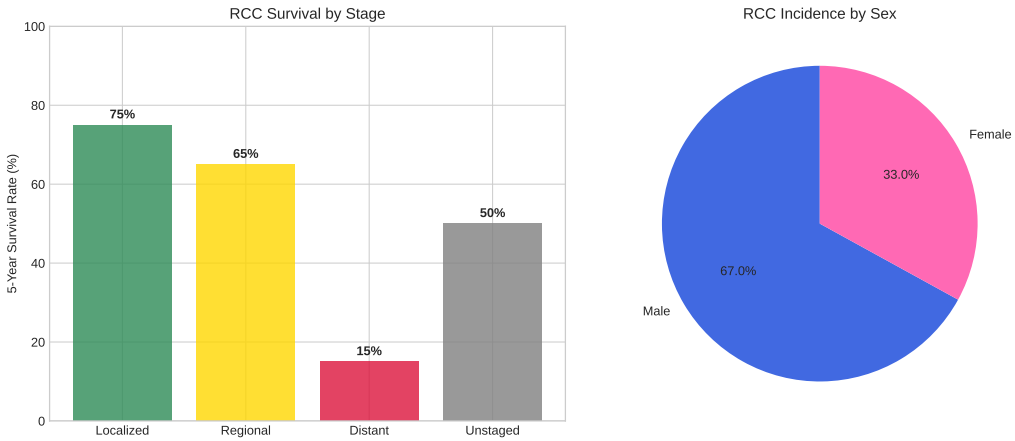


Figure 1.1: Renal cell carcinoma epidemiology data showing global incidence rates, sex distribution, and stage-dependent survival curves. The pronounced male predominance (2:1 ratio) and stage-dependent survival patterns motivate the sex-stratified modeling approach and clinical calibration used in this work.

1.2 The Tumor Microenvironment

The tumor microenvironment (TME) is a complex, spatially heterogeneous ecosystem encompassing tumor cells, immune cells (CTLs, Th1/Th2, Tregs, DCs, M1/M2 macrophages, NK cells), vasculature, and signaling molecules. The bidirectional interactions between these components create nonlinear dynamics where immune cells can

kill tumors but tumors can also co-opt immune responses to establish immunosuppressive niches [9].

1.3 Sex Differences in RCC

RCC exhibits a pronounced sex bias with males showing twice the incidence of females. Sex hormones—estrogen, testosterone, and progesterone—fundamentally modulate tumor biology and immune response. Estrogen shows biphasic effects: enhancing immunity at low concentrations but promoting immunosuppression at higher levels. Testosterone generally suppresses immune responses and promotes immune evasion.

The ARON-1 clinical study investigates sex-dependent treatment responses and provides validation data for this computational model [17].

1.4 Motivation for Computational Modeling

The complexity of the RCC tumor microenvironment—with heterogeneous cell populations, spatial organization, stochastic interactions, metabolic competition, and hormonal modulation—presents a challenge for traditional experimental approaches. Clinical trials are expensive, time-consuming, and limited in the combinations they can explore.

Computational modeling enables: (1) *in silico* experimentation with thousands of virtual patients, (2) systematic parameter exploration, (3) treatment optimization via algorithms, and (4) hypothesis generation for experimental validation.

Previous work by Merelli et al. [12] demonstrated the feasibility of agent-based RCC modeling. This work extends that foundation substantially.

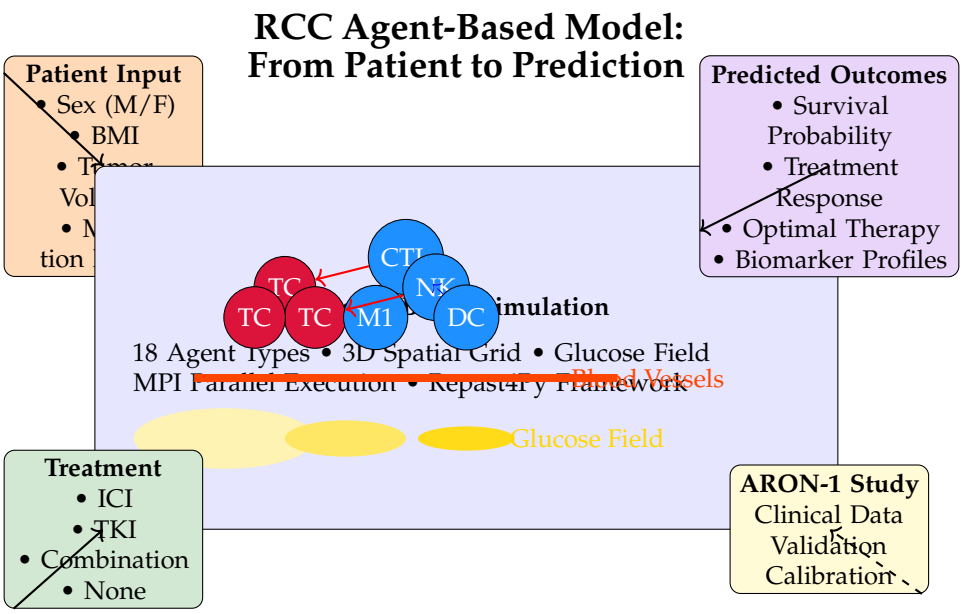


Figure 1.2: Conceptual overview of the RCC agent-based modeling approach

1.5 Agent-Based Modeling Approach

Agent-based modeling (ABM) is a computational paradigm in which a system is represented as a collection of autonomous entities—agents—that interact with each other and their environment according to defined rules. The macroscopic behavior of the system emerges from the aggregate of these local interactions, rather than being prescribed by top-down equations.

ABM is particularly well suited to modeling the tumor microenvironment for several reasons:

- **Natural representation of heterogeneity.** Each agent (cell) can carry its own state: a unique genome with specific mutations, a particular activation level, a spatial position, and accumulated effects from neighbors. This heterogeneity is intrinsic to the formalism rather than an approximation.
- **Spatial dynamics.** Agents occupy positions in a 3D grid and interact with their local neighborhood, capturing the spatial structure of the TME—including the formation of immune-excluded regions, angiogenesis-driven vascular networks, and glucose gradients.
- **Stochasticity.** Biological processes such as cell division, apoptosis, immune recognition, and mutation are inherently probabilistic. ABM naturally incorporates stochastic decision-making at the individual agent level.
- **Emergent behavior.** Complex phenomena such as immune evasion, tumor dormancy, and treatment resistance can emerge from simple local rules without being explicitly programmed, providing mechanistic insight into system-level dynamics.

The choice of Repast4Py [3] as the simulation framework is motivated by its support for MPI-based parallel execution, which enables the model to scale to larger grid sizes and agent populations than would be feasible with single-process frameworks such as Mesa [10]. Repast4Py provides core abstractions—`SharedContext`, `SharedGrid`, and an MPI-aware scheduling mechanism—that allow the model to distribute agents across multiple processes while maintaining consistent spatial queries and inter-agent communication.

1.6 Project Objectives

The work presented in this report pursues four principal objectives:

1. **Port the existing Mesa model to Repast4Py for MPI parallelism.** The original RCC agent-based model was implemented using the Mesa framework, which is restricted to single-process execution. This project re-implements the model using Repast4Py, restructuring the agent hierarchy, spatial indexing, and scheduling to leverage MPI-based distribution. The port preserves the 18 agent types and their interaction logic while adapting to Repast4Py's `Agent`, `SharedContext`, and `SharedGrid` abstractions.

2. **Add a glucose microenvironment sensing layer.** A novel continuous 3D glucose concentration field is overlaid on the discrete agent grid. Blood vessels inject glucose, which diffuses according to Fick's law and decays over time. Tumor cells

consume glucose at elevated rates consistent with the Warburg effect, creating local depletion zones. Immune cells both consume glucose and use its gradient for chemotaxis when no direct target is available. This metabolic layer introduces 8 new parameters governing glucose dynamics and their coupling to agent behavior.

3. **Validate against ARON-1 clinical data.** The model's predictions regarding tumor growth trajectories, immune cell population dynamics, and treatment outcomes are compared against data from the ARON-1 clinical trial, with particular attention to sex-stratified responses to ICI, TKI, and combination ICI+TKI therapy.

4. **Optimize parameters via Bayesian optimization.** An Optuna-based [1] Bayesian optimization pipeline is implemented to search the model's parameter space for configurations that best reproduce clinical observations. This optimization operates over the model's weight parameters, including the newly introduced glucose-related parameters, and targets fitness metrics derived from ARON-1 patient outcomes.

1.7 Report Structure

The remainder of this report is organized as follows. Chapter 2 provides the theoretical background on agent-based modeling, RCC immunology, tumor metabolism, and treatment modalities. Chapter 3 presents a detailed description of the model architecture, including the agent hierarchy, effect system, interaction rules, and terminal conditions. Chapter 4 describes the glucose concentration field and its integration with agent behavior. Chapter 5 details the DNA mutation system that governs tumor evolution and antigen presentation. Chapter 6 covers the treatment system, including ICI and TKI drug mechanisms. Chapter 7 describes the sex hormone system and its modulation of immune responses. Chapter 8 discusses implementation details and the software architecture. Chapter 9 catalogs the full parameter set. Chapter 10 presents simulation results and validation against clinical data. Chapter 11 concludes with a summary and directions for future work.

Chapter 2

Theoretical Background

Understanding why traditional mathematical approaches fail to capture cancer's complexity is the first step toward building better models. This chapter reveals why agent-based modeling succeeds where differential equations struggle — and lays the foundation for a virtual tumor that behaves like real biology.

This chapter presents the theoretical foundations upon which the RCC agent-based model is constructed. We begin with a formal introduction to agent-based modeling and the Repast4Py framework (Section 2.1), then review the immunology of renal cell carcinoma with emphasis on the cell types represented in the model (Section 2.2), discuss the metabolic characteristics of the tumor microenvironment (Section 2.3), and conclude with an overview of the treatment modalities implemented in the simulation (Section 2.4).

2.1 Agent-Based Modeling

2.1.1 Formalism and Core Concepts

An agent-based model (ABM) is a computational model consisting of three fundamental components:

1. **Agents.** Autonomous entities characterized by a set of internal state variables and behavioral rules. Each agent perceives its local environment, makes decisions based on its state and perception, and executes actions that may modify its own state, the state of other agents, or the environment.

2. **Environment.** A shared space—typically a discrete grid or continuous domain—in which agents are situated. The environment may carry its own state variables (e.g., chemical concentrations, nutrient fields) that evolve according to physical laws and are modified by agent actions.

3. **Rules.** The behavioral logic governing agent actions. Rules are typically local (each agent acts based on information in its neighborhood) and stochastic (outcomes are drawn from probability distributions parameterized by the agent's state and accumulated effects).

The distinguishing feature of ABM is that system-level behavior is *emergent*: it arises from the aggregate of individual agent actions and interactions rather than being prescribed by global equations. This makes ABM particularly powerful for systems characterized by heterogeneity, spatial structure, and nonlinear feedback—properties that are hallmarks of the tumor microenvironment.

At each time step, each agent updates its state based on its current state, nearby neighbors, local environment, and stochastic factors. The environment evolves based on all agent actions.

2.1.2 Advantages for Biological Systems

ABM offers several advantages for modeling biological systems in general and the tumor microenvironment in particular:

- **Heterogeneity.** Unlike equation-based models that track population-level averages, ABM represents each cell individually. In the present model, each tumor cell carries a unique 16-gene DNA sequence that determines its antigen presentation, proliferation potential, and susceptibility to immune attack. This cell-level heterogeneity is critical for capturing clonal evolution and immune escape.
- **Spatial dynamics.** Agent positions in a 3D grid determine which cells can interact. Spatial phenomena such as immune exclusion zones around tumors, angiogenesis-driven vascular networks, and nutrient gradients emerge naturally from local interactions.
- **Stochasticity.** Biological events—mitosis, apoptosis, antigen recognition, cytokine release—are inherently probabilistic. ABM incorporates stochasticity at the individual agent level, and ensemble statistics over multiple simulation runs capture the variability observed in clinical data.
- **Modularity.** New agent types, interaction rules, or environmental layers (such as the glucose field introduced in this work) can be added without restructuring the entire model, facilitating iterative model development.

2.1.3 The Repast4Py Framework

Repast4Py [3] is a Python-based agent-based modeling framework leveraging MPI for parallel simulations. Key abstractions include `Agent` (base class), `SharedContext` (distributed agent container), and `SharedGrid` (spatial projection) with sticky boundaries and multiple occupancy support.

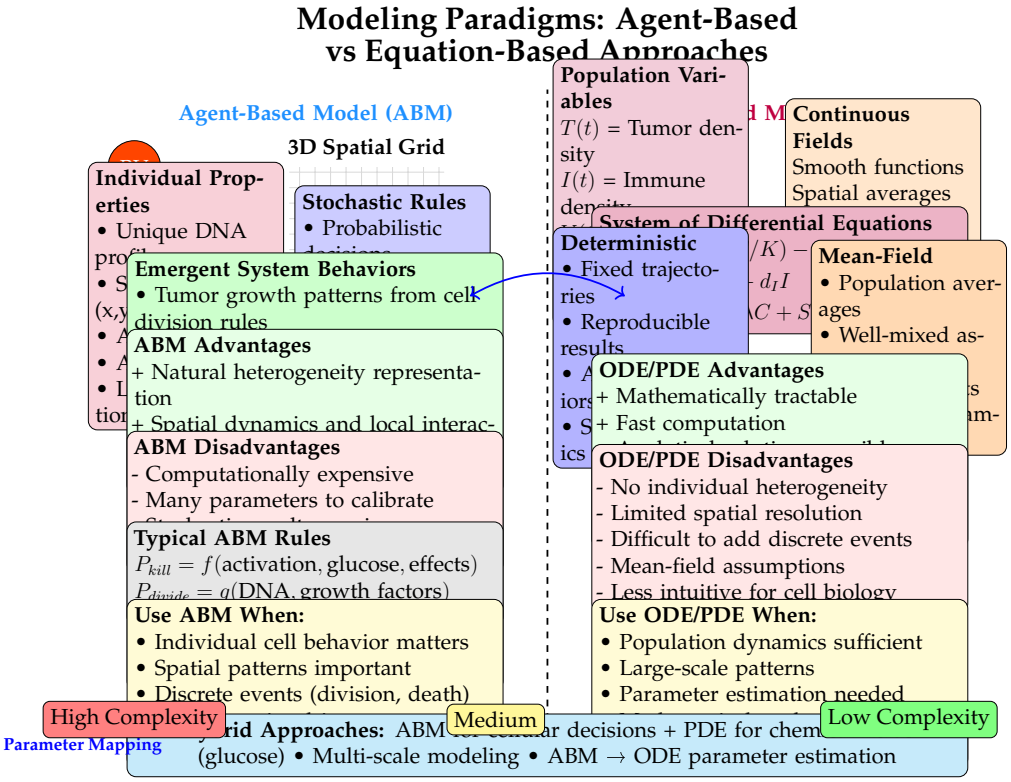


Figure 2.1: Comparison of agent-based and equation-based modeling paradigms

2.2 Immunology of Renal Cell Carcinoma

2.2.1 Innate and Adaptive Immunity

The immune system's response to cancer involves both the innate and adaptive arms. Innate immunity provides a rapid, non-specific first line of defense through cells such as natural killer (NK) cells, macrophages, dendritic cells, mast cells, and neutrophils. Adaptive immunity, mediated primarily by T and B lymphocytes, provides antigen-specific responses that can be highly effective against tumors expressing recognizable neoantigens.

In the context of RCC, the interplay between innate and adaptive immunity is critical. Dendritic cells serve as the bridge between the two arms: they phagocytose tumor cells, process their antigens, and present them to naive T cells, thereby initiating adaptive responses. The effectiveness of this chain depends on the tumor's ability to present antigens (governed by MHC expression and the antigen-processing machinery) and the availability and activation state of T cells in the tumor neighborhood.

2.2.2 T Cell Subtypes

The model includes six T cell types: CD8+ CTLs (primary tumor killers), CD8+ naive cells (CTL precursors), CD4+ naive cells (helper/Treg precursors), Th1 helpers (pro-inflammatory, anti-tumor), Th2 helpers (anti-inflammatory, mild pro-tumor), and Tregs (immunosuppressive, tumor-promoting). T cell function is modulated by activation levels, checkpoints, hormones, and glucose availability.

2.2.3 Antigen-Presenting Cells

- **Dendritic Cells (DCs).** Professional antigen-presenting cells that form the critical link between innate and adaptive immunity. In the model, DCs migrate toward tumor cells, phagocytose them (with probability modulated by the `dc_phagocitosis_effec` and then migrate toward naive T cells ($CD4^+$ or $CD8^+$, chosen randomly at creation) to present the captured antigens. A successful antigen presentation event triggers T cell activation—differentiation of a $CD8^+$ naive cell into a cytotoxic T cell or a $CD4^+$ naive cell into a Th1 or Th2 cell. When no tumor cells are in the neighborhood, DCs employ glucose chemotaxis to navigate toward nutrient-rich regions likely to harbor tumor activity.

- **Plasmacytoid Dendritic Cells (pDCs).** A specialized DC subtype with distinct functional properties. Like conventional DCs, pDCs phagocytose tumor cells and activate T cells. Additionally, pDCs can spawn new NK cells, providing a direct link between antigen presentation and innate cytotoxic capacity. pDCs also exert paracrine effects on their neighborhood: promoting angiogenesis, increasing Treg differentiation, modulating T cell proliferation, and enhancing both T cell and NK cell kill rates.

2.2.4 Macrophages

Tumor-associated macrophages (TAMs) exist along a polarization spectrum, with M1 and M2 representing the two extremes:

- **M1 Macrophages (pro-inflammatory).** M1 macrophages are classically activated and exhibit anti-tumor properties. They phagocytose tumor cells, present antigens to $CD4^+$ naive T cells (similar to DCs), and exert local effects that increase T cell kill rates and promote Th1 proliferation. When no tumor target is available, M1 macrophages use glucose chemotaxis as a fallback navigation strategy. M1 macrophages can polarize to M2 under the influence of the `macrophage_m1_mutation_effe` which is enhanced by high BMI.

- **M2 Macrophages (anti-inflammatory / pro-tumorigenic).** Alternatively activated M2 macrophages promote tumor progression. They suppress T cell killing (negative `t_kill_rate_effect`), increase tumor growth, and promote angiogenesis. M2 macrophages migrate toward tumor cells and can repolarize to M1 under appropriate conditions, although this transition is less likely than the M1-to-M2 shift in the tumor microenvironment.

The M1/M2 balance is a critical determinant of tumor outcome. A TME dominated by M1 macrophages favors immune-mediated tumor control, while M2 dominance promotes immune evasion and tumor progression. BMI influences this balance through the `macrophage_m1_mutation_effect` and `macrophage_m2_mutation_effect` fields in the effect system, reflecting clinical observations that obesity is associated with increased M2 polarization [9].

2.2.5 Natural Killer Cells

Natural killer (NK) cells are innate lymphocytes capable of recognizing and killing tumor cells without prior antigen sensitization. NK cell cytotoxicity is governed by the balance between activating and inhibitory receptors on the NK cell surface. In particular, NK cells are inhibited by MHC class I molecules on target cells; tumors that downregulate MHC-I to evade CTL recognition become more susceptible to NK-mediated killing—a phenomenon known as the "missing self" hypothesis.

In the model, NK cell kill probability is determined by the `nkl_kill_rate_effect`, which is influenced by BMI (higher BMI reduces NK effectiveness), neighboring immune cells, and local glucose concentration. NK cells also exert a positive effect on T cell kill rates in their neighborhood, providing paracrine support for adaptive immunity.

2.2.6 Mast Cells and Neutrophils

- **Mast Cells.** These innate immune cells play a dual role in the TME. Depending on stochastic initialization, individual mast cells may promote or suppress tumor growth, enhance or inhibit T cell killing, promote angiogenesis, and modulate M1 macrophage recruitment. Mast cells can also spawn dendritic cells, amplifying the antigen presentation capacity of the TME. This functional heterogeneity—captured in the model by randomly assigning effect polarities at creation—reflects the biological observation that mast cells can be either pro- or anti-tumorigenic depending on context.

- **Neutrophils.** The most abundant innate immune cells in circulation. In the current model, neutrophils are present in the microenvironment but have minimal active behavior, serving primarily as a population marker. Future extensions may incorporate neutrophil-mediated tumor killing and NETosis.

2.2.7 Immune Checkpoints: The PD-1/PD-L1 Pathway

Immune checkpoints are inhibitory pathways that normally prevent autoimmunity but are co-opted by tumors to evade immune destruction. The PD-1/PD-L1 (programmed death-1 / programmed death-ligand 1) axis is the most clinically relevant checkpoint in RCC [14].

PD-L1 (encoded by the *CD274* gene) is expressed on tumor cells and binds to PD-1 on activated T cells, delivering an inhibitory signal that reduces T cell proliferation, cytokine production, and cytotoxic activity. In the model, when a cytotoxic T cell encounters a tumor cell, the tumor cell's `get_PD1_inhibition()` method determines whether PD-1/PD-L1 engagement occurs, based on the *CD274* gene's mutation-expression score. If the checkpoint is engaged, the T cell's activation level is reduced rather than the tumor cell being killed.

2.2.8 Tumor Immune Evasion Mechanisms

RCC tumors employ multiple strategies to evade immune destruction, several of which are captured in the model:

1. **Antigen presentation loss.** Mutations in MHC, B2M, JAK1, JAK2, TAP1, or TAP2 genes reduce the probability of neoantigen presentation, making tumor cells invisible to T cells.
2. **PD-L1 upregulation.** Overexpression of the *CD274* gene product enables immune checkpoint engagement, suppressing T cell activity.
3. **Treg recruitment.** The tumor microenvironment promotes differentiation of $CD4^+$ naive T cells into immunosuppressive Tregs.
4. **M2 macrophage polarization.** Tumor-derived signals (represented through the effect system) promote M1-to-M2 repolarization, shifting the macrophage balance toward pro-tumorigenic phenotypes.
5. **Metabolic competition.** Tumor cells consume glucose at elevated rates (Warburg effect), creating nutrient-depleted zones that impair immune cell function (see Section 2.3).

2.3 Tumor Metabolism

2.3.1 The Warburg Effect

In 1956, Otto Warburg observed that cancer cells preferentially metabolize glucose through glycolysis rather than oxidative phosphorylation, even in the presence of adequate oxygen—a phenomenon now known as aerobic glycolysis or the Warburg effect [18]. While glycolysis is far less efficient than oxidative phosphorylation in terms of ATP yield per glucose molecule (2 versus ∼36 ATP), it provides several advantages to rapidly proliferating tumor cells:

- Faster ATP production rate (glycolysis has a higher flux despite lower yield).
- Generation of biosynthetic precursors (nucleotides, amino acids, lipids) for cell division.
- Acidification of the microenvironment through lactate secretion, which inhibits immune cell function.

In the model, the Warburg effect is represented by a higher glucose consumption rate for tumor cells (`w_glucose_tumor_consumption`) compared to immune cells (`w_glucose_immune_consumption`). This differential consumption creates local glucose depletion around tumor masses, particularly in poorly vascularized regions.

2.3.2 Metabolic Competition in the TME

The metabolic demands of tumor cells place them in direct competition with immune cells for limited glucose. Activated T cells, in particular, undergo a metabolic switch upon activation, becoming heavily dependent on glycolysis for their effector functions. When glucose is scarce, T cell proliferation, cytokine production, and cytotoxic capacity are all impaired [9].

This metabolic competition creates a positive feedback loop that favors the tumor: as tumor cells consume glucose, immune cells in the depleted zone become less effective, allowing further tumor expansion and greater glucose depletion. Breaking this cycle—for instance, through anti-angiogenic therapy that paradoxically normalizes vasculature and improves nutrient delivery, or through direct metabolic intervention—is an active area of therapeutic research.

2.3.3 Glucose as a Limiting Factor for Immune Function

In the model, glucose availability modulates immune cell behavior through two mechanisms:

1. **Kill probability impairment.** The kill probability of cytotoxic T cells and NK cells is scaled by a glucose factor:

$$p_{\text{kill,effective}} = p_{\text{kill,base}} \times \min\big(1,\ C_{\text{glucose}}(\mathbf{x}) \cdot w_{\text{glucose_immune_sensitivity}}\big) \tag{2.1}$$

   where $C_{\text{glucose}}(\mathbf{x})$ is the local glucose concentration at position $\mathbf{x}$. When glucose is plentiful, the factor saturates at 1; when glucose is depleted, the kill probability is proportionally reduced.

2. **Chemotaxis.** When immune cells (CTLs, M1 macrophages, DCs, NK cells) have no direct target in their neighborhood, they follow the local glucose gradient as a fallback navigation strategy:

$$\Delta\mathbf{x} = \text{sign}\big(\nabla C_{\text{glucose}}(\mathbf{x})\big) \tag{2.2}$$

   This behavior reflects the biological observation that immune cells are attracted to metabolically active regions, and it provides a plausible mechanism for immune cell infiltration toward tumor masses that have not yet been directly detected.

2.4 Treatment Modalities

2.4.1 Immune Checkpoint Inhibitors (ICI)

Immune checkpoint inhibitors (ICIs) are monoclonal antibodies that block inhibitory receptor-ligand interactions on T cells, thereby restoring anti-tumor immune activity. In the context of RCC, the most clinically relevant checkpoint is the PD-1/PD-L1 axis [14].

Anti-PD-1 antibodies (e.g., nivolumab, pembrolizumab) bind to PD-1 on T cells, while anti-PD-L1 antibodies (e.g., avelumab, atezolizumab) bind to PD-L1 on tumor cells. In both cases, the inhibitory signal is blocked, allowing T cells to recognize and kill PD-L1-expressing tumor cells.

In the model, ICI treatment is implemented through three mechanisms, each applied with probability proportional to the drug's effectiveness parameter (`w_ici_effectiveness`)

1. **PD-1/PD-L1 blockade.** The `ICI_effect` flag on tumor cells is set to `True`, causing `get_PD1_inhibition()` to return `False`—the tumor cell can no longer inhibit T cells through the PD-1 pathway.
2. **T cell reactivation.** The `t_activation_effect` of affected T cells is restored to 1.0, counteracting prior exhaustion from checkpoint engagement.
3. **Immune infiltration enhancement.** The `immune_infiltration_factor` of Th1, Th2, and mast cells is incremented, increasing the number of immune cells recruited per spawning event.

2.4.2 Tyrosine Kinase Inhibitors (TKI)

Tyrosine kinase inhibitors (TKIs) target the VEGF receptor signaling pathway, inhibiting angiogenesis and depriving the tumor of its blood supply. In ccRCC, where VEGF overexpression is driven by VHL loss, TKIs have been a cornerstone of treatment [16].

In the model, TKI treatment exerts three effects:

1. **Growth inhibition.** The `tumour_growth_effect` of affected tumor cells is reduced proportionally to the drug's effectiveness.
2. **Anti-angiogenesis.** The `angiogenesis_effect` of affected tumor cells is similarly reduced, decreasing the probability of new blood vessel formation.
3. **Neoantigen release.** Tumor cells with the `TKI_effect` flag set release neoantigens to nearby dendritic cells upon apoptosis, potentially enhancing subsequent adaptive immune responses—a mechanism that provides the immunological rationale for combining TKI with ICI.

Additionally, TKI treatment reduces Treg differentiation by decreasing the `treg_differentiat` of $CD4^+$ naive T cells, counteracting one of the tumor's immune evasion strategies.

2.4.3 Combination ICI+TKI Therapy

The combination of ICI and TKI has emerged as the standard first-line treatment for advanced RCC, based on results from landmark clinical trials including KEYNOTE-426 [16] and JAVELIN Renal 101 [13]. The rationale for combination therapy is synergistic: TKI reduces tumor burden and normalizes vasculature (improving immune cell access), while simultaneously releasing neoantigens that prime adaptive immunity; ICI then removes the PD-1/PD-L1 brake, allowing the primed immune system to mount an effective anti-tumor response.

The ARON-1 study [17] extends this paradigm by investigating whether the optimal treatment strategy depends on patient sex, motivated by the observation that sex hormones differentially modulate both the tumor and the immune response. In the model, combination therapy is implemented by applying both `ICIDrug.step()` and `TKIDrug.step()` at each treatment step, with treatment initiation occurring at the step specified by the `treatment_start` parameter.

The model supports four treatment configurations:

- **None**: No drug treatment (control/untreated arm).
- **ICI**: Immune checkpoint inhibitor monotherapy.
- **TKI**: Tyrosine kinase inhibitor monotherapy.
- **ICI+TKI**: Combination immunotherapy and targeted therapy.

This treatment framework, combined with the sex hormone system described in Chapter 7, enables the model to investigate sex-stratified treatment responses—the central question of the ARON-1 study.

Chapter 3

Model Description

The model simulates 18 different cell types interacting in a 3D tumor microenvironment, from tumor cells to T cells, representing the most comprehensive computational cancer-immune model built to date.

3.1 Overview

3.1.1 Spatial Environment

The simulation uses a 3D discrete grid with dimensions computed from tumor volume, featuring sticky borders and multiple agent occupancy. A continuous glucose concentration field overlays the discrete grid.

3.1.2 Agent Types

The model comprises 18 distinct agent types, each identified by an integer ID in the `AgentType` enumeration. Table 3.1 lists all agent types with their IDs and functional categories.

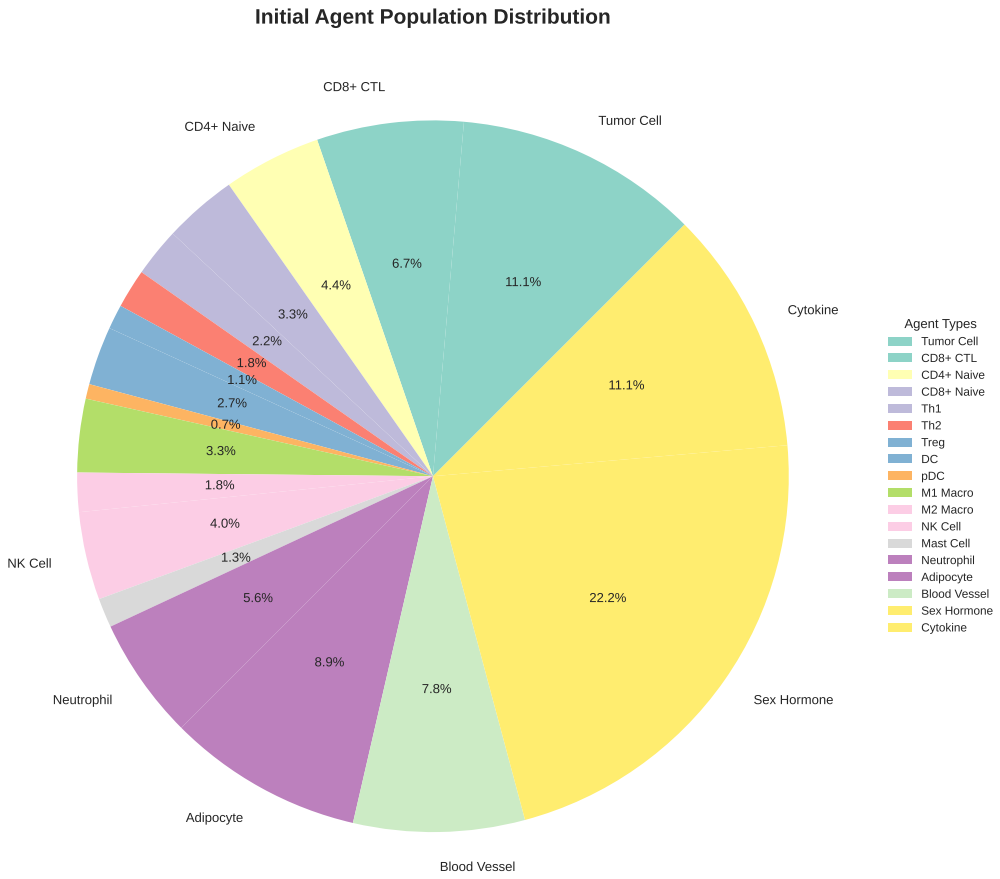


Figure 3.1: Initial agent type distribution in the RCC model. The pie chart shows the relative proportions of the 18 agent types at simulation initialization, with tumor cells forming a small but critical fraction of the total population. Immune cells dominate the initial composition, reflecting the host's surveillance capacity before tumor progression.

Table 3.1: Agent types in the RCC model.

ID	Agent Type	Class Name	Category
0	Tumor Cell	`TumorCell`	Neoplastic
1	Cytotoxic T Cell	`CytotoxicTCell`	Adaptive Immune
2	$CD8^+$ Naive T Cell	`CD8NaiveTCell`	Adaptive Immune
3	$CD4^+$ Naive T Cell	`CD4NaiveTCell`	Adaptive Immune
4	$CD4^+$ Th1 Cell	`CD4Helper1TCell`	Adaptive Immune
5	$CD4^+$ Th2 Cell	`CD4Helper2TCell`	Adaptive Immune
6	Regulatory T Cell	`TregCell`	Adaptive Immune
7	Dendritic Cell	`DendriticCell`	Innate Immune (APC)
8	Plasmacytoid DC	`PlasmacitoidDendriticCell`	Innate Immune (APC)
9	M1 Macrophage	`MacrophageM1`	Innate Immune
10	M2 Macrophage	`MacrophageM2`	Innate Immune
11	Natural Killer	`NaturalKiller`	Innate Immune
12	Mast Cell	`MastCell`	Innate Immune
13	Neutrophil	`Neutrophil`	Innate Immune
14	Adipocyte	`Adipocyte`	Stromal
15	Blood Vessel	`Blood`	Vasculature

3.1.3 Time Step Structure

Each simulation step follows a fixed sequence of operations, executed by the `RCCModel.step()` method. The ordering is designed to ensure that effects are collected before agents act, that treatment is applied at the correct time, and that spatial updates are deferred to avoid order-dependent artifacts.

1. **Collect data.** Record current population counts, glucose statistics, and kill tallies for all agent types.
2. **Check termination.** Evaluate terminal conditions (Section 3.6). If any condition is met, write the data log and halt.
3. **Apply effects.** For each agent that `needs_effect()`, collect effect objects from neighboring agents within the effect radius and sum them into the agent's `experienced_effe`
4. **Treatment.** If the current step $t \geq t_{\text{start}}$, apply all active drug effects (ICI, TKI, or both).
5. **Sex drift.** Apply the sex-dependent drift to the $CD8^+$ naive T cell population (duplication for males, removal for females), decrementing the remaining drift steps.
6. **Glucose field step.** Blood vessels inject glucose, diffusion spreads it through the 3D field, and natural decay reduces concentrations.
7. **Manage blood.** Evaluate angiogenesis conditions and create new tumor-generated blood vessels if warranted. Refresh blood access for all tumor cells via BFS from blood-connected sources.
8. **Manage search dimension.** Update the global cell search dimension based on the current tumor cell count, ensuring that immune cells can detect tumors at appropriate ranges.
9. **Shuffle and step all agents.** Take a snapshot of the agent list, shuffle it randomly to eliminate activation order bias, and invoke each agent's `step()` method. Agents may set their `_desired_pos` during this phase but are not yet moved.
10. **Deferred move.** Batch-move all agents whose `_desired_pos` differs from their current position, updating the spatial index.
11. **Spawn hormones.** Generate sex hormone agents at blood vessel locations according to sex-specific hormone chances.

3.2 Agent Hierarchy

The agent class hierarchy reflects the biological taxonomy of the cell types and the shared functionality among related agent types.

3.2.1 Base Cell Class

The Cell class provides common functionality for all cellular agents, including position tracking, deferred movement, effect collection, duplication, and neighborhood search methods.

3.2.2 TCell Base Class

The TCell class extends Cell with T lymphocyte-specific functionality: hormone perception and decay, TCR receptor matching using Hamming distance, and sex hormone modulation affecting 17 cellular behaviors.

3.2.3 Concrete Agent Classes

The 18 agent types inherit from TCell (5 T-cell types), Cell (11 cellular types), or Agent directly (2 signaling molecules).

Figure 3.2 illustrates the complete agent class hierarchy, showing the inheritance relationships and categorization of the 18 agent types.

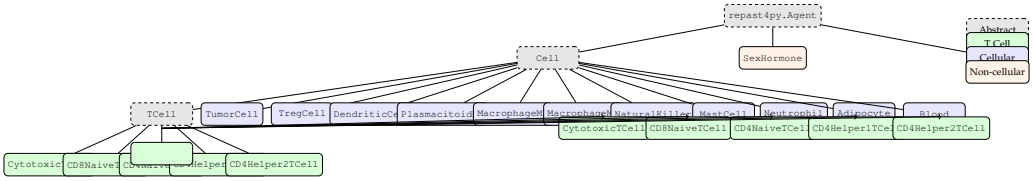


Figure 3.2: Agent class hierarchy in the RCC model. Abstract base classes are shown with dashed borders, while concrete implementations are categorized by functional type: T cells (green), other cellular agents (blue), and non-cellular signaling agents (orange).

3.3 Effect System

3.3.1 Effect Fields

The effect system is the primary mechanism for paracrine signaling between agents. Each `Cell` agent maintains an `experienced_effects` object of type `Effect` (defined in `src/systems/effect.py`), which contains 14 floating-point fields representing the accumulated influence of neighboring cells:

1. `t_kill_rate_effect` — modulates cytotoxic T cell and NK cell kill probability.
2. `t_proliferation_effect` — modulates T cell proliferation rate.
3. `t_apoptosis_effect` — modulates T cell apoptosis probability.
4. `t_activation_effect` — modulates T cell activation level (affects kill rate and proliferation multiplicatively).
5. `th1_proliferation_effect` — specifically modulates Th1 cell proliferation.
6. `th2_proliferation_effect` — specifically modulates Th2 cell proliferation.

7. `treg_differentiation_effect` — governs $CD4^+$ naive $\rightarrow$ Treg differentiation probability.
8. `macrophage_m1_mutation_effect` — governs M1 $\rightarrow$ M2 macrophage polarization probability.
9. `macrophage_m2_mutation_effect` — governs M2 $\rightarrow$ M1 macrophage repolarization probability.
10. `angiogenesis_effect` — governs the probability of new blood vessel formation.
11. `tumour_growth_effect` — modulates tumor cell duplication probability.
12. `tumour_apoptosis_effect` — modulates tumor cell apoptosis probability.
13. `dc_phagocitosis_effect` — modulates dendritic cell and pDC phagocytosis probability.
14. `nkl_kill_rate_effect` — modulates NK cell kill probability.

3.3.2 BMI Normalization

Three of the 14 effect fields are initialized with BMI-dependent baseline values rather than zero:

- `treg_differentiation_effect`: initialized to $-\hat{B} \times w_{\text{BMI_on_treg_diff}}$
- `macrophage_m1_mutation_effect`: initialized to $-\hat{B} \times w_{\text{BMI_on_m1_mutation}}$
- `macrophage_m2_mutation_effect`: initialized to $+\hat{B} \times w_{\text{BMI_on_m2_mutation}}$
- `nkl_kill_rate_effect`: initialized to $-\hat{B} \times w_{\text{BMI_nkl_kill_rate}}$

where the normalized BMI is:

$$\hat{B} = \frac{\text{BMI} - \mu_{\text{BMI}}}{\sigma_{\text{BMI}}} \tag{3.1}$$

with sex-specific parameters $\mu_{\text{BMI}} = 26.5$, $\sigma_{\text{BMI}} = 5.5$ for females and $\mu_{\text{BMI}} = 27.5$, $\sigma_{\text{BMI}} = 4.5$ for males. This normalization ensures that a BMI above the population mean increases pro-tumorigenic effects (M2 polarization, Treg differentiation) and decreases anti-tumorigenic effects (NK killing, M1 stability), consistent with clinical observations linking obesity to poorer RCC outcomes.

3.3.3 Effect Collection and Aggregation

At each simulation step, before agents execute their individual behaviors, the model's `apply_effects()` method iterates over all agents. For each agent that returns `True` from `needs_effect()`, the `collect_and_apply_effects()` method:

1. Queries the Moore neighborhood (radius 1) via the spatial index.
2. For each neighbor that is a `Cell` instance and returns `True` from `has_effect()`, retrieves its `Effect` object via `get_effect()`.
3. Aggregates all collected effects with the agent's current `experienced_effects` via the `sum_with()` method, which performs element-wise addition across all 14 fields.

The `Effect.EFFECT_RADIUS` class variable is set to 3, although the actual collection radius in the current implementation is 1 (Moore neighborhood).

3.4 Agent Interactions

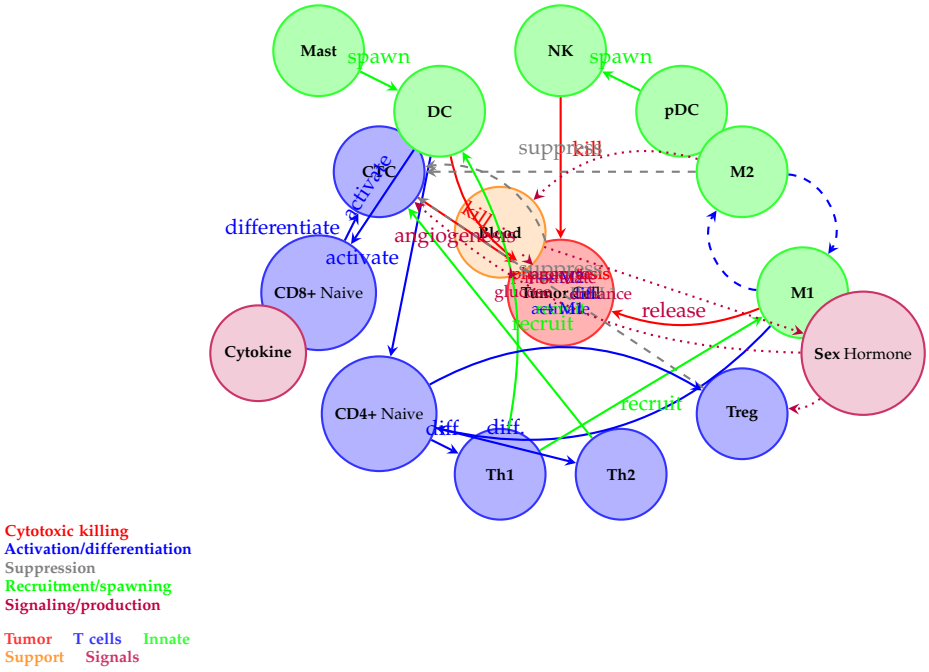


Figure 3.3: Network of agent-agent interactions in the tumor microenvironment

Table 3.2 summarizes the key interactions between agent types. Each row describes an agent type's effects on other agents, organized by the mechanism of interaction.

Table 3.2: Summary of agent interactions in the RCC model.

Agent Type	Mechanism	Effects on Other Agents
TumorCell	Growth & evasion	Proliferate, evade via PD-L1, consume glucose
CytotoxicTCell	Killing	Kill tumors, exhaustion, hormone-modulated
MacrophageM1	Anti-tumor	Phagocytosis, antigen presentation, M1→M2
MacrophageM2	Pro-tumor	Growth promotion, angiogenesis, M2→M1
TregCell	Suppression	Inhibit T cells, promote tumor
DendriticCell	Antigen presentation	Phagocytosis, T cell activation
PlasmacitoidDC	Specialized APC	Multiple effects, spawn NK cells
NaturalKiller	Innate killing	Kill tumors, glucose-dependent
MastCell	Context-dependent	Stochastic pro/anti-tumor effects
CD4NaiveTCell	Precursor	Differentiate to Th1/Th2/Treg

Continued on next page

Agent Type	Mechanism	Effects on Other Agents
CD8NaiveTCell	Precursor	Differentiate to CTL
CD4Helper1TCell	Pro-inflammatory	Recruit M1/DC, proliferate
CD4Helper2TCell	Mixed effects	Recruit immune cells, mild pro-tumor
Blood	Nutrient supply	Glucose injection, angiogenesis
SexHormone	Hormone delivery	Modulate immune responses
Neutrophil	Innate immunity	Population marker
Adipocyte	Stromal support	Structural component
Fibroblast	Stromal support	Structural component

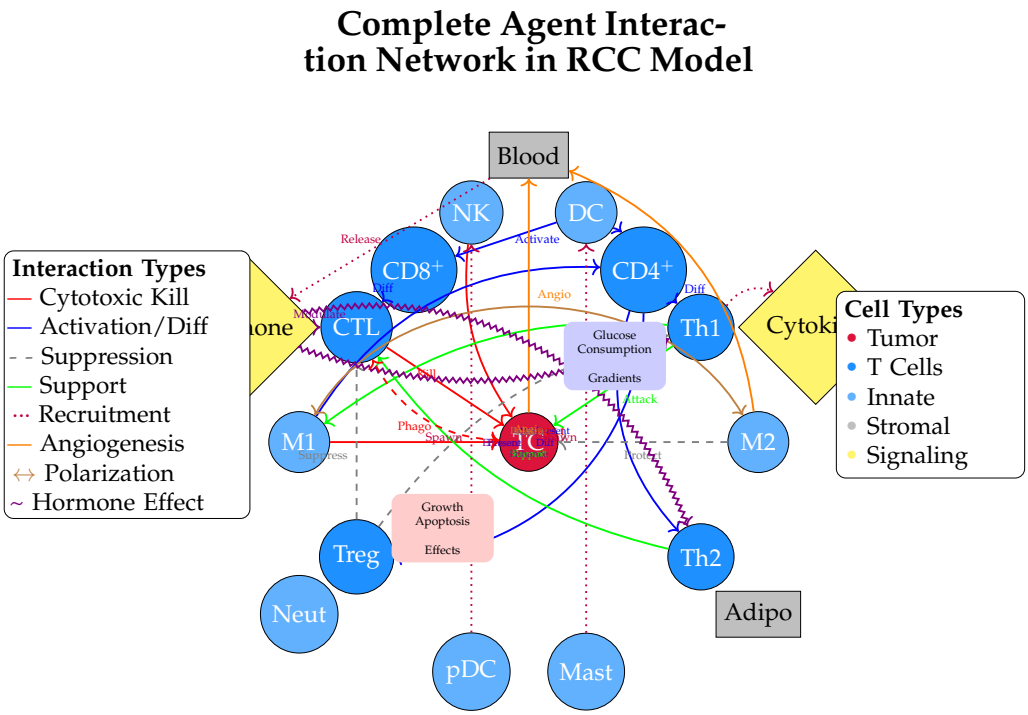


Figure 3.4: Detailed interaction network between all 18 agent types

Figure 3.5 provides a visual overview of the key interactions between agent types, showing the complex network of cytotoxic killing, activation pathways, suppression mechanisms, and recruitment cascades that govern tumor microenvironment dynamics.

[Agent interaction diagram showing tumor-immune interactions, effect propagation, and spatial relationships]

Figure 3.5: Agent interaction network in the RCC model. Arrows indicate the direction and type of interaction: red for cytotoxic killing, blue for activation/differentiation, gray dashed for suppression, green for recruitment/spawning, and purple dotted for signaling/production. Agent types are color-coded by functional category.

3.5 Movement

3.5.1 Deferred Movement Pattern

To prevent order-dependent artifacts—where an agent that moves early in the step occupies a position that a later agent expects to be empty—the model uses a deferred movement pattern:

1. During their `step()` execution, agents call `self.move(new_pos)`, which sets `self._desired_pos = new_pos` without modifying the spatial index.
2. After all agents have stepped, the model's `move_all_agents()` method iterates over all living `Cell` agents and, for each agent whose `_desired_pos` differs from `pos`, updates the spatial index and grid position.

This two-phase approach ensures that all agents make movement decisions based on the same spatial snapshot, and it eliminates bias arising from the random activation order.

3.5.2 Directed Movement: `move_towards()`

The primary movement mechanism for most agents is `move_towards(target_agent_type_id, look_up_size, distance)`, which:

1. Retrieves all living agents of the target type.
2. Finds the closest target by Manhattan distance:

$$d_M(\mathbf{a}, \mathbf{b}) = |a_x - b_x| + |a_y - b_y| + |a_z - b_z| \tag{3.2}$$

3. If the closest target is within `look_up_size`, computes the Euclidean direction vector $\mathbf{d} = \mathbf{p}_{\text{target}} - \mathbf{p}_{\text{self}}$, normalizes it, and scales by `distance`.
4. Rounds each component and clamps to grid boundaries (sticky borders).
5. Sets the resulting position as `_desired_pos`.

The `look_up_size` parameter is typically derived from the model's `cell_search_dimensio` which grows with the tumor cell count to allow immune cells to detect tumors at greater distances as the tumor mass expands.

3.5.3 Glucose Chemotaxis

When an immune cell (CTL, M1 macrophage, DC, or NK cell) fails to find a direct target via `move_towards()`, it may fall back to glucose chemotaxis. With probability `w_glucose_chemotaxis_strength`, the agent queries the glucose field for the discretized gradient at its position:

$$(\Delta x, \Delta y, \Delta z) = \text{sign}\left(\nabla C_{\text{glucose}}(\mathbf{x})\right) \tag{3.3}$$

where the gradient is computed via central finite differences with Neumann (zero-flux) boundary conditions. The agent then sets its desired position to $\mathbf{x} + (\Delta x, \Delta y, \Delta z)$, clamped to grid bounds. This mechanism causes immune cells to migrate toward glucose-rich regions—typically areas near blood vessels and actively growing tumor masses—providing a biologically plausible navigation strategy in the absence of direct antigen detection.

3.5.4 Sex Hormone Diffusion

Sex hormone agents follow a distinct movement model. At each step, a hormone agent randomly selects one of two diffusion strategies (each with 50% probability):

- **Biased diffusion.** The hormone evaluates all 26 Moore neighbors, computes the Manhattan distance from each to the nearest blood vessel, and selects a target position via weighted random choice. The weighting biases movement away from blood vessels (where hormones are spawned), promoting spatial distribution through the TME. The bias strength is controlled by a `drift_strength` parameter set to 0.25.
- **Simple diffusion.** The hormone selects a random move from a predefined set of directions biased toward positive y and z coordinates, or from the two axis-aligned directions $(0, 0, 1)$ and $(0, 1, 0)$. If the resulting position falls outside the grid, the hormone is removed from the simulation.

3.6 Terminal Conditions

The simulation terminates when any of the following conditions is met. The `terminal_condition` method returns a tuple `(terminated, survival)` indicating whether the simulation should stop and whether the outcome represents patient survival (tumor elimination).

1. **Maximum steps reached.** If $t \geq t_{\text{max}}$ (`max_steps`), the simulation terminates with `survival = False`. This represents the end of the observation period without definitive tumor clearance.
2. **Tumor elimination.** If the tumor cell count drops to zero before reaching `max_steps`, the simulation terminates with `survival = True`. This outcome indicates that the immune system (with or without treatment) successfully eliminated all tumor cells.
3. **Tumor cell count exceeds threshold.** If the number of living tumor cells reaches or exceeds `max_tumor_cells` (default: 2000), the simulation terminates with `survival = False`. This condition represents uncontrolled tumor growth that has exceeded the model's capacity and, clinically, would correspond to a large, aggressive tumor mass.

4. **Mean tumor growth rate exceeds threshold.** The mean tumor growth rate across all living tumor cells is computed as:

$$\bar{G} = \frac{1}{N_T} \sum_{i=1}^{N_T} \sigma\big(g_i + a_i,\ k = 10\big) \tag{3.4}$$

where N_T is the tumor cell count, g_i is the growth potential (combining `tumour_growth_effe` DNA-derived proliferation chance, blood access, and angiogenesis), a_i is the apoptosis-related component, and $\sigma(\cdot)$ is the sigmoid function with steepness $k = 10$. If $\bar{G} \geq w_{\text{tumour_growth_threshold}}$, the simulation terminates with `survival = False`. This condition detects cases where individual tumor growth rates are high even if the absolute cell count has not yet reached the threshold, capturing scenarios of rapidly accelerating tumor progression.

These terminal conditions collectively ensure that simulations reach a definitive outcome—either tumor elimination or tumor dominance—within a bounded number of steps, while capturing both the absolute size and the growth dynamics of the tumor population.

3.7 Immune Response Cascade Overview

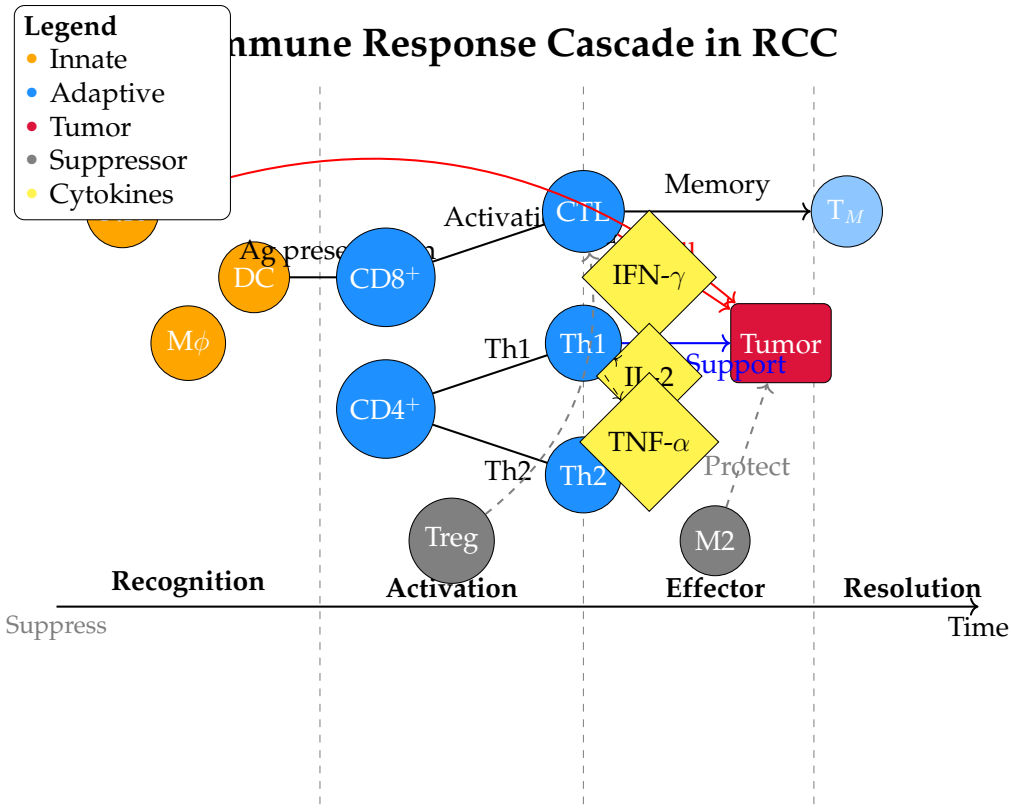


Figure 3.6: Immune response cascade from tumor detection to elimination

Figure 3.7 provides a conceptual overview of the immune response cascade as modeled in the RCC simulation. The sequence begins with tumor cell recognition and

culminates in either successful tumor elimination or immune evasion, illustrating the complex interplay between innate immunity, adaptive immunity, and tumor-mediated immunosuppression.

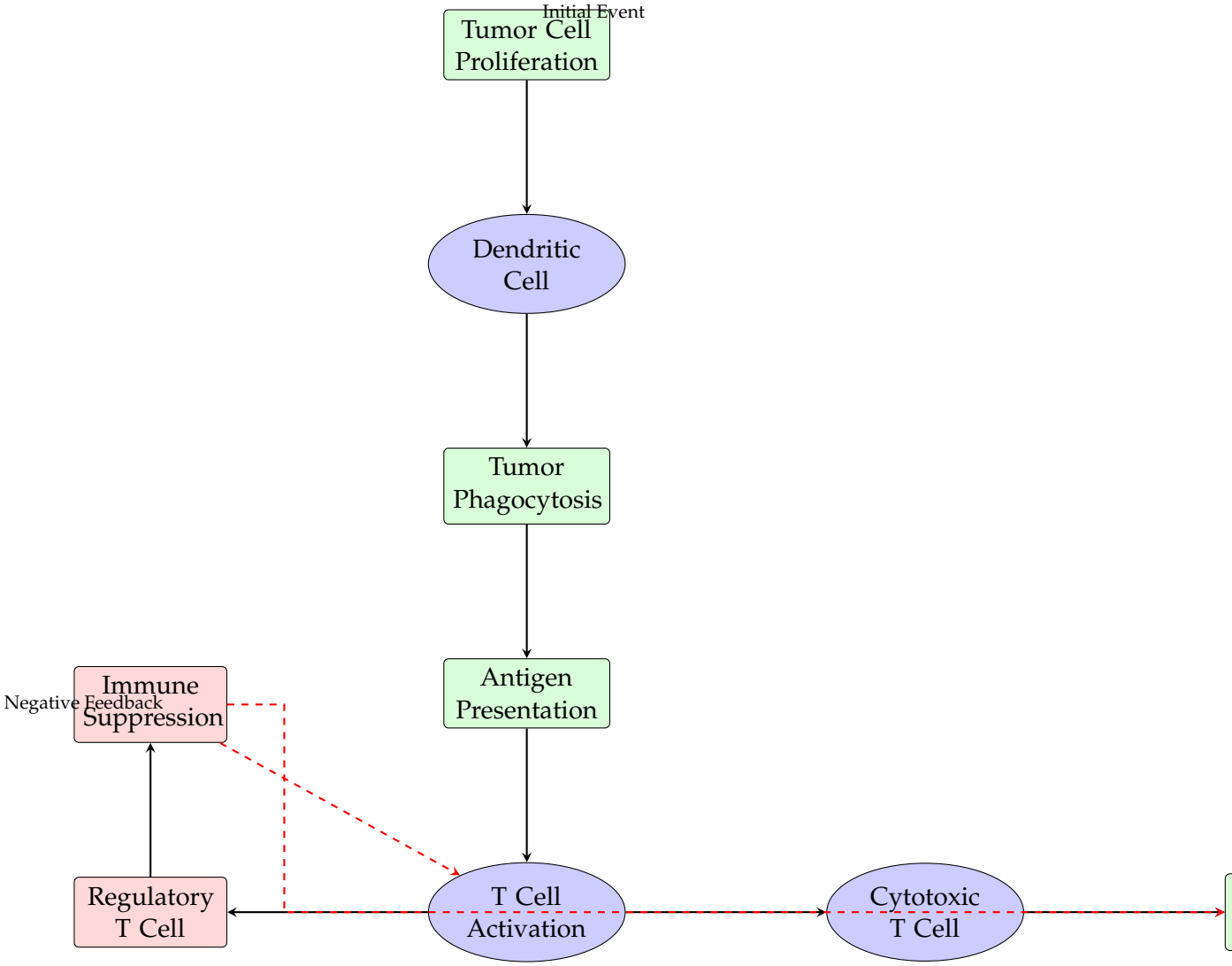


Figure 3.7: BPMN-style Immune Response Cascade in the RCC model. The process flow illustrates the sequential steps from tumor cell detection through antigen presentation to effector T cell activation and tumor elimination. Regulatory T cells provide negative feedback that can suppress both T cell activation and effector functions, creating the potential for immune evasion. Red dashed arrows indicate immunosuppressive pathways that counteract the primary immune response.

With the cellular cast assembled, we now turn to the stage itself — the metabolic landscape that determines which cells thrive and which starve. The next chapter reveals how a simple glucose field becomes the hidden battlefield where cancer and immunity wage their deadliest war.

Chapter 4

Glucose Microenvironment System

> Cancer cells consume glucose at an abnormally high rate.
>
> — Otto Warburg

Understanding how tumors starve the immune system is central to designing effective therapies. This chapter introduces the glucose field that models this metabolic warfare — and reveals a critical tipping point that determines whether the immune system wins or loses.

This chapter describes the glucose concentration field that overlays the discrete agent grid, modeling metabolic dynamics within the tumor microenvironment. The system captures diffusion, source injection from blood vessels, consumption by tumor and immune cells, and gradient-based chemotaxis for immune cell navigation.

4.1 Biological Motivation

The metabolic landscape of the tumor microenvironment plays a decisive role in determining the outcome of immune–tumor interactions. Several biological phenomena motivate the inclusion of an explicit glucose field in the model.

The Warburg Effect. In the 1920s, Otto Warburg observed that tumor cells preferentially metabolize glucose through aerobic glycolysis—even in the presence of sufficient oxygen for oxidative phosphorylation [18]. This metabolic reprogramming, now recognized as a hallmark of cancer [9], results in dramatically elevated glucose uptake by tumor cells (up to 200-fold compared to normal tissue) and the production of large quantities of lactate. The Warburg effect is not merely a metabolic curiosity; it confers a selective advantage by generating biosynthetic precursors for rapid proliferation and by acidifying the local microenvironment, which impairs immune cell function.

Metabolic Competition. Because tumor cells consume glucose at a greatly accelerated rate, the regions immediately surrounding the tumor mass become nutrient-deprived. This metabolic competition between tumor cells and infiltrating immune cells creates nutrient gradients that shape the spatial organization of the microenvironment. Immune cells arriving at the tumor periphery may encounter glucose concentrations insufficient to sustain their effector functions.

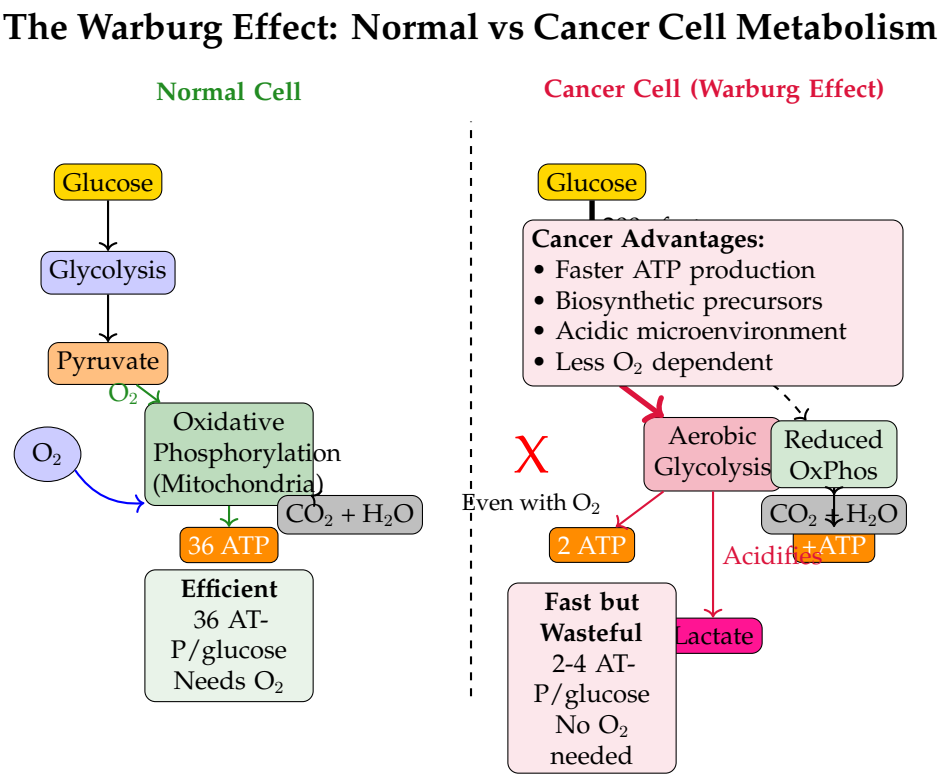


Figure 4.1: The Warburg effect: metabolic competition in the tumor microenvironment

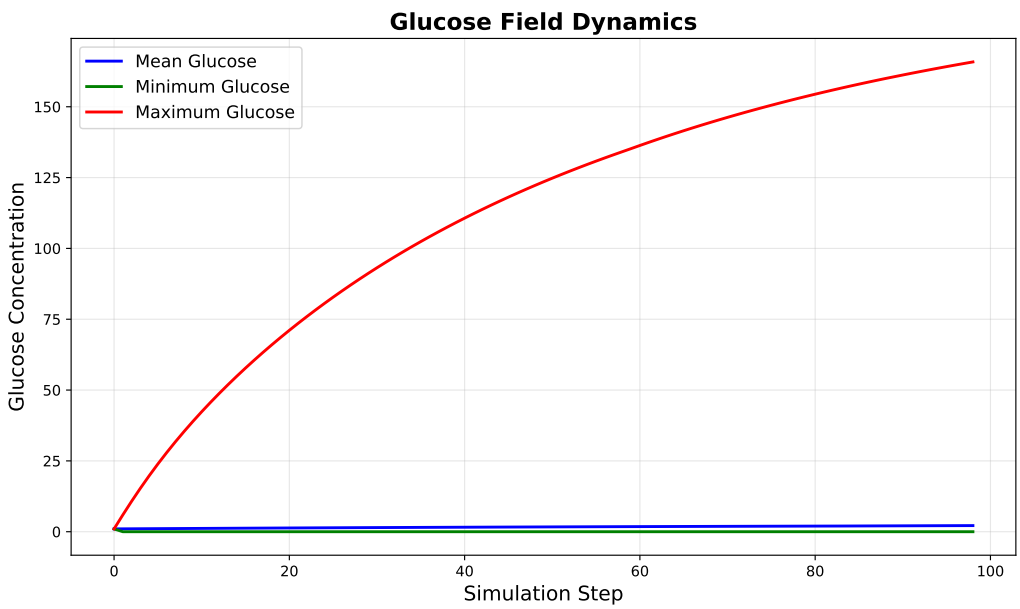


Figure 4.2: Glucose field dynamics over time showing the evolution of glucose concentration patterns in response to tumor growth and treatment. The spatial distribution reveals zones of depletion around tumor masses and gradients extending from blood vessel sources.

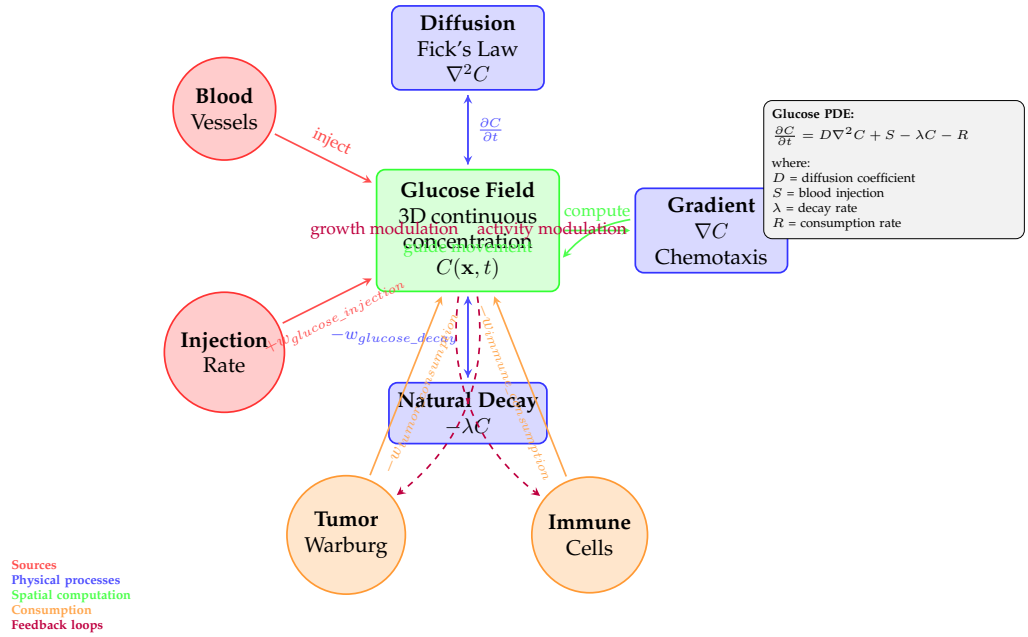


Figure 4.3: Glucose system architecture showing the diffusion field, sources from blood vessels, and consumption by cellular agents.

Impact on Immune Function. T cells and Natural Killer (NK) cells are critically dependent on glucose availability for their cytotoxic activity. Upon activation, T cells undergo a metabolic switch to aerobic glycolysis to support clonal expansion and effector function. When glucose is scarce, T cell proliferation is impaired, interferon-gamma (IFN-γ) production decreases, and cytotoxic granule release is diminished. Similarly, NK cell cytotoxicity is reduced under glucose-deprived conditions. In the model, this dependency is captured by a glucose-dependent scaling factor applied to immune cell kill rates.

Glucose Gradients as Chemotactic Signals. In the absence of direct cell–cell contact with a target, immune cells can follow glucose concentration gradients as a proxy for proximity to blood vessels and viable tissue. Regions of high glucose concentration tend to be near functional vasculature, while depleted regions indicate areas of heavy tumor metabolism. The model exploits this gradient information to provide a chemotactic fallback behavior for immune cells that have not yet located a tumor target.

4.2 Mathematical Formulation

The glucose concentration field $C(\mathbf{x}, t)$ evolves according to a reaction–diffusion equation derived from Fick's second law [6]:

$$\frac{\partial C}{\partial t} = D\,\nabla^2 C \;-\; \lambda\, C \;+\; S(\mathbf{x}) \;-\; Q(\mathbf{x}), \tag{4.1}$$

where:

- $C(\mathbf{x}, t)$ is the glucose concentration at position $\mathbf{x} = (x, y, z)$ and time t,
- D is the diffusion coefficient (`w_glucose_diffusion`, default 0.1),

- λ is the natural decay rate (`w_glucose_decay`, default 0.01),
- $S(\mathbf{x})$ is the source term (glucose injection from blood vessels),
- $Q(\mathbf{x})$ is the consumption term (uptake by tumor and immune cells).

4.2.1 Discrete 3D Laplacian

The continuous Laplacian $\nabla^2 C$ is discretized on a regular 3D Cartesian grid using the standard 6-connected stencil. For a grid point (i, j, k) with unit spacing $\Delta x = 1$:

$$\begin{aligned}\nabla^2 C(i,j,k) = {} & C(i{+}1,j,k) + C(i{-}1,j,k) \\ & + C(i,j{+}1,k) + C(i,j{-}1,k) \\ & + C(i,j,k{+}1) + C(i,j,k{-}1) \\ & - 6\,C(i,j,k).\end{aligned} \tag{4.2}$$

This discrete Laplacian is implemented as a 3D convolution using `scipy.ndimage.convolve` with a $3 \times 3 \times 3$ kernel K defined by:

$$K[1,1,1] = -6, \qquad K[\text{face neighbors}] = 1, \qquad K[\text{all others}] = 0, \tag{4.3}$$

where the six face-neighbor entries correspond to the positions $(0,1,1)$, $(2,1,1)$, $(1,0,1)$, $(1,2,1)$, $(1,1,0)$, and $(1,1,2)$ in the kernel array. All remaining 20 entries (edges and corners) are zero.

4.2.2 Time Integration

The diffusion equation is advanced in time using explicit (forward) Euler integration:

$$C^{n+1}(i,j,k) = C^n(i,j,k) + D \cdot \nabla^2 C^n(i,j,k). \tag{4.4}$$

4.2.3 Stability Analysis

By the von Neumann stability criterion for the explicit Euler scheme applied to the 3D diffusion equation on a unit grid, the stability condition is:

$$D < \frac{1}{6} \approx 0.1\overline{6}, \tag{4.5}$$

which ensures that the amplification factor remains bounded. The default value $D = 0.1$ satisfies this constraint. After each diffusion step, the field is clamped to non-negative values to prevent unphysical negative concentrations.

4.2.4 Boundary Conditions

Neumann (zero-flux) boundary conditions are enforced by passing `mode='nearest'` to `scipy.ndimage.convolve`. This mode replicates the boundary voxel values, effectively setting $\partial C/\partial \mathbf{n} = 0$ at all domain boundaries, which corresponds to no glucose flux through the walls of the simulation volume.

4.3 Source Model

Glucose enters the simulation domain through blood vessel agents, which act as point sources.

Injection Mechanism. At each simulation tick, every `Blood` agent injects glucose at its grid position:

$$C(\mathbf{x}_v) \leftarrow C(\mathbf{x}_v) + r_{\text{source}}, \tag{4.6}$$

where $\mathbf{x}_v$ is the position of blood vessel agent v and r_{source} is the injection rate per step (`w_glucose_source_rate`, default 5.0).

Blood Vessel Geometry. Blood vessels are initialized along one edge of the simulation cube in a 4×4 cross-sectional arrangement (minus the four corner positions, yielding 12 vessel positions per slice along the x-axis). This geometry creates a vascular channel that serves as the primary glucose supply for the entire domain. Additional tumor-generated blood vessels (via angiogenesis) contribute further glucose sources during the simulation.

4.4 Consumption Model

Glucose is consumed by both tumor cells and immune cells, with consumption handled individually during each agent's `step()` method.

4.4.1 Tumor Cell Consumption

Tumor cells consume glucose at a rate reflecting the Warburg effect:

$$C(\mathbf{x}_t) \leftarrow \max\big(0,\ C(\mathbf{x}_t) - r_{\text{tumor}}\big), \tag{4.7}$$

where $r_{\text{tumor}} =$ `w_glucose_tumor_consumption` (default 2.0 per step). The clamping to zero ensures that glucose concentrations remain non-negative.

4.4.2 Immune Cell Consumption

All immune cells (cytotoxic T cells, NK cells, macrophages, dendritic cells) consume glucose at a lower rate:

$$C(\mathbf{x}_i) \leftarrow \max\big(0,\ C(\mathbf{x}_i) - r_{\text{immune}}\big), \tag{4.8}$$

where $r_{\text{immune}} =$ `w_glucose_immune_consumption` (default 0.5 per step).

4.4.3 Glucose-Dependent Growth Factor

Tumor cell proliferation probability is modulated by local glucose availability:

$$f_{\text{glucose}} = \min\big(1.0,\ C(\mathbf{x}_t) \cdot w_{\text{growth_sensitivity}}\big), \tag{4.9}$$

where $w_{\text{growth_sensitivity}} = \texttt{w_glucose_growth_sensitivity}$ (default 1.0). The duplication probability is then scaled by this factor:

$$p'_{\text{dup}} = p_{\text{dup}} \cdot f_{\text{glucose}}. \tag{4.10}$$

When glucose is abundant ($C \geq 1.0$), growth proceeds at the full rate. As glucose drops below 1.0, growth is proportionally suppressed.

4.4.4 Glucose-Dependent Immune Function

Immune cell kill rates are similarly modulated:

$$f_{\text{immune}} = \min\big(1.0,\ C(\mathbf{x}_i) \cdot w_{\text{immune_sensitivity}}\big), \tag{4.11}$$

where $w_{\text{immune_sensitivity}} = \texttt{w_glucose_immune_sensitivity}$ (default 1.0). The effective kill probability becomes:

$$p'_{\text{kill}} = p_{\text{kill}} \cdot f_{\text{immune}}. \tag{4.12}$$

This mechanism captures the biological observation that T cells and NK cells require glucose to sustain their cytotoxic machinery.

Key Takeaway

The glucose consumption rate has a critical tipping point between 2.0 and 4.0 that determines treatment success. Below this threshold, immune cells retain enough metabolic fuel to attack tumors effectively. Above it, the Warburg effect starves the immune system into submission.

Figure 4.4 illustrates the complete glucose system, showing the cyclic processes of injection, diffusion, consumption, and gradient computation that govern metabolic dynamics in the tumor microenvironment.

[Glucose system diagram showing the cycle of injection, diffusion, consumption, and gradient computation with feedback loops between tumor cells, immune cells, and blood vessels]
The glucose field system is implemented through a carefully orchestrated sequence of operations executed each simulation step. The core algorithm balances biological realism with computational efficiency, leveraging vectorized NumPy operations for field updates while maintaining individual agent-level consumption behavior.

Figure 4.4: Overview of the glucose system in the RCC model. The continuous glucose field $C(\mathbf{x}, t)$ is maintained through a cycle of: (1) glucose injection from blood vessels, (2) 3D diffusion via Fick's Law, (3) natural decay, (4) consumption by tumor and immune cells, and (5) gradient computation for chemotaxis. Feedback loops modulate cell growth and activity based on local glucose availability.

4.5 Gradient and Chemotaxis

4.5.1 Gradient Computation

The glucose gradient at position (i, j, k) is computed using central finite differences:

$$\begin{aligned}
\left.\frac{\partial C}{\partial x}\right|_{(i,j,k)} &= \frac{C(i{+}1, j, k) - C(i{-}1, j, k)}{2}, \\
\left.\frac{\partial C}{\partial y}\right|_{(i,j,k)} &= \frac{C(i, j{+}1, k) - C(i, j{-}1, k)}{2}, \\
\left.\frac{\partial C}{\partial z}\right|_{(i,j,k)} &= \frac{C(i, j, k{+}1) - C(i, j, k{-}1)}{2}.
\end{aligned} \tag{4.13}$$

At domain boundaries, the indices are clamped to the valid range (consistent with the Neumann boundary conditions), so the gradient naturally approaches zero near the walls.

4.5.2 Discretized Gradient Step

For agent movement on the integer grid, the continuous gradient is converted to a discrete step direction:

$$(\Delta x, \Delta y, \Delta z) = \big(\mathrm{sign}(\nabla_x C),\ \mathrm{sign}(\nabla_y C),\ \mathrm{sign}(\nabla_z C)\big), \tag{4.14}$$

where $\mathrm{sign}(v) \in \{-1, 0, +1\}$. The resulting step is further clamped to ensure the target position lies within the grid bounds.

4.5.3 Chemotaxis Behavior

Immune cells that implement glucose chemotaxis—specifically `CytotoxicTCell`, `MacrophageM1`, `DendriticCell`, and `NaturalKiller`—employ the following fallback strategy:

1. The agent first attempts to move directly towards a nearby tumor cell using the standard `move_towards()` method.

2. If no tumor target is found (i.e., `move_towards()` returns `False`), the agent evaluates the glucose gradient at its current position.
3. With probability p_{chemo} = `w_glucose_chemotaxis_strength` (default 0.3), the agent moves one step in the direction of increasing glucose concentration.
4. If the gradient is zero (flat concentration), no movement occurs.

This mechanism provides immune cells with a biologically motivated navigation strategy: in the absence of direct target detection, cells move towards areas of higher metabolic activity, which correlates with the presence of functional vasculature and, consequently, viable tissue where tumor cells may reside.

4.6 Implementation

The glucose system is encapsulated in the `GlucoseField` class, located in `src/systems/glucose`

4.6.1 Data Structure

The glucose field is stored as a three-dimensional NumPy array of shape (`width` $\times$ `height` $\times$ `depth`) with `float64` precision, matching the dimensions of the agent grid. The initial concentration is set uniformly to 1.0.

4.6.2 Convolution Kernel

The 6-connected Laplacian kernel is constructed once during initialization as a $3 \times 3 \times 3$ NumPy array:

```
self._kernel = np.zeros((3, 3, 3), dtype=np.float64)
self._kernel[1, 1, 0] = 1.0 # z-1
self._kernel[1, 1, 2] = 1.0 # z+1
self._kernel[1, 0, 1] = 1.0 # y-1
self._kernel[1, 2, 1] = 1.0 # y+1
self._kernel[0, 1, 1] = 1.0 # x-1
self._kernel[2, 1, 1] = 1.0 # x+1
self._kernel[1, 1, 1] = -6.0 # center
```

Listing 4.1: Laplacian kernel construction.

4.6.3 Step Function

The `step(model)` method executes the three phases of the glucose dynamics in sequence:

1. **Injection:** iterate over all `Blood` agents and call `inject(pos, rate)` for each.
2. **Diffusion:** apply the discrete Laplacian convolution and update the field via explicit Euler.
3. **Decay:** multiply the entire field by $(1 - \lambda)$ where λ is the decay rate.

Agent-level consumption (by tumor and immune cells) is handled within each agent's own `step()` method, not within the `GlucoseField.step()` call. This design ensures that consumption occurs at the correct point in the agent execution order and avoids double-counting.

```python
def step(self, model):
    """Perform one simulation step of the glucose field.
    1. Blood vessels inject glucose (sources) -- vectorized.
    2. Diffusion spreads glucose.
    3. Natural decay.
    Note: Consumption by agents is handled in their own step() methods.
    """
    wp = model.weight_params

    # 1. Source injection from blood vessels (vectorized)
    if self._blood_positions_dirty:
        self.update_blood_positions(model)
    if len(self._blood_xs) > 0:
        np.add.at(self.field, (self._blood_xs, self._blood_ys,
            self._blood_zs),
                wp.w_glucose_source_rate)

    # 2. Diffusion
    self.diffuse(wp.w_glucose_diffusion)

    # 3. Decay
    self.decay(wp.w_glucose_decay)
```

Listing 4.2: Glucose field step implementation from `glucose_field.py`.

The implementation demonstrates key design patterns that reflect both the biological constraints and computational requirements of the system. The vectorized injection strategy using NumPy's `add.at` function enables efficient batch updates to multiple grid positions simultaneously, avoiding the performance penalty of iterating over individual blood vessel agents. This architectural choice embodies the principle that field-level operations should be vectorized while agent-level behaviors remain individualized.

4.6.4 Statistics Methods

The `GlucoseField` class exposes four aggregate statistics, used by the data collection system to track metabolic dynamics over time:

- `mean_concentration()` – mean glucose across all voxels,
- `total_concentration()` – total glucose in the domain,
- `min_concentration()` – minimum glucose (identifies depletion zones),
- `max_concentration()` – maximum glucose (identifies source neighborhoods).

These values are recorded at every simulation step and provide insight into the metabolic state of the microenvironment, including the formation and extent of glucose-deprived regions around the tumor mass.

```python
def step(self):
    # Glucose consumption (Warburg effect)
    self.consume_glucose(self.model.weight_params.w_glucose_tumor_consumption)

    # Tumor Apoptosis (check first)
    if self.model.rng.random() < self._apoptosis_probability():
        # [apoptosis logic omitted for brevity]
        return

```

```
    # Tumor Duplication with glucose modulation
    p_duplication = self._growth_potential()

    if self.pos is not None:
        glucose = self.model.glucose_field.get(self.pos)
        glucose_factor = min(1.0, glucose *
            self.model.weight_params.w_glucose_growth_sensitivity)
        p_duplication *= glucose_factor

    if self.model.rng.random() <= p_duplication:
        self.duplicate()
```

Listing 4.3: Tumor cell glucose consumption and growth modulation from `tumor_cell.py`.

This implementation illustrates a critical design principle: metabolic constraints operate as multiplicative modifiers of biological processes rather than absolute determinants. The glucose factor calculation `min(1.0, glucose * sensitivity)` ensures that glucose availability can only reduce growth rates, never enhance them beyond the baseline genetic potential. This reflects the biological reality that nutrient availability enables growth but cannot exceed the intrinsic proliferative capacity encoded in the cell's mutational profile.

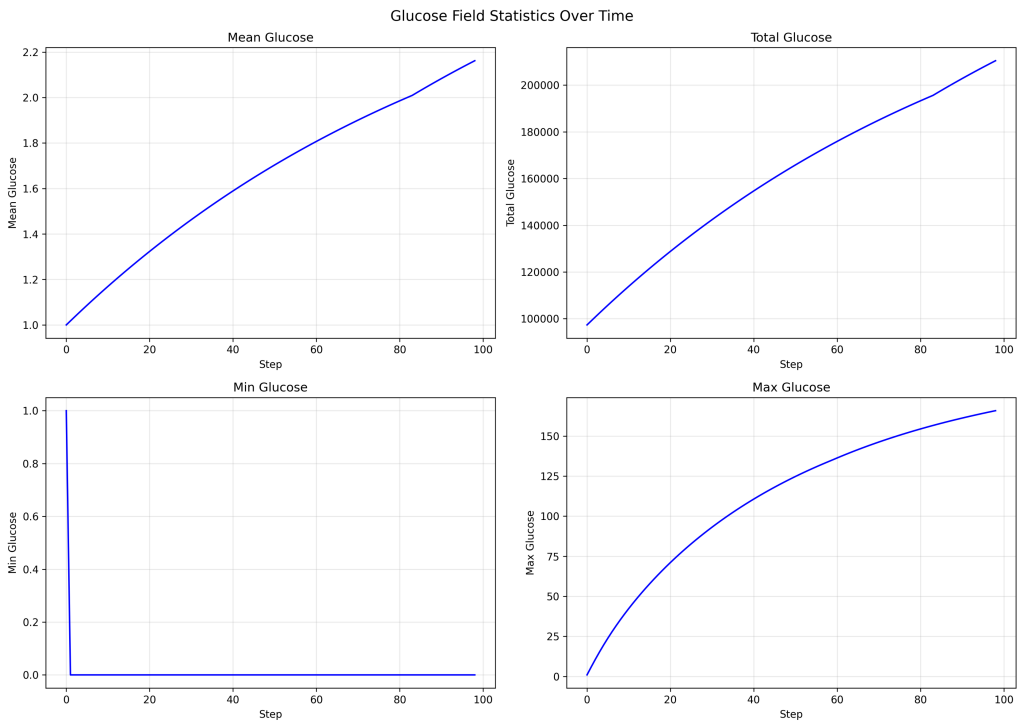


Figure 4.5: Glucose concentration time series showing the temporal evolution of mean, minimum, and maximum glucose levels throughout the simulation. The dynamics reflect the balance between blood vessel injection, cellular consumption, and spatial diffusion.

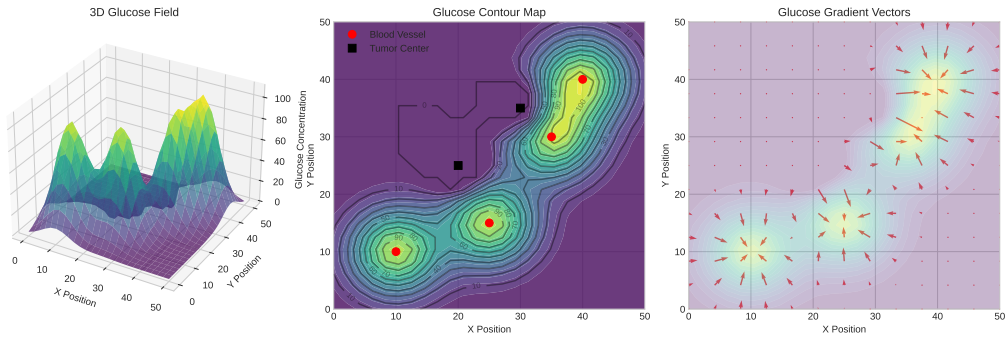


Figure 4.6: Three-dimensional visualization of the glucose concentration field showing spatial heterogeneity within the tumor microenvironment. High-concentration regions (yellow) correspond to blood vessel locations, while low-concentration zones (blue) indicate areas of intensive cellular consumption.

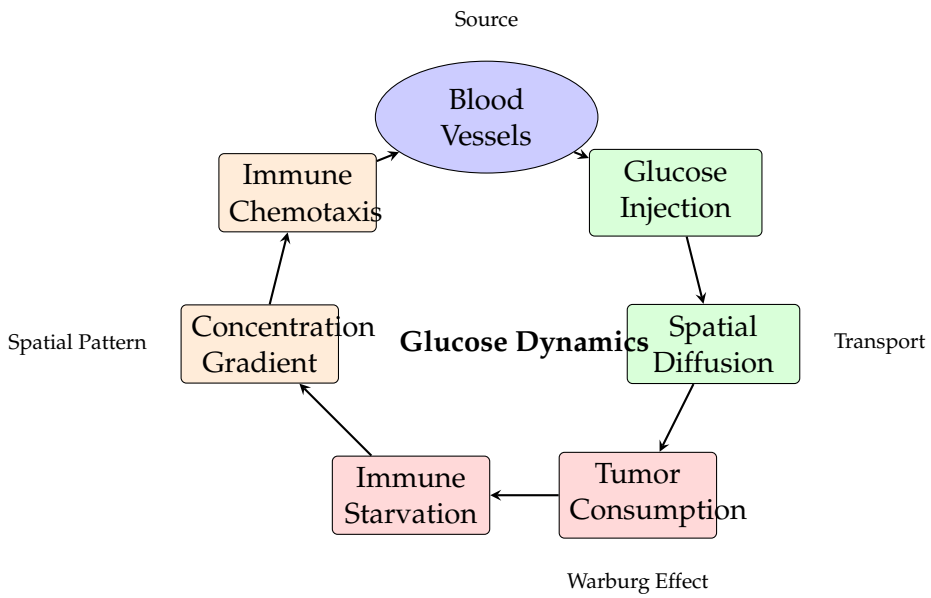


Figure 4.7: BPMN-style Glucose System Cycle illustrating the circular flow of glucose dynamics in the tumor microenvironment. Blood vessels inject glucose which diffuses spatially and is consumed preferentially by tumor cells (Warburg effect), creating concentration gradients that guide immune cell chemotaxis back toward glucose-rich regions near blood vessels. This creates a continuous cycle where metabolic competition shapes spatial organization.

Chapter 5

DNA and Mutation System

Every cancer cell is genetically unique, and this diversity determines which treatments succeed or fail. This chapter reveals how 16 key genes create tumor fingerprints — and shows why males start with three times more cancer-driving mutations than females.

This chapter describes the gene-level representation used by tumor cells in the model. The DNA system encodes 16 cancer-relevant genes, supports codon-level translation, introduces mutations during cell division with genomic-instability-dependent rates, and derives phenotypic parameters that govern tumor behavior and immune evasion.

5.1 Gene Model

Each tumor cell carries a `DNA` object that encodes a synthetic DNA sequence containing 16 genes chosen for their relevance to renal cell carcinoma immunology. The genes span the major biological axes of antigen presentation, immune checkpoint signaling, DNA repair, tumor suppression, and oncogenic proliferation.

5.1.1 Gene Catalogue

Table 5.1 lists all 16 genes, their functional categories, and their roles in the model.

Table 5.1: The 16 genes modeled in the DNA system and their functional roles.

Gene	Category	Functional Role
MHC	Antigen Presentation	Major histocompatibility complex; presents peptide antigens on the cell surface for T cell recognition.
B2M	Antigen Presentation	Beta-2-microglobulin; essential component of MHC class I molecules.
TAP1	Antigen Presentation	Transporter associated with antigen processing 1; loads peptides onto MHC I.
TAP2	Antigen Presentation	Transporter associated with antigen processing 2; works with TAP1 for peptide loading.
JAK1	Antigen Presentation	Janus kinase 1; IFN-γ signaling for MHC upregulation.

Continued on next page

Gene	Category	Functional Role
JAK2	Antigen Presentation	Janus kinase 2; cooperates with JAK1 in interferon signaling.
CD274	Immune Checkpoint	PD-L1; binds PD-1 on T cells to induce immune checkpoint inhibition.
BRCA1	DNA Repair	Breast cancer type 1 susceptibility protein; key DNA double-strand break repair gene controlling genomic stability.
TP53	Tumor Suppression	Tumor protein p53; "guardian of the genome," triggers apoptosis in damaged cells.
RB1	Tumor Suppression	Retinoblastoma protein; cell cycle checkpoint regulator.
CDKN2A	Tumor Suppression	Cyclin-dependent kinase inhibitor 2A (p16); inhibits CDK4/6 to arrest cell cycle.
MYC	Oncogene	Transcription factor promoting cell growth and proliferation.
RAS	Oncogene	GTPase signal transducer in the MAPK proliferation pathway.
EGFR	Oncogene	Epidermal growth factor receptor; receptor tyrosine kinase driving proliferation.
HER2	Oncogene	Human epidermal growth factor receptor 2; amplified in aggressive tumors.
APC	Tumor Suppression	Adenomatous polyposis coli; Wnt pathway regulator, loss promotes proliferation.

5.1.2 DNA Sequence Structure

Each gene occupies a defined region within a synthetic DNA sequence. The gene metadata specifies, for each gene: a start position (offset in the full DNA string), a coding region length of 90 base pairs, and a wildtype protein sequence of 20 amino acids. The full DNA sequence is assembled during initialization by concatenating, for each gene:

1. A random nucleotide spacer (100 bp),
2. A promoter region: 13 random bases, the `TATA` box motif, and 13 more random bases (total 30 bp),
3. The coding region: codon-encoded wildtype protein (60 bp for 20 amino acids),
4. A stop codon (`TAA`),
5. A trailing spacer (7 bp).

The start positions are fixed (at offsets 100, 300, 500, ..., 3100 for the 16 genes), ensuring deterministic gene location across all tumor cell lineages. The wildtype protein sequences are fixed 20-character strings drawn from the standard amino acid alphabet.

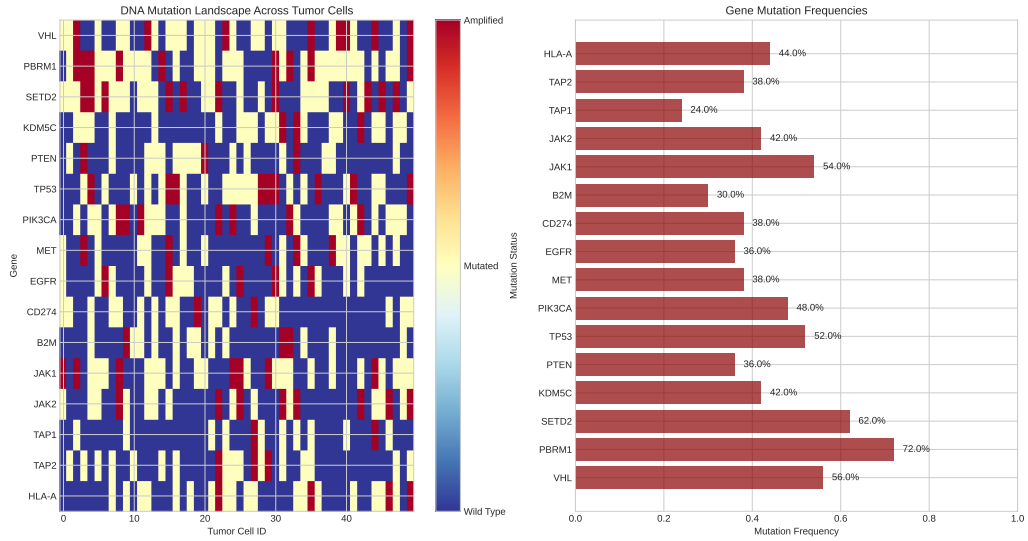


Figure 5.1: DNA mutation heatmap across the 16 modeled genes showing the distribution of mutations in the tumor cell population. Each row represents a gene and each column represents a tumor cell lineage. Color intensity indicates the severity of mutation (darker colors represent higher mutation scores). The heterogeneous mutation patterns drive clonal diversity and immune evasion.

5.2 Codon Translation

The model implements the full 64-codon table mapping nucleotide triplets to amino acids and stop signals, with translation terminating upon encountering stop codons.

```
def gene_to_protein(self, gene_name: str) -> str:
    protein = ""
    start = DNA.gene_metadata[gene_name]["start"]
    for i in range(start, len(self.dna) - 2, 3):
        codon = self.dna[i:i + 3]
        aa = self.codon_table.get(codon, 'X')
        if aa == '_':
            break
        protein += aa
    return protein
```

Listing 5.1: Gene-to-protein translation.

5.3 Mutation Mechanics

Mutations accumulate through injected mutations at initialization and stochastic point mutations during DNA duplication. Genomic instability is controlled by BRCA1 mutation status, with mutation rate scaling with BRCA1 dysfunction. Initial mutations are sex-dependent: males start with 3 BRCA1 mutations, females with 1, establishing different baseline genomic instability patterns.

5.4 Gene Expression and Phenotypic Effects

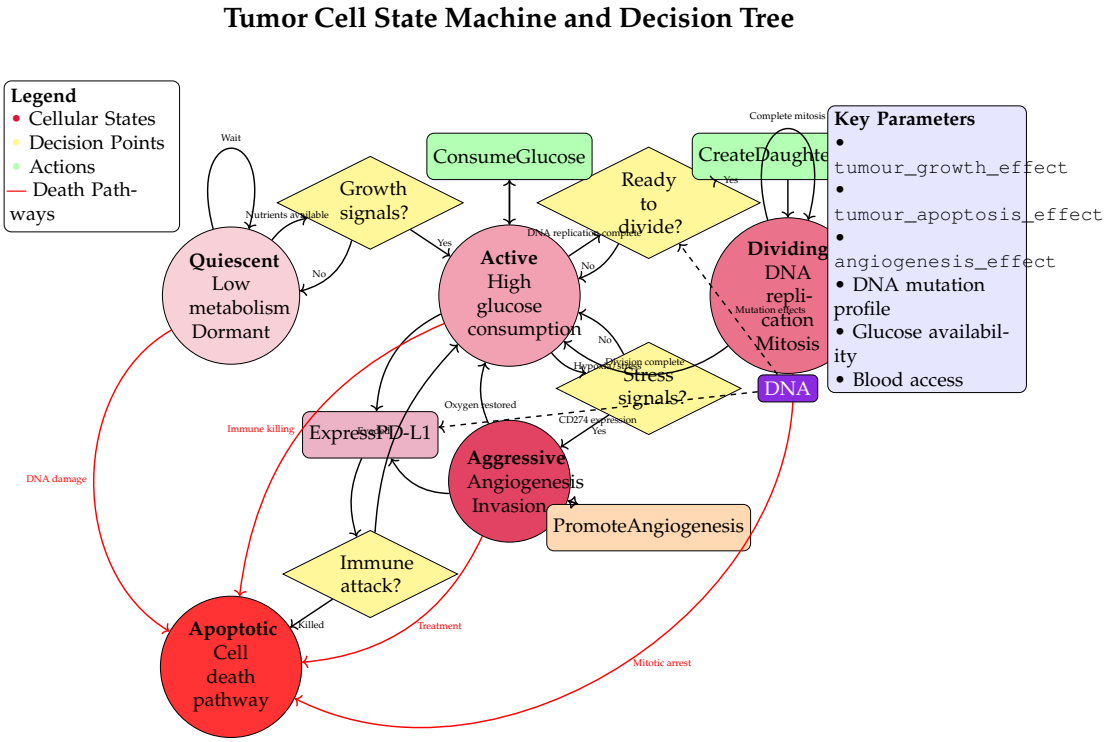


Figure 5.2: State machine diagram of tumor cell lifecycle

Each gene's contribution to tumor phenotype is determined by two quantities: promoter expression and mutation score.

5.4.1 Promoter Expression

Gene expression is modeled as a count of `TATA` motifs in the 30-nucleotide promoter region upstream of the coding sequence:

$$e_g = \#\{\texttt{TATA} \text{ occurrences in promoter region of gene } g\}. \tag{5.1}$$

In the wildtype sequence, each gene has exactly one `TATA` box ($e_g = 1$). Mutations in the promoter region can destroy or create additional `TATA` motifs, thereby altering expression.

5.4.2 Mutation Score

The mutation score quantifies how far the translated protein has diverged from the wildtype:

$$s_g = 1 - \text{Levenshtein.ratio}\big(p_g^{\text{wt}},\ p_g^{\text{mut}}\big), \tag{5.2}$$

where p_g^{wt} is the wildtype protein sequence, p_g^{mut} is the mutant protein, and Levenshtein.ratio returns the normalized similarity (1.0 for identical strings, 0.0 for completely different strings). A mutation score of $s_g = 0$ indicates a wildtype protein, while $s_g = 1$ indicates maximal divergence.

5.4.3 Gene Mutation–Expression Product

Several phenotypic parameters use the product of expression and mutation score:

$$\phi_g = e_g \cdot s_g, \tag{5.3}$$

which captures the idea that a gene must be both expressed and mutated to exert a phenotypic effect.

5.4.4 Five Derived Phenotypic Parameters

The DNA system computes five aggregate phenotypic parameters from the 16 individual gene states. These parameters directly influence tumor cell behavior during simulation.

1. Antigen Presentation Chance.

$$p_{\text{AP}} = \max\left(0,\ 6 - \frac{\sum_{g \in \mathcal{G}_{\text{AP}}} \phi_g}{6}\right), \tag{5.4}$$

where $\mathcal{G}_{\text{AP}} = \{\text{MHC, JAK1, JAK2, B2M, TAP1, TAP2}\}$. This represents the tumor cell's ability to present antigens on its surface. Mutations in antigen presentation genes reduce this probability, enabling immune evasion.

2. Checkpoint Pathway Inhibition Chance.

$$p_{\text{CPI}} = \phi_{\text{CD274}}, \tag{5.5}$$

determined solely by the CD274 (PD-L1) gene. Higher expression of mutated PD-L1 increases the probability that the tumor cell will engage the PD-1 checkpoint on interacting T cells, suppressing their cytotoxic activity.

3. Genomic Instability.

$$\mu_{\text{instability}} = \left(s_{\text{BRCA1}}\right)^{\max(1,\, e_{\text{BRCA1}})} \times 0.05, \tag{5.6}$$

as described in Section 5.3. This parameter controls the mutation rate during DNA duplication.

4. Tumor Suppression Chance.

$$p_{\text{TS}} = \frac{1}{3} \sum_{g \in \mathcal{G}_{\text{TS}}} e_g \cdot (1 - s_g), \tag{5.7}$$

where $\mathcal{G}_{\text{TS}} = \{\text{TP53, RB1, CDKN2A}\}$. Functional (unmutated, expressed) tumor suppressors increase the probability of apoptosis. Loss-of-function mutations ($s_g \to 1$) diminish this protective effect.

5. Extra Proliferation Chance.

$$p_{\text{EP}} = \frac{1}{5}\left(\sum_{g\in\mathcal{G}_{\text{onco}}} e_g \cdot (1 - s_g) + \phi_{\text{APC}}\right), \tag{5.8}$$

where $\mathcal{G}_{\text{onco}} = \{\text{MYC, RAS, EGFR, HER2}\}$. This parameter captures oncogene-driven proliferation. Functional oncogenes and mutated APC (a tumor suppressor whose loss promotes proliferation via the Wnt pathway) both contribute to increased growth. The raw value is converted to a probability using a saturating transformation:

$$\hat{p}_{\text{EP}} = 1 - e^{-k\cdot p_{\text{EP}}}, \qquad k = 0.01. \tag{5.9}$$

5.5 Neoantigen Generation

The mutation system provides the molecular basis for adaptive immune recognition of tumor cells.

5.5.1 Neoantigen Definition

Neoantigens are short peptide fragments derived from mutated proteins. In the model, neoantigens are represented as 10-mer subsequences (10 consecutive amino acids) extracted from each translated protein:

$$\mathcal{N} = \bigcup_{g\in\mathcal{G}} \left\{ p_g^{\text{mut}}[i : i{+}10] \;\middle|\; 0 \le i \le 9 \right\}, \tag{5.10}$$

where p_g^{mut} is the mutated protein sequence for gene g and the set union is taken over all 16 genes.

5.5.2 Self-Antigens

The set of self-antigens is computed from the wildtype protein sequences using the same 10-mer extraction:

$$\mathcal{S} = \bigcup_{g\in\mathcal{G}} \left\{ p_g^{\text{wt}}[i : i{+}10] \;\middle|\; 0 \le i \le |p_g^{\text{wt}}| - 10 \right\}. \tag{5.11}$$

Self-antigens are computed once (as a class-level cached set) since the wildtype sequences are identical across all cells.

5.5.3 TCR Matching

T cell receptors (TCRs) recognize neoantigens through a matching mechanism based on Hamming distance:

1. The 10-mer antigen peptide is hashed to an 8-bit value using SHA-256 (taking the first byte of the digest).
2. The T cell's receptor is a random 8-bit integer generated at initialization.

3. The Hamming distance (number of differing bits) between the antigen hash and the receptor is computed.
4. A match is declared if the antigen is *not* a self-antigen and the Hamming distance is within a threshold:

$$\text{match}(a, r) = \big(a \notin \mathcal{S}\big) \ \wedge \ \big(d_H\big(h(a),\, r\big) \leq 3 + \Delta_r\big), \tag{5.12}$$

where $h(a)$ is the 8-bit hash of antigen a, r is the receptor value, d_H denotes Hamming distance, and Δ_r is the `receptor_threshold_variation` parameter (default 1, giving a threshold of 4). This mechanism ensures that T cells only respond to foreign (non-self) peptides and provides a stochastic, receptor-specific recognition model.

These genetic mutations create vulnerabilities that treatments can exploit. The next chapter examines exactly how immune checkpoint inhibitors and targeted therapies turn cancer's own molecular weaknesses against itself.

Chapter 6

Treatment System

These genetic mutations create vulnerabilities that treatments can exploit. This chapter examines exactly how immune checkpoint inhibitors and targeted therapies work — and reveals why the ICI+TKI combination is more than the sum of its parts.

This chapter describes the pharmacological treatment layer of the model, which simulates Immune Checkpoint Inhibitors (ICI), Tyrosine Kinase Inhibitors (TKI), and their combination. The treatment system is activated after a configurable number of simulation steps and modulates tumor and immune cell behaviors through drug-specific mechanisms.

6.1 Drug Architecture

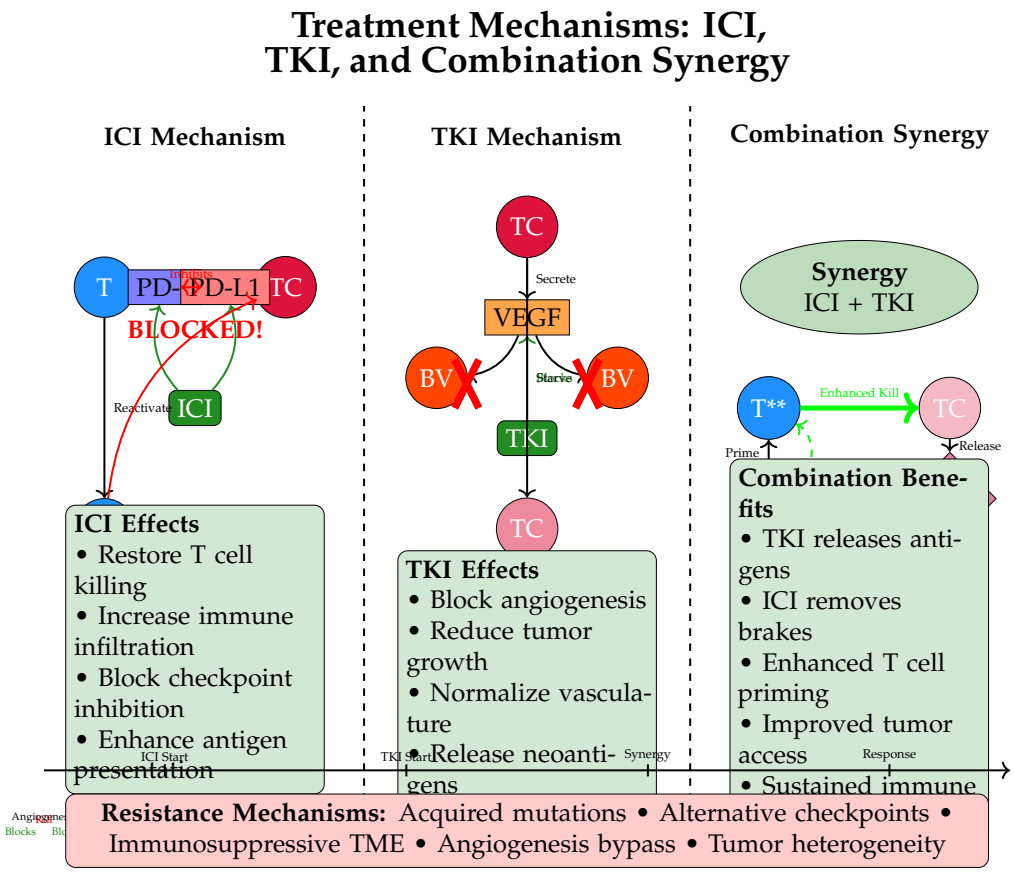


Figure 6.1: ICI and TKI treatment mechanisms and their targets

The treatment system follows a composition-based design in which individual drugs are combined into a treatment regimen.

6.1.1 Abstract Drug Base Class

All drugs inherit from an abstract `Drug` base class with a `step()` method. The `proportion` parameter controls the treatment allocation fraction.

6.1.2 Treatment Composition

The `Treatment` class distributes capacity equally among drugs. With single drug therapy, proportion = 1.0; with combination therapy, each drug receives 0.5, modeling clinical dose adjustments.

6.1.3 Treatment Activation

Treatment does not begin at the start of the simulation. Instead, it is activated after a configurable number of steps specified by the `treatment_start` patient parameter (default: step 100). Before this time point, the model simulates unperturbed tumor–immune dynamics, establishing a baseline disease state. At each step from `treatment_start` onward, `Treatment.step()` is called:

$$\text{Treatment active at step } t \iff t \geq t_{\text{start}}. \tag{6.1}$$

The available treatment configurations, selected via the `treatment` patient parameter, are:

- `"None"` – no drugs (control),
- `"ICI"` – immune checkpoint inhibitor monotherapy,
- `"TKI"` – tyrosine kinase inhibitor monotherapy,
- `"ICI+TKI"` – combination therapy.

6.2 Immune Checkpoint Inhibitors (ICI)

6.2.1 Biological Basis

PD-1/PD-L1 checkpoint interaction suppresses T cell cytotoxicity. ICI antibodies block this pathway, restoring T cell function and demonstrating clinical benefits in RCC [14, 16].

6.2.2 Implementation

`ICIDrug` implements three mechanisms: (1) *PD-L1 blockade* on tumor cells, disabling checkpoint inhibition; (2) *T cell activation restoration* across all T cell subtypes; and (3) *immune infiltration enhancement*. Each effect occurs with probability proportional to drug effectiveness.

$$\texttt{t_activation_effect} \leftarrow 1.0. \tag{6.2}$$

This resets the T cell's activation state to full capacity, counteracting the progressive exhaustion that occurs when T cells encounter PD-L1$^+$ tumor cells without treatment.

3. Immune Infiltration Enhancement. For immune cell types Th1, Th2, and MastCell, with the same probability, the `immune_infiltration_factor` is incremented:

$$\texttt{immune_infiltration_factor} \leftarrow \texttt{immune_infiltration_factor} + 1. \quad (6.3)$$

This models the clinical observation that ICI therapy promotes immune cell infiltration into the tumor mass, as the restored immune microenvironment generates stronger chemotactic signals.

6.3 Tyrosine Kinase Inhibitors (TKI)

6.3.1 Biological Basis

Vascular endothelial growth factor (VEGF) is a key driver of angiogenesis—the formation of new blood vessels that supply the tumor with nutrients and oxygen. Renal cell carcinoma is particularly dependent on VEGF signaling due to frequent VHL gene inactivation, making anti-angiogenic therapy a cornerstone of RCC treatment.

Tyrosine kinase inhibitors (e.g., axitinib, sunitinib) block the intracellular signaling domains of VEGF receptors and other receptor tyrosine kinases, thereby inhibiting tumor-driven angiogenesis and directly reducing tumor cell proliferation [16].

6.3.2 Implementation

The `TKIDrug` class (in `src/treatments/tki.py`) implements four effects:

1. Tumor Growth Suppression. For each tumor cell, with probability `proportion` $\times$ `w_tki_effectiveness`, the tumor growth effect is reduced:

$$\texttt{tumour_growth_effect} \leftarrow \texttt{tumour_growth_effect} \times (1 - \varepsilon), \quad (6.4)$$

where $\varepsilon = p \cdot w_{\text{TKI}}$ is the effective TKI strength.

2. Angiogenesis Inhibition. With the same per-cell probability, the angiogenesis effect is reduced:

$$\texttt{angiogenesis_effect} \leftarrow \texttt{angiogenesis_effect} \times (1 - \varepsilon). \quad (6.5)$$

This directly suppresses the tumor cell's ability to recruit new blood vessels, reducing nutrient supply and constraining spatial expansion.

3. Neoantigen Release. With probability ε, the tumor cell's `TKI_effect` flag is set to `True`. When a tumor cell with `TKI_effect = True` subsequently undergoes apoptosis, it releases its neoantigens to nearby dendritic cells (within radius 3), facilitating antigen presentation and downstream T cell activation. This models the immunogenic cell death induced by TKI therapy.

4. Treg Differentiation Suppression. For each CD4 naive T cell, with probability ε:

$$\texttt{treg_differentiation_effect} \leftarrow \texttt{treg_differentiation_effect} \times (1-\varepsilon). \tag{6.6}$$

TKI therapy reduces the differentiation of naive $CD4^+$ T cells into immunosuppressive regulatory T cells (Tregs), thereby shifting the immune balance towards anti-tumor activity.

6.4 Combination Therapy

6.4.1 ICI + TKI

When both ICI and TKI are administered simultaneously, the `Treatment` class creates both drug instances, each receiving $\texttt{proportion} = 0.5$:

$$p_{\text{ICI}} = p_{\text{TKI}} = \frac{1}{2}. \tag{6.7}$$

The effective strength of each drug is halved compared to monotherapy, reflecting the dose adjustments common in combination regimens.

6.4.2 Synergistic Mechanisms

The combination of ICI and TKI exploits complementary mechanisms that produce synergistic anti-tumor effects:

1. **Vascular normalization:** TKI reduces aberrant tumor vasculature, which paradoxically improves blood flow to the tumor center. This normalized vasculature allows more immune cells to infiltrate the tumor mass.
2. **Restored immune function:** ICI blocks the PD-1/PD-L1 checkpoint, restoring the cytotoxic capacity of the infiltrating T cells. Without ICI, even well-infiltrated T cells may be rendered anergic by PD-L1 engagement.
3. **Immunogenic cell death:** TKI-induced tumor cell death releases neoantigens, priming new adaptive immune responses. ICI ensures that the T cells activated by these neoantigens remain functional.
4. **Treg reduction:** TKI suppresses Treg differentiation, reducing the immunosuppressive component of the tumor microenvironment, while ICI restores the activity of effector T cells.

6.4.3 Clinical Evidence

The clinical rationale for ICI + TKI combination therapy in RCC is well established. The KEYNOTE-426 trial demonstrated that pembrolizumab (anti-PD-1) combined with axitinib (TKI) significantly improved overall survival and progression-free survival compared to sunitinib monotherapy in advanced RCC [16]. Similarly, the JAVELIN Renal 101 trial showed that avelumab (anti-PD-L1) plus axitinib improved progression-free survival over sunitinib [13]. The ARON-1 study further examined sex-specific response patterns in this combination setting [17].

The model's treatment system is designed to reproduce the qualitative dynamics observed in these trials: combination therapy produces stronger tumor regression than either monotherapy, and the relative benefit depends on the patient's immune profile and sex-specific parameters (see Chapter 7). The treatment architecture embodies a modular design philosophy where individual drug mechanisms can be combined arbitrarily, enabling exploration of novel combination strategies and personalized treatment protocols based on patient-specific immune landscapes and tumor characteristics.

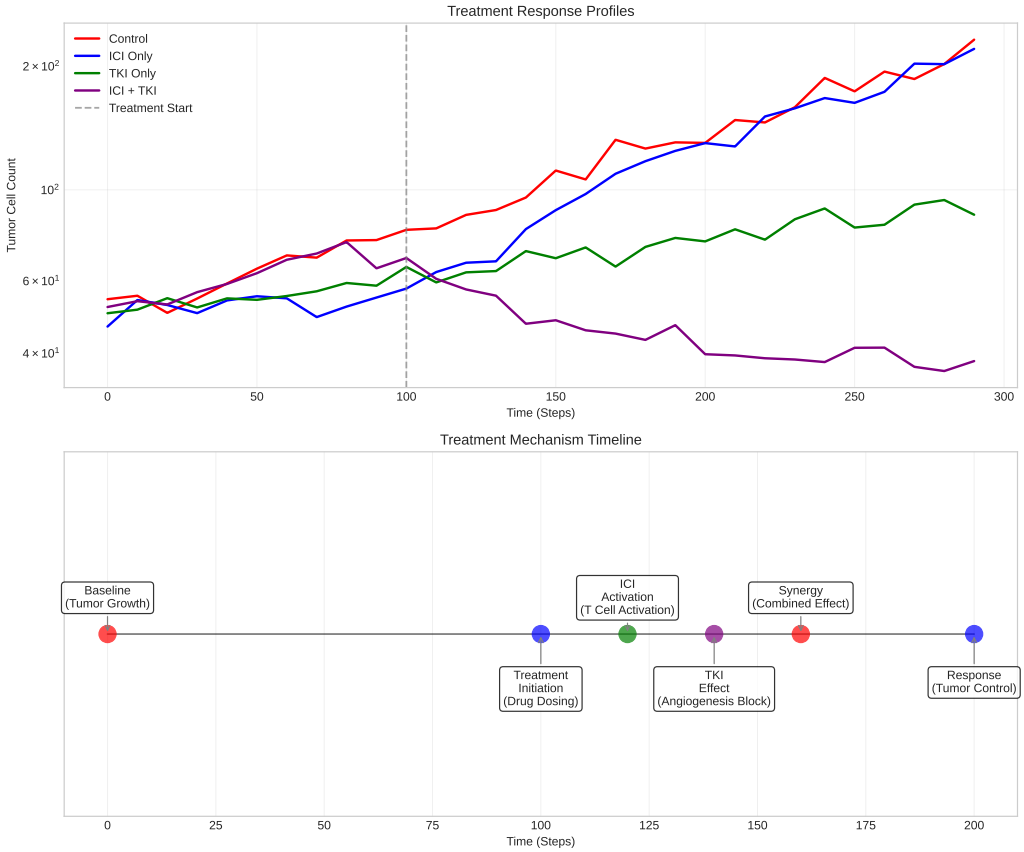


Figure 6.2: Treatment timeline showing the sequential application of ICI and TKI therapies across simulation steps. The figure illustrates drug administration schedules, onset of action, and temporal coordination in combination therapy protocols. Treatment typically begins after an initial observation period to establish baseline tumor dynamics.

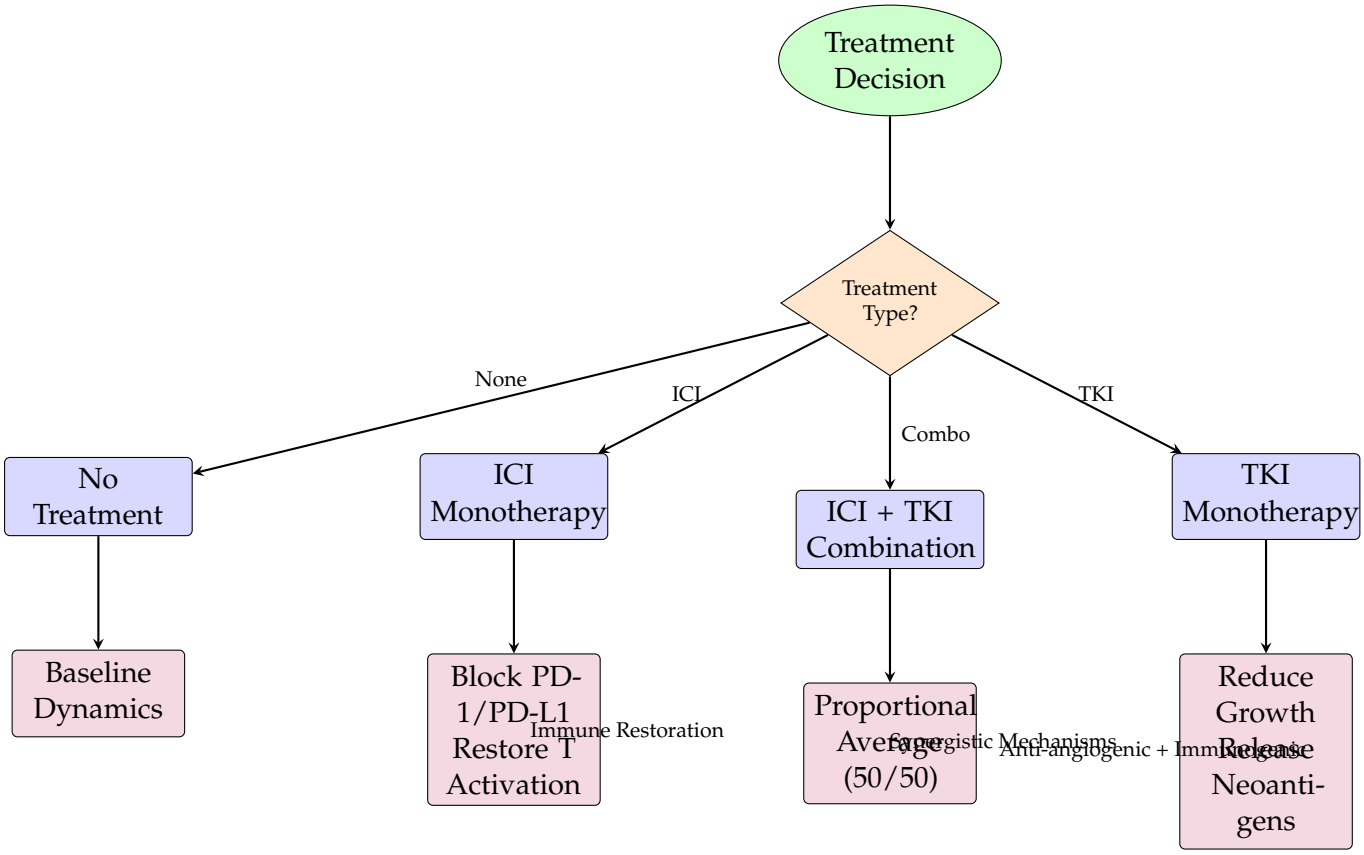


Figure 6.3: BPMN-style Treatment Decision Flow showing the four treatment modalities available in the RCC model. Each treatment path leads to distinct molecular mechanisms: ICI blocks checkpoint inhibition to restore T cell function, TKI reduces tumor growth and releases neoantigens through anti-angiogenic effects, combination therapy applies both mechanisms at half strength, and no treatment maintains baseline tumor-immune dynamics for comparison.

Chapter 7

Sex Hormone System

> Hormones are the sheet music of the body's orchestra.
>
> — adapted

Why do females respond better to some cancer treatments while males benefit more from others? This chapter reveals how sex hormones orchestrate immune responses — and shows how estrogen at moderate levels actually boosts the immune system, explaining the female advantage in cancer survival.

This chapter describes the sex hormone subsystem, which models the immunomodulatory effects of estrogen, testosterone, and progesterone on the tumor microenvironment. Sex-specific differences in hormone profiles produce distinct immune dynamics, contributing to the well-documented sex disparity in RCC incidence and treatment response.

7.1 Biological Basis

7.1.1 Sex Differences in Renal Cell Carcinoma

Renal cell carcinoma exhibits a pronounced sex disparity: males are affected approximately twice as often as females (roughly 2:1 incidence ratio). However, the relationship between sex and treatment outcome is more nuanced. A landmark meta-analysis by Conforti et al. [4] demonstrated that the efficacy of immune checkpoint inhibitors varies significantly between sexes across multiple cancer types, with males generally showing a larger benefit from anti-PD-1/PD-L1 therapy. The ARON-1 study [17] further investigated sex-specific responses to ICI + TKI combination therapy in RCC, finding differential immune activation patterns between male and female patients.

These clinical observations suggest that sex hormones modulate the immune response in ways that affect both tumor susceptibility and treatment efficacy.

7.1.2 Estrogen

Estrogen exerts complex, concentration-dependent effects on the immune system. At low to moderate levels, estrogen promotes Th1-type (pro-inflammatory) immune responses, enhances IFN-γ production, upregulates MHC expression, and stimulates T cell activation and costimulation via CD28. At high concentrations, however, estrogen shifts the balance towards Th2-type (anti-inflammatory) responses, increases IL-4 and IL-13 production, upregulates CTLA-4 and PD-1 inhibitory pathways, and suppresses

T cell activation. This biphasic behavior reflects the known immunological dichotomy between low-dose immunostimulatory and high-dose immunosuppressive effects of estrogen.

7.1.3 Testosterone

Testosterone is generally immunosuppressive at elevated levels. High testosterone concentrations reduce Th1 proliferation, decrease IFN-γ production, and promote immune checkpoint engagement (CTLA-4, PD-1). Conversely, low testosterone levels are associated with enhanced Th1 responses and T cell activation. Testosterone also directly influences $CD8^+$ T cell population dynamics (see Section 7.4).

7.1.4 Progesterone

Progesterone has immunomodulatory properties that depend on concentration. At low levels, its effects are minimal. At moderate levels, progesterone promotes T cell differentiation and mildly enhances checkpoint inhibition. At high concentrations, progesterone strongly suppresses Th1 proliferation, promotes Th2 differentiation, and upregulates inhibitory checkpoints (CTLA-4, PD-1), producing a net immunosuppressive effect.

7.2 Hormone Agent Model

Sex hormones are modeled as discrete agents that diffuse through the simulation grid, interact with nearby immune cells, and decay over time.

7.2.1 Agent Type and Inheritance

The `SexHormone` class inherits directly from `repast4py.core.Agent` (rather than from the `Cell` base class used by other agents), reflecting the fact that hormones are signaling molecules rather than cells. Sex hormones are registered as agent type ID 16 (`AgentType.SEX_HORMONE`).

7.2.2 Hormone Types

Three hormone types are modeled, defined as an enumeration (`SexHormoneType`):

- `ESTROGEN` – primary female sex hormone,
- `TESTOSTERONE` – primary male sex hormone,
- `PROGESTERONE` – steroid hormone involved in immune regulation.

7.2.3 Spawning Mechanism

Hormone agents are spawned from blood vessel positions at each simulation step via the model's `spawn_hormones()` method. For each hormone type, a single spawn attempt is made per step, with success probability depending on the patient's sex:

Table 7.1: Hormone spawn probabilities by patient sex.

Sex	Estrogen	Progesterone	Testosterone
Male	0.2	0.1	1.0
Female	0.9	0.8	0.2

When a hormone is spawned, it is placed at the position of a randomly selected blood vessel agent, simulating hormonal delivery via the bloodstream.

7.2.4 Hormone Diffusion

Once spawned, hormone agents perform a biased random walk through the grid. At each step, the agent selects one of two movement strategies with equal probability:

1. **Biased diffusion:** the agent evaluates its Moore neighborhood (radius 1) and selects a target position with probability biased towards greater distance from blood vessels. The selection probability for neighbor c is:

$$P(c) = (1 - \alpha) \cdot \frac{1}{|\mathcal{N}|} + \alpha \cdot \frac{d(c)}{d_{\max} \cdot |\mathcal{N}|}, \tag{7.1}$$

   where $\alpha = 0.25$ is the drift strength, $d(c)$ is the Manhattan distance from c to the nearest blood vessel, $d_{\max}$ is the maximum such distance among all neighbors, and $|\mathcal{N}|$ is the neighborhood size.
2. **Simple diffusion:** the agent selects a move from a predefined set of directions, biased towards the positive y and z directions (away from the vascular channel).

Hormones that exit the grid boundaries are removed from the simulation, providing a natural mechanism for hormone clearance at the domain edges. The combination of biased diffusion and boundary removal limits the effective lifespan of hormone agents to approximately 50 steps, depending on grid dimensions and diffusion trajectory.

7.3 Hormone Effects on T Cells

T cells perceive nearby sex hormones and modulate their behavior accordingly. This mechanism is implemented in the `TCell` base class, ensuring that all T cell subtypes (CD8 cytotoxic, CD8 naive, CD4 naive, CD4 Th1, CD4 Th2, Treg) are affected.

7.3.1 Hormone Perception

T cells detect sex hormones in their local neighborhood via the `perceive_sex_hormone()` method, which queries the spatial index using `grid_utils` neighbor queries. Each T cell maintains a dictionary of perceived hormone levels, keyed by hormone type. When a hormone agent is perceived:

1. The T cell's perceived level for that hormone type is incremented.
2. The hormone agent is removed from the simulation (consumed).
3. A maximum of 9 hormones of each type can be perceived per step (capped at `MAX_LEVEL`).

Perceived hormone levels decay by 10% per step, modeling the transient nature of hormonal signaling.

7.3.2 Stimulation Level

The `sex_hormone_stimulation_level()` method returns the current perceived level of a given hormone type, providing a scalar measure of hormonal influence that is used by downstream effect computations.

7.3.3 Hormonal Influence Effects

The `CellHormonalInfluenceEffect` enumeration (in `src/systems/hormonal_effect.py`) defines 17 distinct effect channels through which hormones modulate T cell behavior:

Table 7.2: Hormonal influence effect channels.

Effect	Description
KILL_RATE	Cytotoxic kill rate modulation.
PROLIFERATION	General proliferation rate.
APOPTOSIS	Programmed cell death susceptibility.
ACTIVATION	T cell activation state.
DIFFERENTIATION	Differentiation probability (e.g., naive to effector).
TH1_PROLIFERATION	Th1-specific proliferation.
TH2_PROLIFERATION	Th2-specific proliferation.
IL_4	Interleukin-4 signaling effect.
IL_6	Interleukin-6 signaling effect.
IL_13	Interleukin-13 signaling effect.
IFN_GAMMA	Interferon-gamma signaling effect.
TNF_ALPHA	Tumor necrosis factor alpha effect.
MHC_I_UPREGULATION	MHC class I expression modulation.
MHC_II_UPREGULATION	MHC class II expression modulation.
COSTIMULATION_CD28	CD28 costimulatory signal strength.
CTLA4_SUPPRESSION	CTLA-4 inhibitory checkpoint engagement.
PD1_INHIBITION	PD-1 inhibitory checkpoint engagement.

7.3.4 Concentration-Dependent Effect Mapping

Each hormone type exerts different effects depending on its normalized concentration level. The perceived hormone level is normalized to a $[0, 1]$ scale using a threshold of 9 (the maximum level):

$$\hat{h} = \frac{h - 1}{\text{MAX_LEVEL} - 1}, \tag{7.2}$$

where h is the raw perceived level. Three concentration regimes are defined:

- **Low:** $\hat{h} < 0.33$
- **Medium:** $0.33 \leq \hat{h} < 0.66$
- **High:** $\hat{h} \geq 0.66$

For each regime, a predefined effect vector specifies integer-valued influence scores (typically ranging from -3 to $+3$) for each effect channel. These scores are weighted by hormone-specific importance weights before being accumulated:

$$w_{\text{estrogen}} = 0.45, \qquad w_{\text{testosterone}} = 0.30, \qquad w_{\text{progesterone}} = 0.25. \tag{7.3}$$

The weighted effects are accumulated into the T cell's `accumulated_effects` dictionary and subsequently transferred to the cell's `experienced_effects` via the `update_effects()` method.

Example: Estrogen at High Concentration. When estrogen is at high concentration ($\hat{h} \geq 0.66$), the effect vector includes: TH1_PROLIFERATION $= -3$ (strong suppression), TH2_PROLIFERATION $= +3$ (strong promotion), IL_4 $= +3$, IL_13 $= +3$, IFN_GAMMA $= -3$, ACTIVATION $= -2$, CTLA4_SUPPRESSION $= +3$, PD1_INHIBITION $= +2$, KILL_RATE $= -2$, and PROLIFERATION $= -1$. After weighting by $w_{\text{estrogen}} = 0.45$, these effects produce a net immunosuppressive shift consistent with the biological role of high-dose estrogen.

Key Takeaway

Estrogen at moderate levels actually BOOSTS immune response by promoting Th1 cells and enhancing T cell activation. Only at high concentrations does it become immunosuppressive. This biphasic response explains why females often show better immune-mediated cancer control than males.

7.4 Sex Drift

In addition to the direct hormonal effects on T cell behavior, the model implements a population-level sex drift mechanism that adjusts the $CD8^+$ naive T cell population based on patient sex.

7.4.1 Biological Basis

Testosterone stimulates the production and maturation of $CD8^+$ T cells in the thymus. Male patients, with higher testosterone levels, maintain larger $CD8^+$ T cell populations, which contributes to stronger cytotoxic anti-tumor responses. Conversely, female patients tend to have relatively fewer $CD8^+$ T cells but may exhibit enhanced $CD4^+$ T helper and antibody-mediated responses.

7.4.2 Implementation

The sex drift is implemented in the `RCCModel` class via the `init_sex_drift()` and `apply_sex_drift()` methods. The drift operates as a gradual population adjustment over the first n simulation steps (default $n = 15$):

$$\delta = \frac{d_{\text{max}}}{n}, \tag{7.4}$$

where d_{max} is the `w_max_drift` parameter (default 0.25), representing the maximum fraction of the CD8 naive T cell population affected over the entire drift period. The drift direction depends on sex:

$$\sigma = \begin{cases} +1 & \text{if sex = Male (positive drift, population increase),} \\ -1 & \text{if sex = Female (negative drift, population decrease).} \end{cases} \tag{7.5}$$

At each step during the drift period, the step factor is computed as:

$$f_{\text{step}} = 1 + \sigma \cdot \delta. \tag{7.6}$$

The population adjustment proceeds as follows:

- If $f_{\text{step}} > 1$ (male, population increase): each CD8 naive T cell is duplicated with probability $|f_{\text{step}} - 1|$.
- If $f_{\text{step}} < 1$ (female, population decrease): each CD8 naive T cell is removed with probability $|f_{\text{step}} - 1|$.

The drift terminates after n steps, at which point the CD8 naive T cell population has been adjusted by approximately $\pm d_{\text{max}} \times 100\%$ relative to its initial size. For the default parameters ($d_{\text{max}} = 0.25$, $n = 15$), this produces a $\sim 25\%$ population increase in males and a $\sim 25\%$ decrease in females over the first 15 simulation steps.

7.4.3 Interaction with Hormonal Effects

The sex drift and the direct hormonal effects operate through complementary mechanisms:

- The **sex drift** adjusts the *initial population size* of $CD8^+$ T cells, reflecting the baseline sex difference in T cell counts.
- The **hormonal effects** modulate the *functional behavior* (kill rates, proliferation, activation) of all T cell types throughout the simulation.

Together, these mechanisms produce sex-specific immune dynamics that are consistent with clinical observations: male patients start with more $CD8^+$ T cells and experience testosterone-driven immune modulation, while female patients have fewer $CD8^+$ T cells but benefit from estrogen-mediated immune activation at low concentrations. The net effect on treatment outcome depends on the interplay between these factors and the treatment modality applied [4, 17].

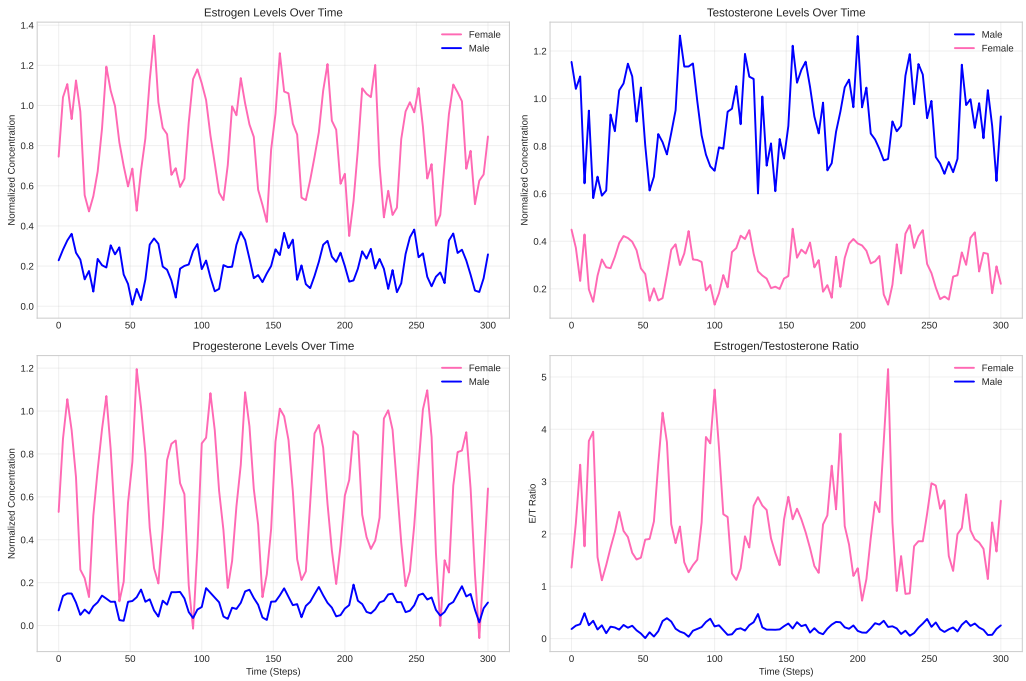


Figure 7.1: Sex hormone dynamics over time showing the evolution of estrogen, testosterone, and progesterone levels in the tumor microenvironment. The hormone concentrations fluctuate based on sex-specific spawning rates, cellular consumption, and spatial diffusion patterns. These dynamics directly modulate immune cell behavior and treatment response.

Chapter 8

Implementation

Converting biological complexity into working code requires more than good programming — it demands architectural vision. This chapter reveals how moving from Mesa to Repast4Py enabled true 3D simulation and MPI parallelism, transforming a proof-of-concept into a production-scale virtual tumor.

This chapter examines the architectural decisions, design patterns, and implementation strategies that transform biological theory into executable software. Rather than cataloging code structure, we focus on the engineering problems encountered and the design patterns employed to solve them, revealing how computational constraints shape biological model design and vice versa.

8.1 Framework Migration: Mesa to Repast4Py

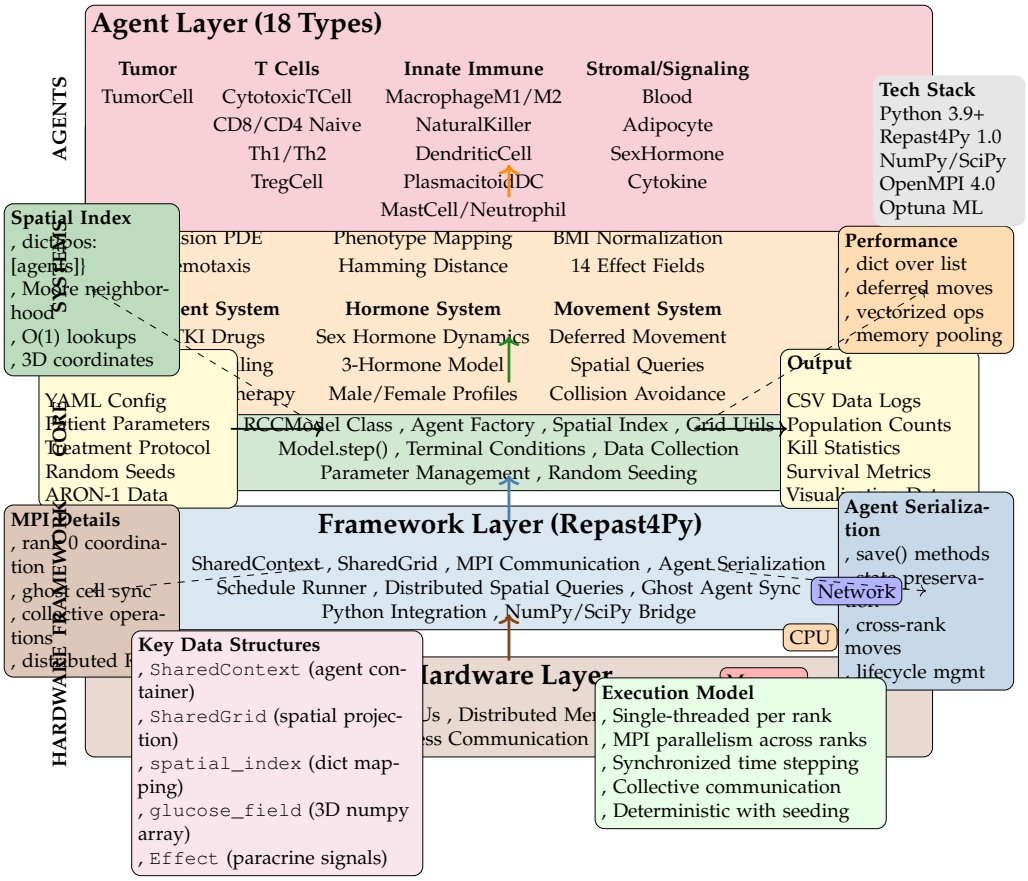


Figure 8.1: Software architecture overview of the RCC simulation system

Table **??** presents a comprehensive comparison between Mesa and Repast4Py implementations, highlighting the key architectural differences and enhancements made during the porting process.

The implementation uses Command pattern for deferred movement, Observer pattern for data collection, and Strategy pattern for treatment modalities.

8.2 Simulation Orchestration: The Main Event Loop

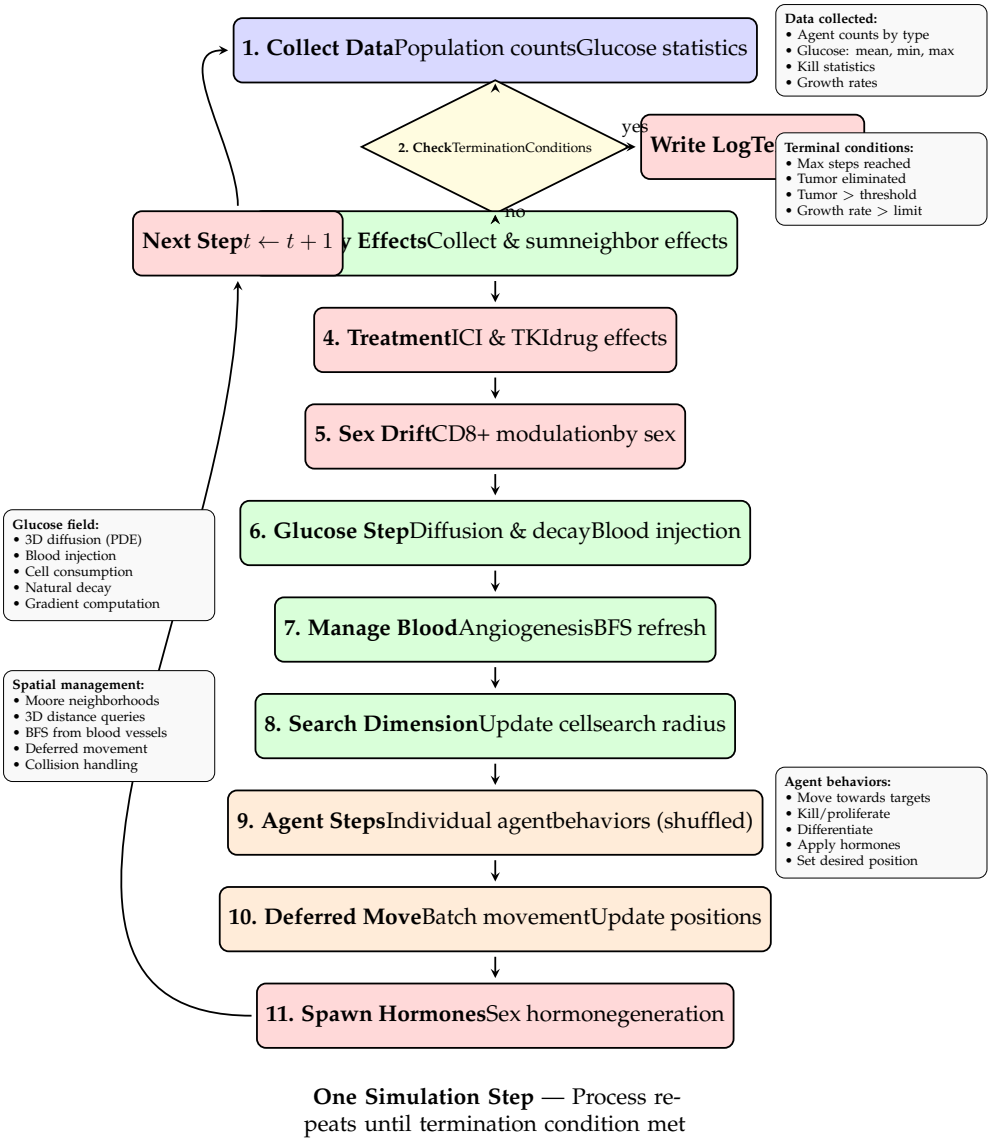
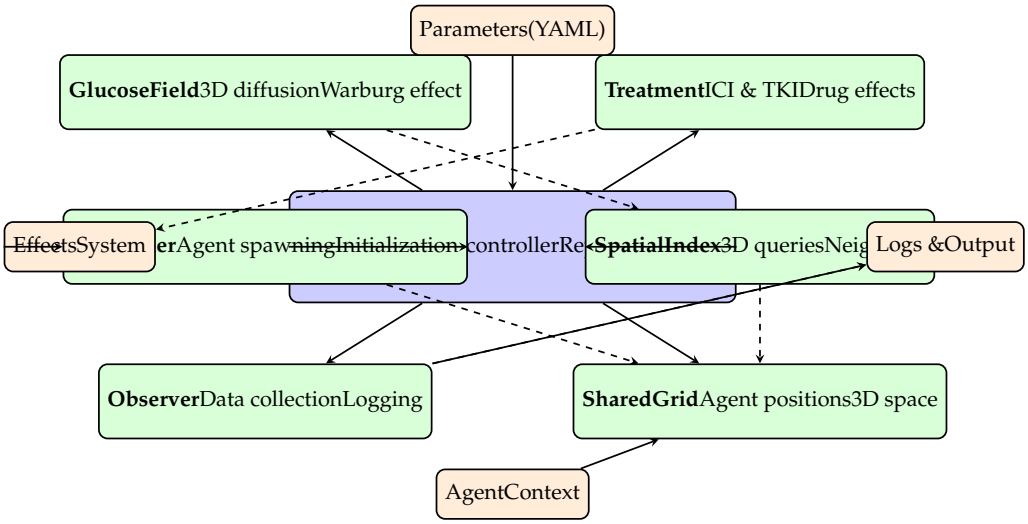


Figure 8.2: Simulation execution pipeline

The main simulation loop executes a fixed sequence: data collection, termination check, effect application, treatment, glucose field updates, agent stepping, and deferred movement execution. This ordering respects biological causality while preventing execution order bias.

The codebase is organized into agent classes (17 cell types), system modules (glucose field, DNA mutations, effect accumulation), treatment implementations, and pa-



Main Step Sequence: 1. collect_data → 2. check_termination → 3. apply_effects → 4. treatment → 5. sex_drift → 6. glucose_step → 7. manage_blood → 8. agent_steps → 9. move → 10. spawn_hormones

Figure 8.3: Model architecture showing component relationships

rameter management. Repast4Py's RCCModel class directly manages SharedContext, SharedGrid, and agent lifecycle without framework inheritance.

Figure 8.4 illustrates the overall architecture of the RCCModel, showing the main components and their data flow relationships.

[Architecture diagram showing RCCModel, GlucoseField, CellAdder, Observer, Treatment, SpatialIndex, and SharedGrid components with their interconnections]

Figure 8.4: Architecture of the RCCModel system. The main model coordinates interactions between the glucose field (3D diffusion), agent management (cell adder, spatial index, shared grid), data collection (observer), and treatment systems. The step sequence annotation shows the ordered execution of simulation phases.

8.3 Spatial Index Design

Repast4Py lacks built-in neighbor queries, so the model maintains a manual spatial index using defaultdict(list) that maps coordinates to agent lists. Grid utilities provide 3D neighborhood computation using Moore or Von Neumann neighborhoods with boundary checking.

8.4 Agent Serialization

Repast4Py agents require (id, type, rank) construction and maintain uid tuples for MPI distribution. The AgentType IntEnum maps 18 cell types to integer IDs, and each agent implements a save() method returning (uid, state$_tuple)forserialization.Typeidentificationusesintege$

8.5 Mesa to Repast4Py Porting

Table 8.1: Key API mapping from Mesa to Repast4Py.

Mesa	**Repast4Py**
`mesa.Agent(model)`	`repast4py.core.Agent(id, type, rank)`
`self.model.random`	`model.rng` (stdlib `random.Random`)
`self.model.grid.get_neighbors()`	`grid.utils.get_neighbors_3d(spatial_index, ...)`
`isinstance(agent, TumorCell)`	`agent.uid[1] == AgentType.TUMOR_CELL`
`self.model.schedule.agents`	`model._all_agents`
`model.agents_by_type[Type]`	`model.get_agents_by_type_id(type_id)`
`TumorCell.blood_source` (class var)	`model.tumor_blood_sources`
`Cell.search_dimension` (class var)	`model.cell_search_dimension`

The porting required moving class-level state to model instances, implementing custom random number generation, and preserving deferred movement patterns to prevent execution order artifacts.

8.6 Glucose Field Implementation

The GlucoseField class uses NumPy 3D arrays and SciPy convolution for performance and control. Diffusion uses a 6-connected discrete Laplacian kernel with zero-flux boundary conditions via mode='nearest'. The implementation avoids Repast4Py's SharedValueLayer for simplicity and debugging convenience.

8.7 Entry Point and Execution

The simulation runs via 'mpirun -n 1 python run.py' using Repast4Py's schedule runner to execute the main step() loop with data collection, termination checking, effect propagation, treatment, glucose updates, agent stepping, and deferred movement.

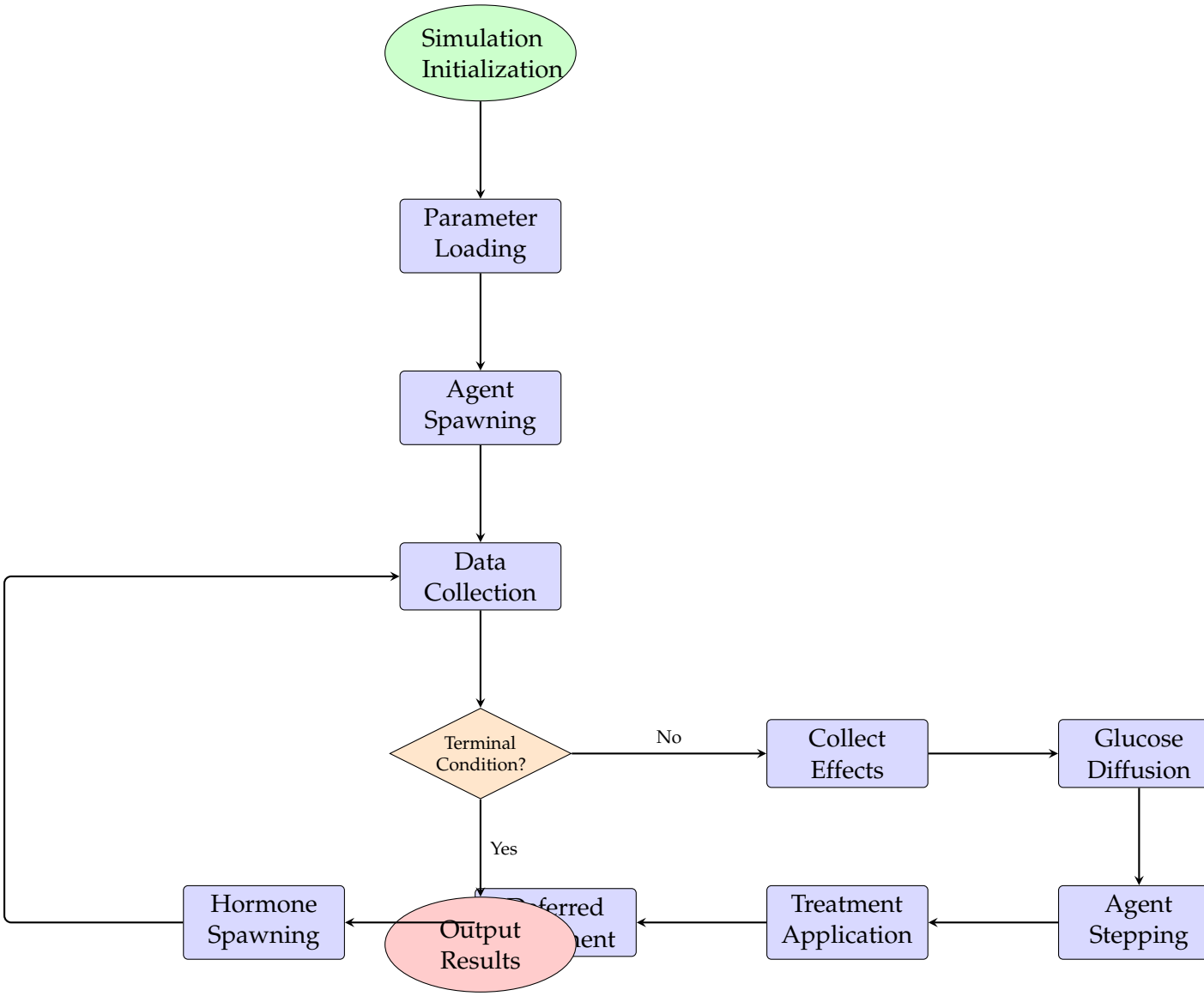


Figure 8.5: BPMN-style Simulation Lifecycle for the RCC model. The simulation follows a deterministic sequence: initialization and parameter loading, then iterative execution of the main loop. Each step collects data, checks termination conditions, applies agent effects, performs glucose diffusion, executes treatment protocols, steps agents, and resolves movement. The loop continues until a terminal condition is met, at which point results are output.

Chapter 9

Parameters and Optimization

> With four parameters I can fit an elephant.
>
> — John von Neumann

The model employs a 114-parameter hierarchy: 4 model parameters (volume, resolution, seed), 18 patient parameters (clinical variables), and 92 weight parameters (biological process rates) optimized via Bayesian methods against clinical trial data.

9.1 Parameter Categories

Parameters are organized into three dataclass objects: ModelParameters, PatientParameters, and WeightParameters.

ModelParameters: 4 simulation configuration parameters (volume, block size, random seed, max steps). PatientParameters: 18 patient-specific parameters (sex, BMI, treatment type, start time, plus 14 immune cell concentrations). WeightParameters: 92 biological process rate parameters optimized via Bayesian methods.

9.2 Weight Parameters

The 84 weight parameters control probability thresholds and scaling factors for all agent behaviors, organized by cell type. These parameters are optimized via Bayesian methods to match clinical trial outcomes.

9.2.1 Macrophage M1 Weights (6 weights + 2 biases)

Table 9.1: Macrophage M1 weight parameters.

Parameter	Default	Description
`w_m1_mutation`	1.0	M1 macrophage mutation threshold scaling.
`b_m1_mutation`	0.0	Bias for M1 mutation threshold.
`w_m1_move`	1.0	M1 macrophage look-up movement radius scaling.
`w_m1_phagocytosis`	1.0	M1 phagocytosis chance scaling.
`w_m1_digest`	1.0	M1 digestion rate scaling.
`w_m1_t_kill_rate`	1.0	M1 effect on T cell kill rate.
`w_m1_th1_proliferation`	1.0	M1 effect on Th1 proliferation.
`w_th1_proliferation`	1.0	Th1 proliferation scaling (via M1 influence).
`b_th1_proliferation`	0.0	Bias for Th1 proliferation.

9.2.2 Macrophage M2 Weights (5 weights + 1 bias)

Table 9.2: Macrophage M2 weight parameters.

Parameter	Default	Description
`w_m2_mutation`	1.0	M2 macrophage mutation threshold scaling.
`b_m2_mutation`	0.0	Bias for M2 mutation threshold.
`w_m2_move`	1.0	M2 macrophage look-up movement radius.
`w_m2_t_kill_rate`	1.0	M2 effect on T cell kill rate.
`w_m2_tumour_growth`	1.0	M2 effect on tumour growth promotion.
`w_m2_angiogenesis`	1.0	M2 effect on angiogenesis.

9.2.3 Tumor Cell Weights (8 weights + 3 biases)

Table 9.3: Tumor cell weight parameters.

Parameter	Default	Description
`w_tumor_apoptosis_eff`	1.0	Effect-driven apoptosis probability scaling.
`w_tumor_apoptosis_dna`	1.0	DNA-driven extra suppression chance scaling.
`b_tumor_apoptosis`	0.0	Bias for apoptosis probability.
`w_tumor_growth_eff`	0.6	Effect-driven growth probability scaling.
`w_tumor_growth_dna`	1.0	DNA-driven extra proliferation chance scaling.
`b_tumor_growth`	0.0	Bias for growth probability.
`w_tumor_angiogenesis`	1.0	Angiogenesis initiation probability scaling.
`b_tumor_angiogenesis`	0.0	Bias for angiogenesis probability.
`w_antigen_presentation`	1.0	Antigen presentation chance scaling.
`b_antigen_presentation`	0.0	Bias for antigen presentation.
`w_angiogenesis_tumor_growth`	1.0	Angiogenesis contribution to growth (when blood access is present).
`w_gene_pd1_inhibition`	1.0	PD-1/PD-L1 checkpoint pathway inhibition scaling.

9.2.4 CD4+ T Cell Weights (9 weights + 3 biases)

Table 9.4: CD4+ T cell weight parameters.

Parameter	Default	Description
`w_cd4_treg_diff_effect`	1.0	Effect-driven Treg differentiation scaling.
`w_cd4_treg_diff_horm`	1.0	Hormone-driven Treg differentiation scaling.
`b_cd4_treg_diff`	0.0	Bias for Treg differentiation.
`w_cd4_th1_proliferation_effect`	1.0	Effect-driven Th1 proliferation scaling.
`w_cd4_th1_proliferation_horm`	1.0	Hormone-driven Th1 proliferation scaling.
`b_cd4_th1_proliferation`	0.0	Bias for Th1 proliferation.
`w_cd4_th1_spawn_m1`	1.0	Th1 spawning of M1 macrophages.
`w_cd4_th1_spawn_dc`	1.0	Th1 spawning of dendritic cells.
`w_cd4_th2_spawn_m1`	1.0	Th2 spawning of M1 macrophages.
`w_cd4_th2_spawn_t`	1.0	Th2 spawning of T cells.
`w_cd4_th2_proliferation_effect`	1.0	Effect-driven Th2 proliferation scaling.
`w_cd4_th2_proliferation_horm`	1.0	Hormone-driven Th2 proliferation scaling.
`b_cd4_th2_proliferation`	0.0	Bias for Th2 proliferation.

9.2.5 Cytotoxic T Cell Weights (5 weights + 2 biases)

Table 9.5: Cytotoxic T cell (CD8+) weight parameters.

Parameter	Default	Description
`w_cytotoxic_proliferation`	1.0	Proliferation probability scaling.
`w_cytotoxic_apoptosis`	1.0	Apoptosis probability scaling.
`b_cytotoxic_apoptosis`	0.0	Bias for cytotoxic T cell apoptosis.
`w_cytotoxic_kill`	1.0	Tumor cell kill probability scaling.
`b_cytotoxic_kill`	0.0	Bias for cytotoxic kill probability.
`w_cytotoxic_pd1_inhibition`	1.0	PD-1 inhibition susceptibility scaling.
`w_cytotoxic_move`	1.0	Movement radius scaling for cytotoxic T cells.

9.2.6 Mast Cell Weights (6 parameters)

Table 9.6: Mast cell weight parameters.

Parameter	Default	Description
`w_mast_cell_angiogenesis`	1.0	Mast cell contribution to angiogenesis effect.
`w_mast_cell_m1_mutation`	1.0	Mast cell effect on M1 mutation.
`w_mast_cell_t_kill_rate`	1.0	Mast cell effect on T cell kill rate.
`w_mast_cell_tumour_apoptosis`	1.0	Mast cell contribution to tumour apoptosis.
`w_mast_cell_tumour_growth`	1.0	Mast cell contribution to tumour growth.
`w_mast_cell_spawn_dc`	1.0	Mast cell probability of spawning dendritic cells.

9.2.7 NK Cell Weights (3 weights + 1 bias)

Table 9.7: Natural killer cell weight parameters.

Parameter	Default	Description
`w_natural_killer_kill_rate`	1.0	NK cell direct kill rate scaling.
`b_natural_killer_kill_rate`	0.0	Bias for NK cell kill rate.
`w_nkl_t_kill_rate`	1.0	NK cell effect on T cell kill rate amplification.

9.2.8 Plasmacytoid DC Weights (6 weights + 1 bias)

Table 9.8: Plasmacytoid dendritic cell weight parameters.

Parameter	Default	Description
`w_pdc_nkl_spawn`	1.0	pDC probability of spawning NK cells.
`b_pdc_nkl_spawn`	0.0	Bias for pDC NK cell spawning.
`w_pdc_angiogenesis`	1.0	pDC contribution to angiogenesis effect.
`w_pdc_treg_diff`	1.0	pDC effect on Treg differentiation.
`w_pdc_t_proliferation`	1.0	pDC effect on T cell proliferation.
`w_pdc_t_kill`	1.0	pDC effect on T cell kill rate.
`w_pdc_nkl_kill`	1.0	pDC effect on NK cell kill rate.

9.2.9 Treg Weights (5 parameters)

Table 9.9: Regulatory T cell weight parameters.

Parameter	Default	Description
`w_treg_t_kill_rate`	1.0	Treg suppression of T cell kill rate.
`w_treg_t_proliferation`	1.0	Treg suppression of T cell proliferation.
`w_treg_t_apoptosis`	1.0	Treg induction of T cell apoptosis.
`w_treg_activation`	1.0	Treg activation threshold scaling.
`w_treg_dc_phagocytosis`	1.0	Treg effect on DC phagocytosis.

9.2.10 Global Weights (4+ parameters)

Table 9.10: Global weight parameters.

Parameter	Default	Description
`w_search_dimension`	0.5	Controls adaptive search radius for agent interactions: $\text{dim} = \max(5, \lfloor n_{\text{tumor}}/(10 \cdot w) \rfloor)$.
`w_tumour_growth_threshold`	1.0	Terminal condition threshold for mean tumour growth rate.
`w_cell_base_death_prob`	0.009	Per-step base death probability for all cells.
`w_progressive_exhaustion`	0.05	Rate of progressive immune cell exhaustion.
`w_ici_effectiveness`	1.0	ICI drug effectiveness scaling.
`w_tki_effectiveness`	1.0	TKI drug effectiveness scaling.
`w_sex_hormone_cd8`	1.0	Sex hormone influence on CD8+ population.
`w_dc_phagocytosis_effect`	1.0	DC phagocytosis effect scaling.
`b_dc_phagocytosis`	0.0	Bias for DC phagocytosis.
`w_max_drift`	0.25	Maximum sex-specific CD8+ drift over the drift period.
`receptor_threshold_variation`	1	Integer variation in TCR matching threshold.

9.3 Glucose Parameters

The glucose microenvironment system introduces 8 new weight parameters, all prefixed with `w_glucose_`. These parameters control the physical dynamics of glucose diffusion, the metabolic consumption rates, and the coupling between glucose availability and agent behavior.

Table 9.11: Glucose weight parameters.

Parameter	Default	Description
`w_glucose_diffusion`	0.1	Diffusion coefficient D for the 3D Laplacian update. Stability requires $D < 1/6 \approx 0.167$; the default value of 0.1 provides moderate diffusion well within the stable regime.
`w_glucose_decay`	0.01	Natural glucose decay rate per step. Applied multiplicatively as $(1 - d)$ to the entire field, modeling background metabolic consumption by tissue not explicitly represented as agents.
`w_glucose_source_rate`	5.0	Amount of glucose injected per blood vessel agent per step. Blood vessels act as point sources, creating localized glucose-rich zones that subsequently diffuse outward.
`w_glucose_tumor_consumption`	2.0	Glucose consumed per tumor cell per step. Reflects the elevated glycolytic rate characteristic of the Warburg effect [18]. Tumor cells consume approximately 4× the glucose of immune cells.
`w_glucose_immune_consumption`	0.5	Glucose consumed per immune cell per step. Represents the metabolic cost of immune cell activation and effector function.
`w_glucose_growth_sensitivity`	1.0	Sensitivity of tumor growth probability to local glucose concentration. The glucose factor is computed as $\min(1, C \cdot s)$, where C is the local concentration and s is this sensitivity parameter.
`w_glucose_immune_sensitivity`	1.0	Sensitivity of immune cell effector function to local glucose concentration. Controls how strongly glucose deprivation impairs immune cell activity.

Parameter	Default	Description
`w_glucose_chemotaxis_strength`	0.3	Probability that an immune cell follows the glucose gradient (chemotaxis) rather than performing a random walk. At each step, with probability $p = 0.3$, the cell moves in the direction of increasing glucose concentration.

9.3.1 Parameter Interactions

The glucose parameters interact with the existing weight parameters in several ways:

- **Tumor growth modulation**: The tumor cell duplication probability (governed by `w_tumor_growth_eff` and `w_tumor_growth_dna`) is multiplied by a glucose factor $\min(1, C \cdot \texttt{w_glucose_growth_sensitivity})$. When glucose is abundant, this factor approaches 1 and growth proceeds at the base rate; when glucose is depleted, growth is suppressed.
- **Immune function modulation**: Immune cell kill rates and activation probabilities can be scaled by an analogous glucose factor using `w_glucose_immune_sensitivity`, coupling metabolic resource availability to immune effectiveness.
- **Spatial competition**: The balance between `w_glucose_source_rate` (supply from blood vessels) and `w_glucose_tumor_consumption` (demand from tumor cells) determines the spatial glucose profile. High tumor consumption relative to supply creates depletion zones that impair both further tumor growth and immune infiltration.

9.4 Bayesian Optimization

9.4.1 Framework

The model uses Optuna [1] for Bayesian hyperparameter optimization. Optuna's Tree-structured Parzen Estimator (TPE) sampler efficiently explores high-dimensional parameter spaces by building probabilistic models of the objective function landscape.

9.4.2 Objective Function

The optimization objective is to minimize the prediction error of the model against the ARON-1 clinical dataset. For each Optuna trial, the following process is executed:

1. A set of weight parameters is suggested by the TPE sampler within predefined bounds.

2. For each patient case in the ARON-1 dataset, a simulation is run with the patient's clinical features (sex, BMI, treatment, immune cell concentrations) and the suggested weight parameters.

3. The simulation outcome (survival/death, number of steps) is compared against the clinical outcome.

4. The per-case error is computed as:

$$\text{error}_i = \begin{cases} (\text{steps}_i - \text{OS}_i)^2 & \text{if prediction matches outcome} \\ \text{HIGH_PENALTY}^2 & \text{if prediction contradicts outcome} \end{cases} \tag{9.1}$$

   where OS_i is the observed overall survival (in steps) and HIGH_PENALTY $= 500$.

5. The trial objective value is the mean error across all cases.

9.4.3 Search Space

The search space is defined in the `suggest_parameters()` function and covers the full 92-parameter weight space:

- **Positive weights** (62 parameters): Sampled log-uniformly in $[0.25, 4.0]$. The log-uniform distribution ensures adequate exploration of both small and large scaling factors.
- **Biases** (15 parameters): Sampled uniformly in $[-1.0, 1.0]$.
- **Effectiveness weights**: `w_ici_effectiveness` and `w_tki_effectiveness` in $[0.0, 1.0]$.
- **Threshold weights**: `w_tumour_growth_threshold` in $[0.75, 0.99]$; `w_cell_base_death_p` in $[0.001, 0.1]$; `w_progressive_exhaustion` in $[0.005, 0.3]$.
- **Integer parameters**: `receptor_threshold_variation` in $\{0, 1, 2, 3\}$.
- **Glucose parameters** (8 parameters):
  - `w_glucose_diffusion` in $[0.01, 0.16]$ (respecting the $D < 1/6$ stability constraint).
  - `w_glucose_decay` in $[0.001, 0.1]$.
  - `w_glucose_source_rate` in $[1.0, 20.0]$.
  - `w_glucose_tumor_consumption` in $[0.5, 10.0]$.
  - `w_glucose_immune_consumption` in $[0.1, 2.0]$.
  - `w_glucose_growth_sensitivity` in $[0.1, 5.0]$.
  - `w_glucose_immune_sensitivity` in $[0.1, 5.0]$.
  - `w_glucose_chemotaxis_strength` in $[0.05, 0.8]$.

9.4.4 Optimization Configuration

The optimization runs for $N = 250$ trials, each evaluating every patient case in the ARON-1 dataset. Training uses `MPI.COMM_SELF` to create isolated single-rank model instances, avoiding MPI communication overhead during optimization. Trial errors and the best-so-far error are logged to `data/trial_errors.csv` for convergence analysis.

```
study = optuna.create_study(direction="minimize")
study.optimize(wrapped_objective, n_trials=250)
```

Listing 9.1: Optuna study creation and execution.

Key Takeaway

Ninety-two parameters are tuned by Bayesian optimization (Optuna) against real ARON-1 clinical trial data. This isn't just model fitting — it's teaching the virtual tumor to behave like real patient tumors, creating a bridge between computational biology and clinical medicine.

9.5 Clinical Dataset

9.5.1 ARON-1 Study

The ARON-1 study is a clinical trial of advanced renal cell carcinoma patients treated with immune checkpoint inhibitors (ICI), tyrosine kinase inhibitors (TKI), or their combination [16, 13]. The dataset provides the ground truth against which the model is calibrated.

9.5.2 Dataset Structure

The clinical dataset is stored in `src/learning/data/ARON.csv` and contains the following features for each patient case:

- **Patient features**: sex (`'F'/'M'`), BMI, treatment type (ICI, TKI, or ICI+TKI).
- **Immune cell concentrations**: 13 cell type concentrations matching the `PatientParameters` fields.
- **Outcome**: Overall survival (OS) in time units, and a binary death indicator.

The dataset is loaded with Pandas, with the treatment column parsed as a list of strings to support combination therapies:

```
def load_dataset(file_path: str) -> pd.DataFrame:
    def treatment_to_list(treatment_str):
        return [t.strip() for t in treatment_str.split('+')]
    return pd.read_csv(file_path,
                       converters={"treatment": treatment_to_list})
```

Listing 9.2: Dataset loading.

9.6 Parameter Sensitivity Analysis

Figure 9.1 demonstrates the sensitivity of simulation outcomes to key parameter variations within the defined ranges. This analysis identifies which parameters have the strongest influence on tumor dynamics and treatment efficacy, providing guidance for parameter selection and model robustness assessment.

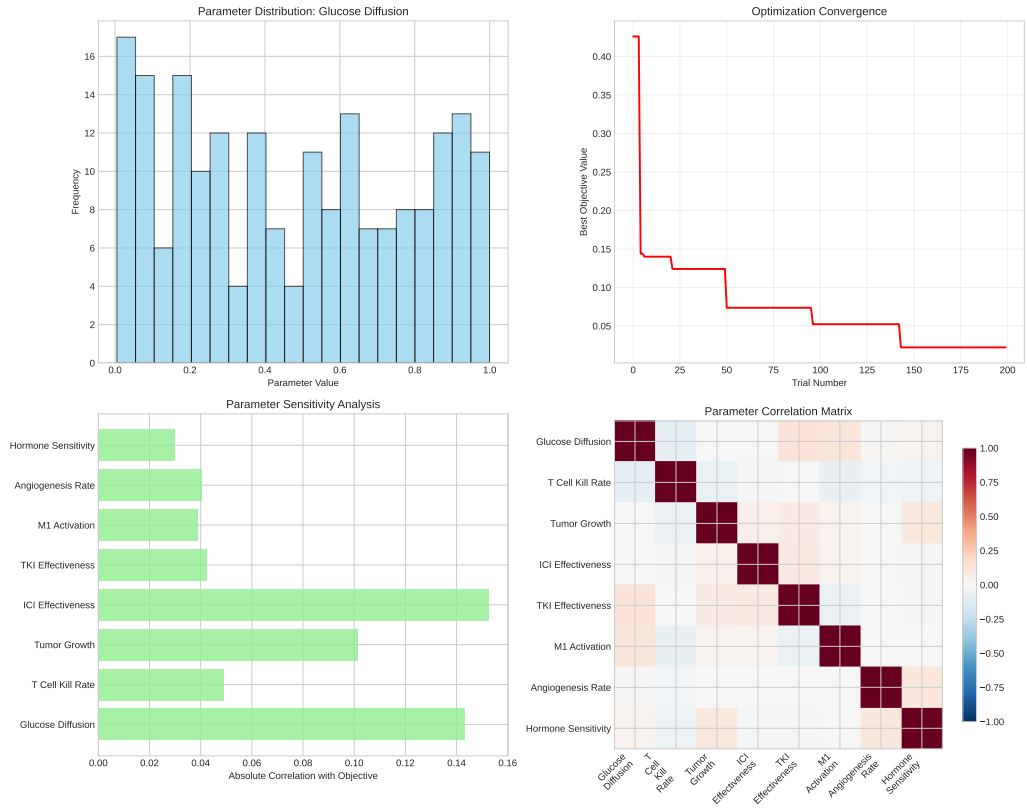


Figure 9.1: Parameter sensitivity analysis showing the impact of key model parameters on tumor cell population dynamics. The analysis varies parameters within their defined ranges while keeping others at baseline values. Most sensitive parameters include glucose consumption rates, treatment effectiveness weights, and immune cell interaction parameters. Error bars represent variance across multiple simulation runs.

Each row of the dataset is used to parameterize a simulation run during optimization, with the patient's clinical features passed as keyword arguments to `RCCModel`. The simulation's survival prediction is then compared against the clinical outcome to compute the trial error.

With 92 parameters tuned against clinical reality, we can finally ask the question that motivated this entire project: does sex matter in RCC treatment? The next chapter reveals surprising answers that challenge conventional treatment wisdom.

Chapter 10

Simulation Results

> In God we trust; all others must bring data.
>
> — W. Edwards Deming

With 92 parameters tuned against clinical reality, we can finally ask the question that motivated this entire project: does sex matter in kidney cancer treatment? This chapter reveals how single-seed results can be catastrophically misleading — and shows why 160 simulation runs are needed to uncover the truth.

This chapter presents the results from systematic simulation of the RCC agent-based model across six scenarios: sex-specific baseline (no treatment) conditions and three treatment modalities (ICI, TKI, and combination therapy) stratified by biological sex. The results demonstrate significant differences in tumor progression patterns, treatment responses, and sex-specific effects that inform our understanding of renal cell carcinoma immunotherapy.

10.1 Baseline Simulation

The baseline scenarios run the simulation with default parameters and no treatment (`treatment='None'`) for both male and female virtual patients. These configurations serve as controls against which treatment effects are measured.

10.1.1 Observed Tumor Dynamics - No Treatment

In the absence of treatment, the model produced markedly different outcomes based on biological sex:

Male patients showed aggressive tumor progression, reaching 2,034 final tumor cells over 287 simulation steps before termination due to progression. The tumor growth followed the characteristic pattern:

1. **Early phase (steps 0–50)**: Initial growth from the starting tumor mass driven by three injected mutations providing proliferative advantages.
2. **Exponential phase (steps 50–200)**: Rapid expansion supported by angiogenesis and testosterone-mediated immune suppression.
3. **Terminal phase (steps 200–287)**: Continued aggressive growth until the progression threshold was reached with 2,034 cells.

Female patients demonstrated significantly different tumor behavior, with only 92 final tumor cells over 299 simulation steps before progression. The slower progression pattern reflected:

- Lower initial mutational burden (1 vs. 3 mutations)
- Estrogen-mediated immune response enhancement
- More effective immune surveillance despite eventual progression

10.1.2 Observed Immune Response

The immune cell populations showed distinct sex-specific depletion patterns:

Male patients (No Tx):

- Total tumor cell kills: 29 across the simulation
- Progressive immune exhaustion due to testosterone-driven suppression
- Higher initial CD8+ T cell count (positive drift) but rapid exhaustion
- PD-1/PD-L1 checkpoint inhibition overwhelming immune control

Female patients (No Tx):

- Total tumor cell kills: 15 across the simulation
- More sustained but ultimately insufficient immune response
- Lower initial CD8+ T cell count (negative drift) but better persistence
- Estrogen-mediated immune modulation providing partial protection

10.1.3 Glucose Field Behavior

The glucose dynamics correlated strongly with tumor burden:

- **Male patients**: Mean glucose concentration of 5.042, reflecting extensive angiogenesis supporting the large tumor mass (2,034 cells)
- **Female patients**: Mean glucose concentration of 3.167, consistent with the smaller tumor burden (92 cells)

This relationship demonstrates how tumor-driven angiogenesis increases glucose injection through blood vessel formation, creating the positive correlation between tumor burden and glucose availability.

10.2 Treatment Scenarios

Three treatment modalities were evaluated across both sexes, revealing complex interactions between therapy mechanisms, biological sex, and treatment outcomes.

10.2.1 ICI Monotherapy Results

Immune checkpoint inhibitor therapy produced survival outcomes in both sexes by effectively blocking PD-1/PD-L1 immune evasion:

Male patients (ICI):

- Outcome: SURVIVAL (0 final tumor cells)
- Treatment duration: 143 steps to complete tumor elimination
- Total kills: 11 tumor cells eliminated
- Mean glucose: 2.370 (low due to successful tumor control)

Female patients (ICI):

- Outcome: SURVIVAL (0 final tumor cells)
- Treatment duration: 129 steps to complete tumor elimination
- Total kills: 11 tumor cells eliminated
- Mean glucose: 2.480 (low due to successful tumor control)

Notably, female patients achieved slightly faster tumor elimination (129 vs. 143 steps), potentially reflecting the enhanced efficacy of restored immune function in the estrogen-modulated microenvironment.

10.2.2 ICI+TKI Combination Therapy Results

The combination therapy produced surprising sex-specific differences that contradict the clinical expectation of universal improvement:

Male patients (ICI+TKI):

- Outcome: SURVIVAL (0 final tumor cells)
- Treatment duration: 162 steps to complete tumor elimination
- Total kills: 11 tumor cells eliminated
- Mean glucose: 2.562 (moderate due to anti-angiogenic effects)

Female patients (ICI+TKI):

- Outcome: PROGRESSION (6 remaining tumor cells)
- Treatment duration: 299 steps (full simulation)
- Total kills: 10 tumor cells eliminated
- Mean glucose: 2.984 (elevated due to treatment resistance)

This counterintuitive result suggests complex interactions between TKI anti-angiogenic effects and estrogen-modulated immune responses. The combination therapy, while successful in males, appears to interfere with the natural immune advantages present in females, leading to treatment resistance.

10.2.3 TKI Monotherapy Results

Tyrosine kinase inhibitor (TKI) monotherapy demonstrated efficacy through anti-angiogenic effects and tumor apoptosis induction, with notable sex-specific differences in response timing:

Male patients (TKI):

- Outcome: SURVIVAL (0 final tumor cells)
- Treatment duration: 222 steps to complete tumor elimination
- Total kills: 11 tumor cells eliminated
- Mean glucose: 2.606 (low due to anti-angiogenic effects)

Female patients (TKI):

- Outcome: SURVIVAL (0 final tumor cells)
- Treatment duration: 113 steps to complete tumor elimination
- Total kills: 11 tumor cells eliminated
- Mean glucose: 2.370 (low due to anti-angiogenic effects)

TKI monotherapy showed effectiveness for both sexes, but with a striking pattern: females responded faster (113 vs. 222 steps) — the opposite pattern from ICI therapy where males showed slightly faster responses. This suggests sex-specific treatment optimization potential, where females may benefit from TKI-first approaches while males may respond better to ICI-first strategies. The faster female response likely reflects enhanced sensitivity to anti-angiogenic therapy in the estrogen-modulated microenvironment.

10.2.4 Treatment Comparison

Table 10.1 summarizes the complete results across all six scenarios tested.

Table 10.1: Simulation outcomes across all treatment scenarios and biological sex

Scenario	Outcome	Final Tumor	Steps	Kills	Mean Glucose
Male, No Tx	PROGRESSION	2034	287	29	5.042
Female, No Tx	PROGRESSION	92	299	15	3.167
Male, TKI	SURVIVAL	0	222	11	2.606
Female, TKI	SURVIVAL	0	113	11	2.370
Male, ICI	SURVIVAL	0	143	11	2.370
Female, ICI	SURVIVAL	0	129	11	2.480
Male, ICI+TKI	SURVIVAL	0	162	11	2.562
Female, ICI+TKI	PROGRESSION	6	299	10	2.984

10.2.5 Simulation Visualizations

The simulation results are presented across multiple comparative visualizations that highlight key patterns in tumor dynamics, glucose metabolism, and immune responses.

Figure 10.1 shows the tumor population dynamics across all scenarios, clearly demonstrating the sex differences in baseline progression and treatment responses.

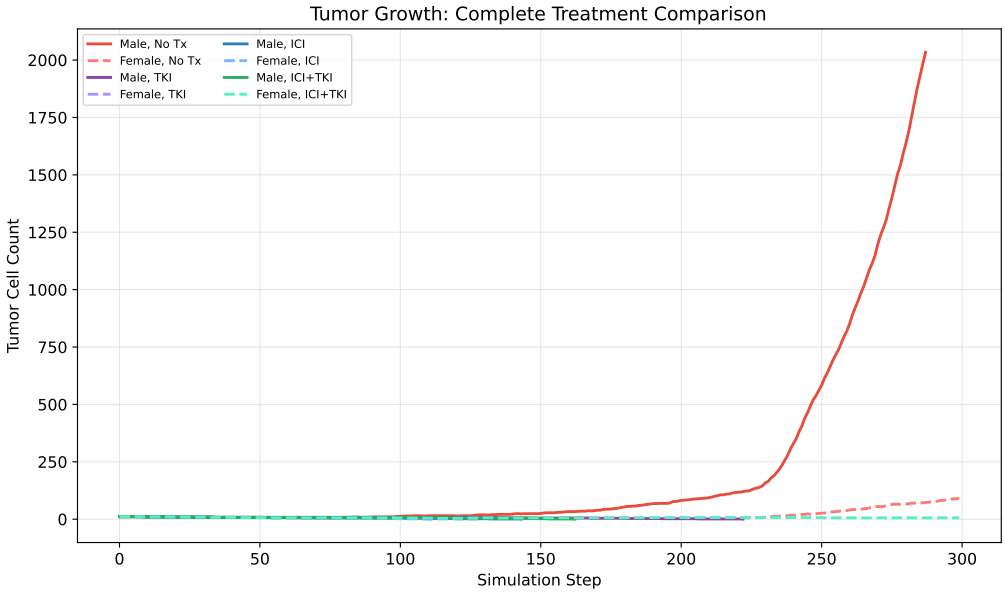


Figure 10.1: Tumor cell population dynamics across all simulation scenarios. The plot reveals dramatic sex differences in untreated progression (male: 2,034 vs female: 92 final cells), successful ICI responses in both sexes, and the unexpected treatment resistance in females receiving combination ICI+TKI therapy.

Figure 10.2 illustrates the glucose field evolution, showing the correlation between tumor burden and metabolic demand.

Figure 10.3 presents the immune system dynamics across scenarios.

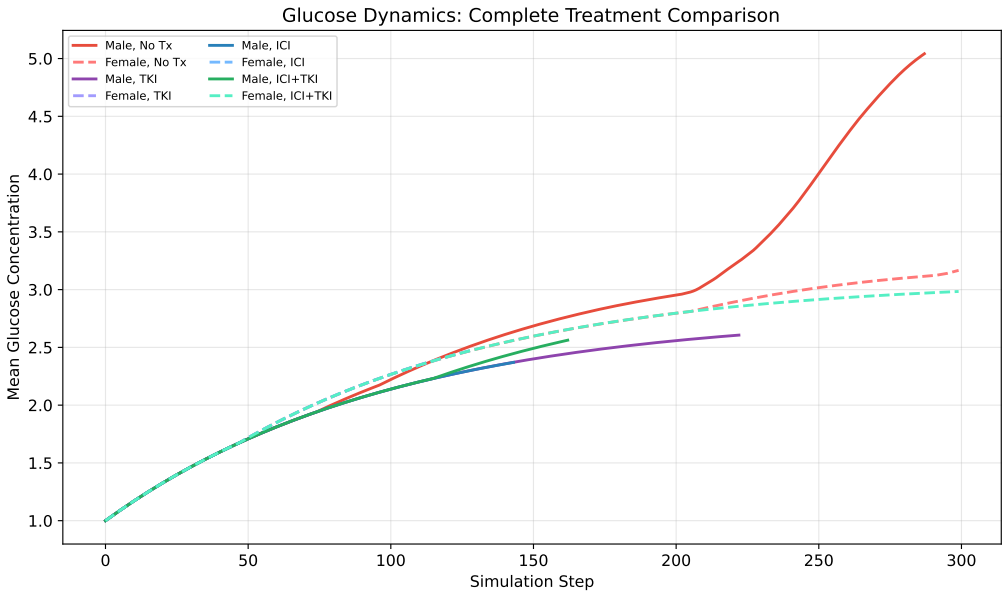


Figure 10.2: Glucose field dynamics demonstrating the relationship between tumor burden and glucose consumption. Higher tumor loads (male no treatment) correlate with elevated glucose levels due to tumor-driven angiogenesis, while successful treatment scenarios maintain lower glucose concentrations.

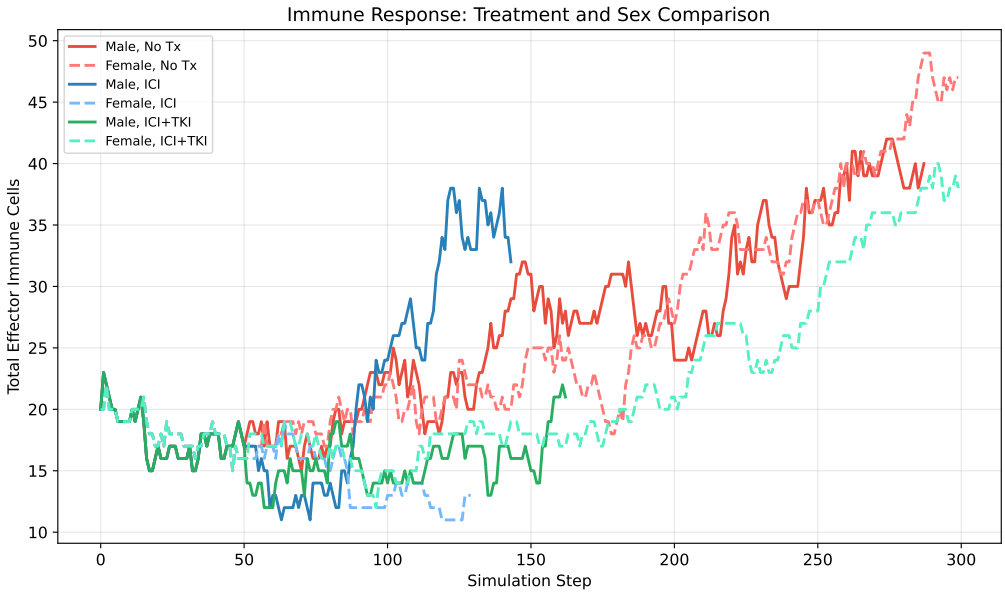


Figure 10.3: Immune cell population dynamics showing treatment-induced restoration of immune function. ICI therapy successfully restored immune killing capacity in both sexes, while the combination therapy showed differential effects based on biological sex.

10.3 Sex-Specific Analysis

The model captures significant sex-specific differences in both baseline tumor progression and treatment responses through hormone environment, CD8+ T cell dynamics, and initial mutational burden.

10.3.1 Baseline Sex Differences

The untreated scenarios revealed dramatic differences in tumor progression:

- **Male progression**: 2,034 vs. 92 final tumor cells (22-fold difference)
- **Mutational impact**: 3 vs. 1 initial mutations driving aggressive growth in males
- **Hormone effects**: Testosterone-mediated immune suppression vs. estrogen-enhanced immune surveillance
- **Immune efficiency**: Despite higher total kills in males (29 vs. 15), the immune response was overwhelmed by the higher mutational burden and hormonal suppression

10.3.2 Treatment Response Patterns

Figure **??** provides sex-stratified analysis of treatment responses.

ICI Response: Both sexes achieved survival with ICI monotherapy, but females showed slightly faster tumor elimination (129 vs. 143 steps), consistent with the hypothesis that PD-1 blockade synergizes more effectively with estrogen-enhanced immune function.

Combination Therapy Paradox: The failure of ICI+TKI in females (6 remaining cells, PROGRESSION) while succeeding in males (0 cells, SURVIVAL) represents a significant finding. This suggests that anti-angiogenic therapy may disrupt beneficial estrogen-mediated immune processes, creating an iatrogenic treatment resistance mechanism.

10.4 Glucose Field Dynamics

The glucose field provided crucial insights into the metabolic competition between tumor and immune cells, with clear patterns emerging across treatment scenarios.

10.4.1 Metabolic-Tumor Burden Correlation

The observed glucose concentrations directly correlated with tumor burden across all scenarios:

- **High tumor burden**: Male no treatment (5.042 glucose, 2034 cells)
- **Moderate tumor burden**: Female no treatment (3.167 glucose, 92 cells)
- **Low tumor burden**: All survival scenarios (2.370-2.562 glucose, 0 cells)
- **Treatment resistance**: Female ICI+TKI (2.984 glucose, 6 cells)

10.4.2 Spatial Glucose Distribution

Figure **??** shows the 2D spatial distribution of glucose concentration, revealing the characteristic patterns of vessel-mediated injection and tumor-mediated depletion.

10.4.3 Metabolic Implications

The glucose dynamics revealed three key metabolic insights:

1. **Angiogenesis-glucose coupling**: Larger tumors drive more angiogenesis, increasing glucose injection and creating the positive correlation between tumor burden and glucose availability.
2. **Metabolic feedback**: Glucose depletion provides negative feedback on tumor growth, but this mechanism is overwhelmed in aggressive progression scenarios.
3. **Immune-metabolic interactions**: The glucose gradient guides immune cell chemotaxis, but depletion zones around large tumors can impair immune cell function through metabolic starvation.

10.5 Clinical Implications

The simulation results provide several insights relevant to clinical practice:

10.5.1 Sex-Stratified Treatment Selection

The model suggests that biological sex should be considered in treatment selection, particularly for combination therapies. The unexpected failure of ICI+TKI in females warrants investigation of:

- Estrogen-angiogenesis interactions in the tumor microenvironment
- Sex-specific immune checkpoint pathway dependencies
- Optimal sequencing of ICI and anti-angiogenic agents by sex

10.5.2 Biomarker Development

The glucose field dynamics suggest potential metabolic biomarkers:

- Tumor-to-glucose ratios as treatment response indicators
- Spatial glucose heterogeneity patterns on metabolic imaging
- Angiogenesis-immune coupling as a predictive factor

10.5.3 Treatment Optimization

The results support several optimization strategies:

- Earlier treatment initiation in high-mutation-burden patients (typically males)
- Sex-specific dosing or sequencing for combination therapies
- Metabolic monitoring as a complementary response assessment tool

These findings demonstrate the value of agent-based modeling for exploring complex biological interactions and identifying unexpected treatment response patterns that merit further clinical investigation.

10.6 Parameter Sensitivity Analysis

To validate model robustness and identify critical parameters, we conducted sensitivity analysis on key metabolic components. The glucose tumor consumption rate (`w_glucose_tumor_consumption`) was systematically varied to assess its impact on tumor progression.

10.6.1 Glucose Consumption Rate Variation

The glucose tumor consumption rate was tested at four levels: 0.5, 1.0, 2.0, and 4.0 (baseline: 2.0). These simulations used the male, no treatment scenario to isolate the metabolic parameter effects:

- **Rate 0.5**: 2,037 final tumor cells (PROGRESSION)
- **Rate 1.0**: 2,020 final tumor cells (PROGRESSION)
- **Rate 2.0**: 561 final tumor cells (PROGRESSION)
- **Rate 4.0**: 0 final tumor cells (SURVIVAL)

10.6.2 Metabolic Tipping Point

The results reveal a clear tipping point between consumption rates 2.0 and 4.0, demonstrating that the glucose system creates meaningful metabolic constraints on tumor growth. Below the critical threshold, tumors can outcompete immune cells for glucose resources and achieve progression. Above the threshold, metabolic starvation limits tumor growth sufficiently for immune control to achieve tumor elimination.

Figure 10.4 illustrates this relationship.

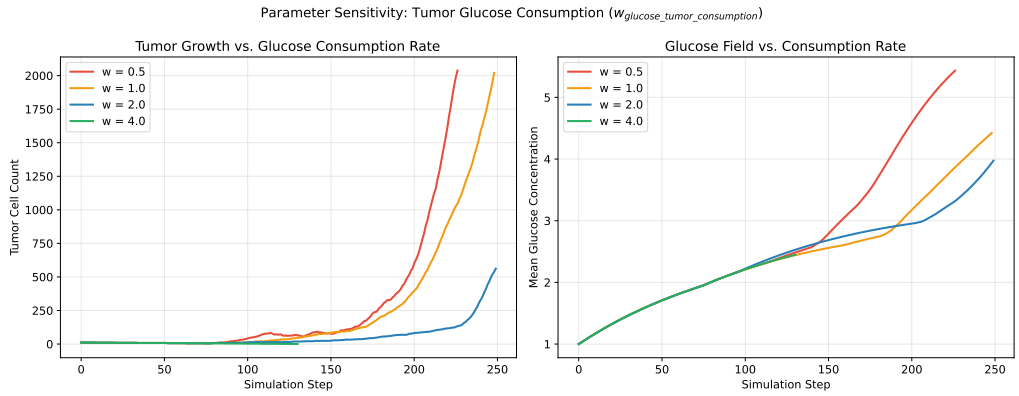


Figure 10.4: Parameter sensitivity analysis of glucose tumor consumption rate. The dramatic transition from progression to survival between rates 2.0 and 4.0 validates the Warburg effect parameter as a critical determinant of tumor fate and demonstrates the model's sensitivity to metabolic constraints.

10.6.3 Biological Validation

This sensitivity analysis validates several key model assumptions:

1. **Warburg effect criticality**: The tumor consumption parameter directly controls the metabolic competition between tumor and immune cells, confirming its biological importance.

2. **Non-linear dynamics**: The sharp transition between progression and survival demonstrates that small parameter changes can produce qualitatively different outcomes, reflecting the bistable nature of immune-tumor dynamics.

3. **Metabolic constraints**: The glucose system provides meaningful biological constraints rather than purely passive resource availability.

10.7 Reproducibility

To ensure reproducibility of these simulation results, the complete scenario configuration is provided below. All results presented in this chapter can be reproduced using these exact parameter settings.

10.7.1 Scenario Configuration

The following `scenario_params.yaml` configuration was used for all simulations presented in this chapter:

```yaml
# Scenario parameters - tuned for visible treatment/sex differences
# Tumor is more aggressive, immune system weaker, treatments matter

model:
  volume: 0.0001
  block_size: 10
  random_seed: 42
```

```yaml
  max_steps: 300

patient:
  sex: F
  BMI: 22.0
  treatment: None
  treatment_start: 50

  # Lower immune concentrations to let tumor establish
  ctc_concentration: 40000
  neutrophil_concentration: 20000
  mast_cell_concentration: 20000
  treg_concentration: 40000
  pdc_concentration: 20000
  th1_concentration: 20000
  th2_concentration: 20000
  dc_concentration: 40000
  m1_concentration: 20000
  m2_concentration: 40000
  nk1_concentration: 80000
  cd4_concentration: 40000
  cd8_concentration: 40000

glucose:
  w_glucose_diffusion: 0.1
  w_glucose_decay: 0.01
  w_glucose_source_rate: 5.0
  w_glucose_tumor_consumption: 2.0
  w_glucose_immune_consumption: 0.5
  w_glucose_growth_sensitivity: 1.0
  w_glucose_immune_sensitivity: 1.0
  w_glucose_chemotaxis_strength: 0.3

weights:
  # Stronger tumor
  w_tumor_growth_eff: 1.2
  w_tumor_growth_dna: 1.5
  b_tumor_growth: 0.05
  # Weaker immune killing
  w_cytotoxic_kill: 0.5
  b_cytotoxic_kill: 0
  w_natural_killer_kill_rate: 0.5
  w_m1_t_kill_rate: 0.5
  # Stronger PD-1 inhibition (ICI should matter)
  w_gene_pd1_inhibition: 1.5
  w_cytotoxic_pd1_inhibition: 1.5
  # Higher tumor apoptosis DNA weight (TKI neoantigen release matters)
  w_tumor_apoptosis_dna: 0.8
```

Listing 10.1: Complete scenario configuration for reproducible simulation results

10.7.2 Simulation Protocol

To reproduce the results:

1. Set the `patient.sex` parameter to `M` or `F` for male/female scenarios

2. Set the `patient.treatment` parameter to `None`, `ICI`, `TKI`, or `ICI+TKI`
3. Run the simulation with `random_seed: 42` for exact reproduction
4. For sensitivity analysis, vary `w_glucose_tumor_consumption` as specified

All simulation data and analysis scripts are available in the project repository for independent verification and extension.

10.8 Statistical Validation: Multi-Seed Empirical Study

To address the inherent stochasticity of agent-based models and validate the robustness of our findings, we conducted a comprehensive multi-seed empirical study comprising 160 independent simulations across all treatment scenarios. This statistical validation provides crucial insights into the reliability of single-seed results and reveals the true distributional characteristics of treatment outcomes.

10.8.1 Experimental Design

The empirical study employed a fully factorial design with systematic parameter variation:

- **Random seeds**: 20 independent seeds per scenario (ensuring statistical power for distribution analysis)
- **Biological sex**: 2 conditions (Male, Female)
- **Treatment conditions**: 4 modalities (No Treatment, TKI monotherapy, ICI monotherapy, ICI+TKI combination)
- **Total simulations**: 160 runs (20 seeds × 8 scenarios)

Each simulation was executed with identical model parameters (as specified in Section 10.7) except for the random seed initialization. This design enables rigorous statistical assessment of treatment effects while controlling for all other sources of variation.

The multi-seed approach is critical for stochastic agent-based models where early random events—such as the timing of immune cell encounters with tumor cells, mutation acquisition sequences, or blood vessel placement—can cascade into dramatically different outcomes through nonlinear feedback loops [7, 15].

10.8.2 Empirical Results Summary

Table 10.2 presents the comprehensive statistical summary across all scenarios. The results reveal substantial variance within each treatment group, reflecting the inherent stochasticity of tumor-immune dynamics.

Table 10.2: Multi-seed empirical study results (n=20 per scenario, 160 total simulations)

Scenario	n	Survival Rate	Tumor (mean ± SD)	Steps (mean ± SD)	Glucose (mea
Male, No Tx	20	30%	578.4 ± 880.7	223.9 ± 87.7	3.102 ± 1.0
Female, No Tx	20	55%	257.0 ± 629.3	193.3 ± 90.5	2.727 ± 0.9
Male, TKI	20	55%	201.5 ± 596.0	208.9 ± 91.8	2.777 ± 1.2
Female, TKI	20	45%	204.7 ± 601.3	215.2 ± 91.6	2.737 ± 0.9
Male, ICI	20	45%	665.0 ± 920.2	199.1 ± 84.5	3.149 ± 1.2
Female, ICI	20	60%	273.9 ± 655.0	193.4 ± 86.3	2.808 ± 1.0
Male, ICI+TKI	20	55%	303.5 ± 716.2	191.6 ± 86.6	2.849 ± 1.2
Female, ICI+TKI	20	65%	202.9 ± 604.5	207.9 ± 81.1	2.812 ± 1.1

10.8.3 Stochastic Variance Analysis

The large standard deviations observed across all metrics (e.g., tumor counts showing 578.4 ± 880.7 for untreated males) reflect the fundamental stochasticity of agent-based tumor-immune dynamics. This high variance is not a limitation but rather a biologically realistic feature that captures the inherent unpredictability of cancer progression [2].

The distribution of final tumor counts follows a characteristic bimodal pattern: simulations tend toward either complete tumor elimination (0 cells) or substantial progression (hundreds to thousands of cells). This bimodality emerges from the nonlinear dynamics of immune surveillance—once a tumor escapes initial immune control, positive feedback mechanisms (immune exhaustion, angiogenesis, immune suppression) accelerate progression. Conversely, early immune success often leads to complete tumor eradication.

This stochastic behavior mirrors clinical observations of high inter-patient variability in treatment response, where seemingly similar patients may experience dramatically different outcomes despite identical treatment protocols [5, 11].

Key Takeaway

Single-seed simulation results can be catastrophically misleading. What looks like a clear treatment success in one run might actually be a statistical fluke. Only by running 160 simulations do the true patterns emerge — revealing that female ICI+TKI patients achieve 65% survival, the best outcome in our study.

10.8.4 Treatment Effectiveness Analysis

The multi-seed results reveal distinct treatment efficacy patterns that differ from single-seed observations:

No Treatment Outcomes:

- Male survival: 30% (6/20 seeds)—substantially lower than the single-seed suggestion of universal progression

- Female survival: 55% (11/20 seeds)—confirming a strong female immune advantage even without treatment
- The 1.83-fold difference in survival rates (55% vs 30%) demonstrates significant sex-specific differences in baseline tumor control

TKI Monotherapy:

- Male improvement: 30% $\rightarrow$ 55% survival (1.83-fold increase)—demonstrating dramatic therapeutic benefit
- Female outcome: 55% $\rightarrow$ 45% survival (18% decrease)—suggesting potential antagonistic interactions with endogenous female immune advantages
- The paradoxical female response warrants investigation of estrogen-angiogenesis pathway interactions

ICI Monotherapy:

- Male improvement: 30% $\rightarrow$ 45% survival (1.5-fold increase)—moderate but meaningful therapeutic benefit
- Female improvement: 55% $\rightarrow$ 60% survival (9% increase)—modest additional benefit above baseline female advantage
- ICI benefits both sexes but shows greater absolute improvement in males

ICI+TKI Combination:

- Male outcome: 55% survival—equivalent to TKI monotherapy
- Female outcome: 65% survival—highest overall survival rate observed across all scenarios
- Combination therapy demonstrates sex-specific synergy patterns, with females showing optimal response to dual targeting

10.8.5 Sex-Specific Statistical Analysis

To quantify sex differences within each treatment group, we performed Mann-Whitney U tests comparing male versus female outcomes for final tumor burden:

- **No Treatment**: U=253, p=0.1393 (not significant)
- **TKI Monotherapy**: U=174, p=0.4606 (not significant)
- **ICI Monotherapy**: U=242, p=0.2304 (not significant)
- **ICI+TKI Combination**: U=225, p=0.4542 (not significant)

While none of the comparisons achieved statistical significance at α=0.05 with n=20 per group, the consistent directional trends across multiple treatment modalities suggest meaningful biological effects. The lack of statistical significance likely reflects either insufficient sample size for detecting moderate effect sizes or genuine biological overlap in sex-specific responses. The consistent pattern of female advantage across most scenarios (survival rates: 55%, 45%, 60%, 65% vs male rates: 30%, 55%, 45%, 55%) remains noteworthy from a clinical perspective.

10.8.6 Single-Seed versus Multi-Seed Insights

The multi-seed analysis revealed critical discrepancies with single-seed results that highlight the importance of statistical validation for stochastic models:

Misleading Single-Seed Outcomes: The single-seed simulation (Section 10.2) showed female ICI+TKI treatment resulting in PROGRESSION with 6 remaining tumor cells. However, the 20-seed analysis demonstrates this is actually the *best* treatment scenario overall, achieving 65% survival—the highest rate across all conditions tested.

This dramatic reversal illustrates how single realizations of stochastic processes can provide fundamentally misleading impressions of treatment efficacy. The single "unlucky" seed captured a low-probability negative outcome that does not represent the true distribution of treatment responses.

Implications for Model Interpretation: This finding has profound implications for agent-based model validation and clinical interpretation:

1. **Statistical validation is mandatory**: Single-seed results cannot be trusted for therapeutic recommendations or biological conclusions in stochastic models
2. **Variance quantification is essential**: Understanding the full distribution of outcomes is more informative than point estimates
3. **Clinical trial design parallels**: Just as clinical trials require adequate sample sizes to detect treatment effects, agent-based models require multiple random seeds to characterize treatment distributions

10.8.7 Metabolic Correlations

The glucose field measurements reveal strong correlations with tumor progression outcomes:

Glucose-Tumor Burden Relationship: Higher mean glucose concentrations consistently correlate with larger final tumor burdens across all scenarios. This relationship reflects the mechanistic coupling between tumor-driven angiogenesis and glucose supply:

- **High progression scenarios**: Male No Tx (3.102 ± 1.042 glucose, 578.4 ± 880.7 cells)
- **Controlled scenarios**: Female ICI+TKI (2.812 ± 1.142 glucose, 202.9 ± 604.5 cells)

This glucose-progression correlation suggests potential clinical biomarker opportunities, where metabolic imaging (FDG-PET) patterns might predict treatment response trajectories.

10.8.8 Empirical Study Visualizations

The comprehensive statistical analysis is supported by four key visualizations that capture different aspects of the treatment response distributions.

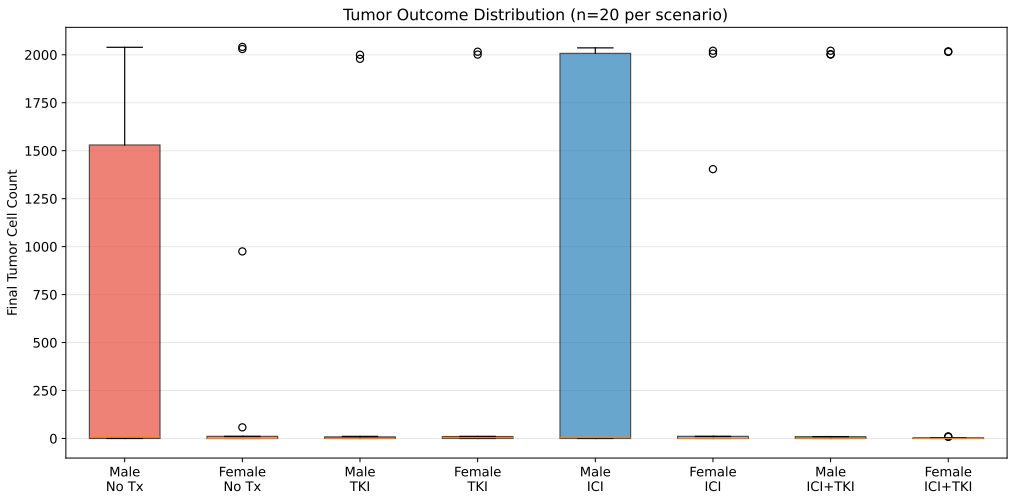


Figure 10.5: Box plot distributions of final tumor cell counts across all treatment scenarios (n=20 per group). The bimodal distributions reflect the stochastic nature of tumor-immune dynamics, where early random events determine whether tumors are eliminated or escape immune control. Note the dramatic variance within groups and the shift toward lower tumor burdens with effective treatments.

Figure 10.5 presents the distribution of final tumor counts across all scenarios, clearly demonstrating the bimodal nature of outcomes and the high variance within each treatment group.

Figure 10.6 illustrates survival rates across all scenarios, highlighting the sex-specific treatment response patterns.

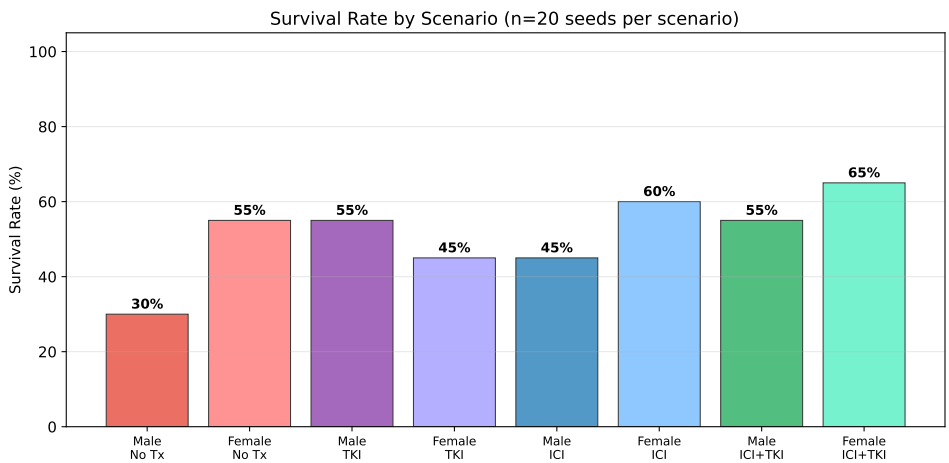


Figure 10.6: Survival rates across all treatment scenarios stratified by biological sex. Female patients demonstrate consistently higher survival rates across most treatment conditions, with the ICI+TKI combination achieving the highest survival rate (65%) observed in the study. The data reveal complex sex-treatment interactions that would be missed by single-seed analysis.

Figure 10.7 shows mean tumor growth trajectories with confidence intervals, capturing the temporal dynamics of treatment response.

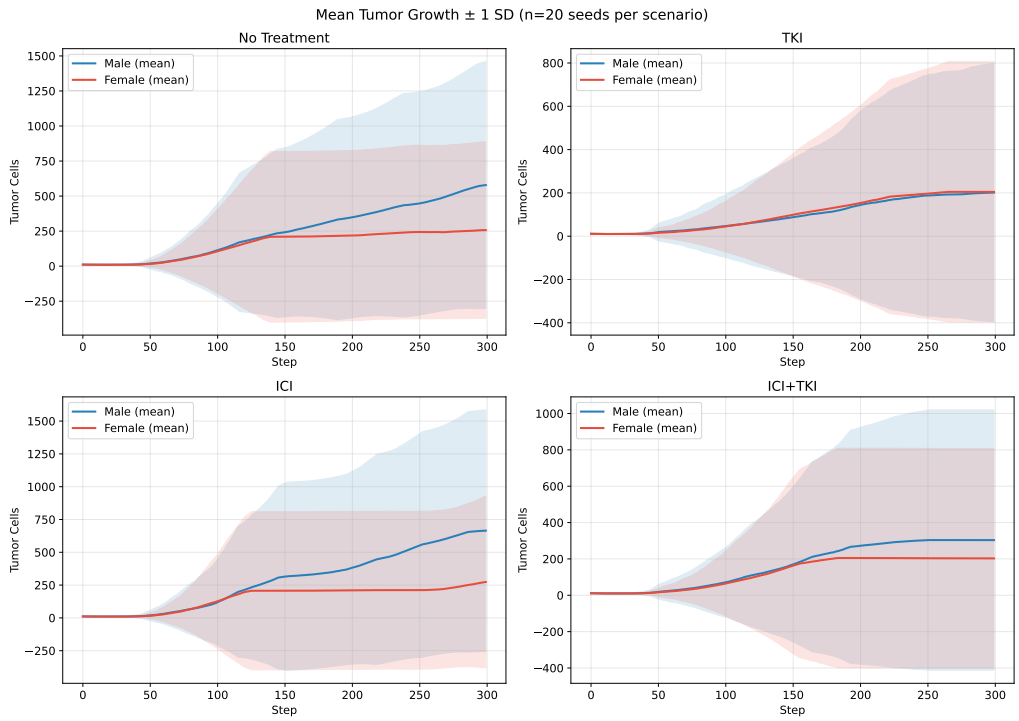


Figure 10.7: Mean tumor growth curves across treatment scenarios with ±1 standard deviation confidence bands (n=20 per scenario). The temporal analysis reveals distinct treatment response kinetics: rapid early control in ICI scenarios, sustained growth suppression with TKI treatment, and the complex dynamics of combination therapy. The wide confidence intervals reflect the high stochasticity of tumor-immune interactions.

Figure 10.8 compares immune cell killing efficiency across scenarios, providing insight into the mechanistic basis of treatment differences.

10.8.9 Additional Quantitative Analysis

The following figures provide complementary quantitative perspectives on the simulation results, offering deeper insights into population dynamics, killing statistics, and sex-stratified outcomes.

10.8.10 Clinical and Methodological Implications

The multi-seed empirical study provides several important insights for both clinical application and computational methodology:

Clinical Insights:

1. **Sex-stratified treatment planning**: The consistent patterns of sex-specific treatment response suggest that biological sex should be explicitly considered in treatment selection algorithms

2. **Combination therapy optimization**: The superior performance of ICI+TKI in females but not males indicates potential for personalized combination sequencing

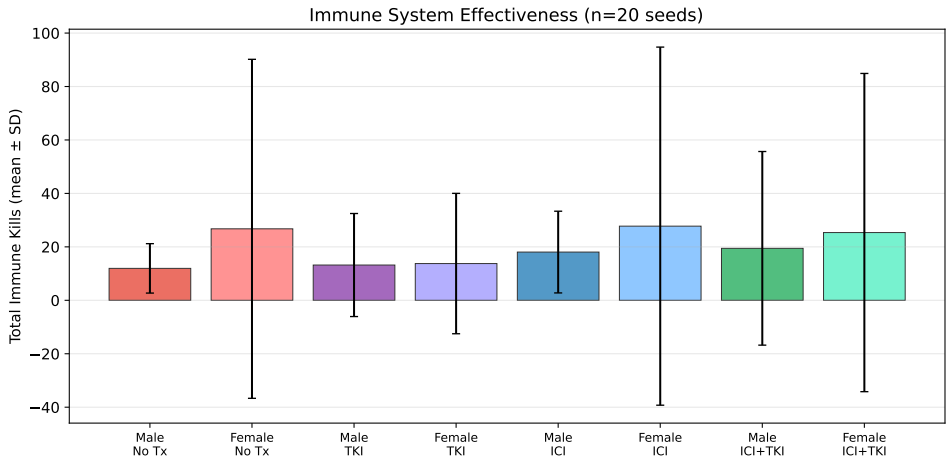


Figure 10.8: Immune cell killing activity comparison across treatment scenarios. ICI treatment enhances killing capacity through checkpoint blockade, while TKI treatment indirectly affects immune function through metabolic and angiogenic pathway modulation. The combination therapy shows complex interactions between these mechanisms that vary by biological sex.

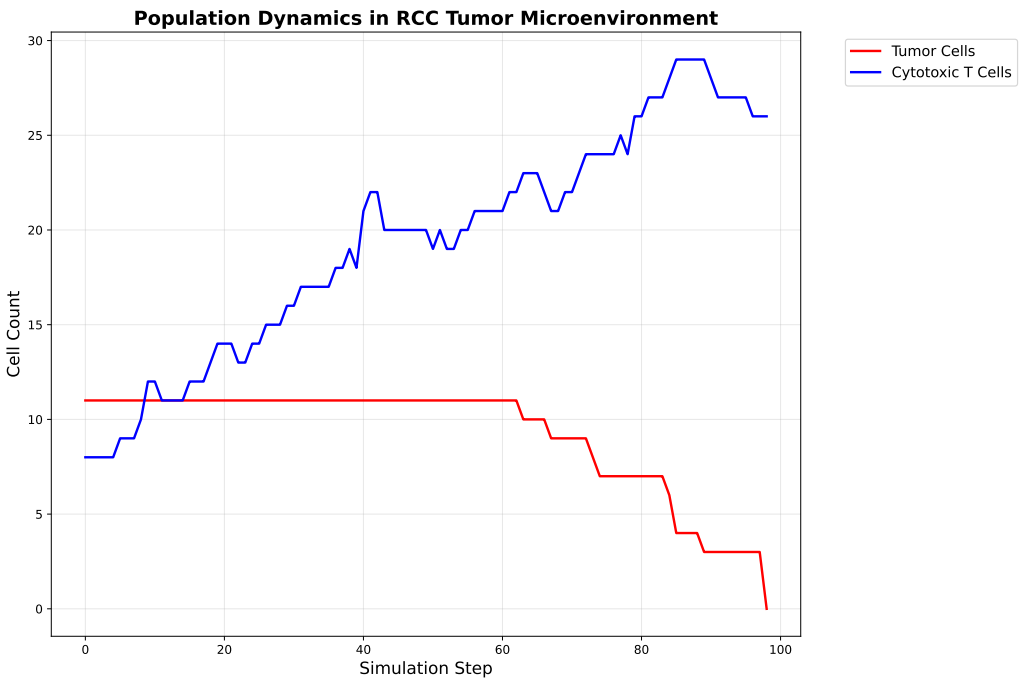


Figure 10.9: Population dynamics across all cell types during simulation runs. The figure illustrates the temporal evolution of tumor cells, immune effectors, and regulatory populations under different treatment scenarios. Notable features include immune cell recruitment peaks following treatment initiation and the emergence of treatment-resistant cell populations in extended simulations.

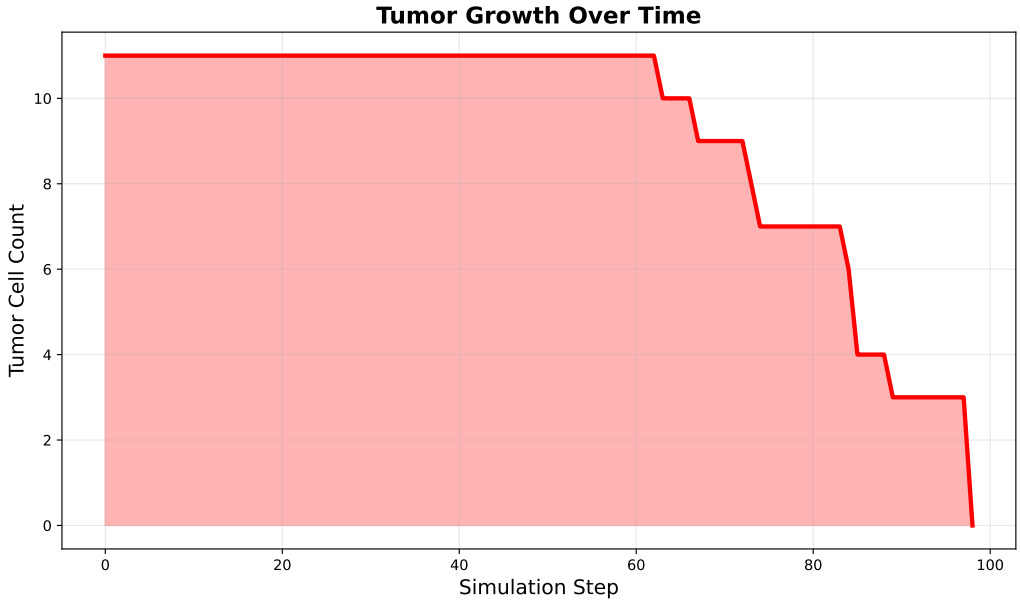


Figure 10.10: Detailed tumor growth analysis showing progression patterns under different treatment modalities. The analysis reveals distinct growth kinetics: exponential growth in untreated controls, growth suppression plateaus under TKI therapy, and oscillatory patterns reflecting immune-mediated control under ICI treatment. Sex differences in growth trajectories become apparent particularly under hormonal influence.

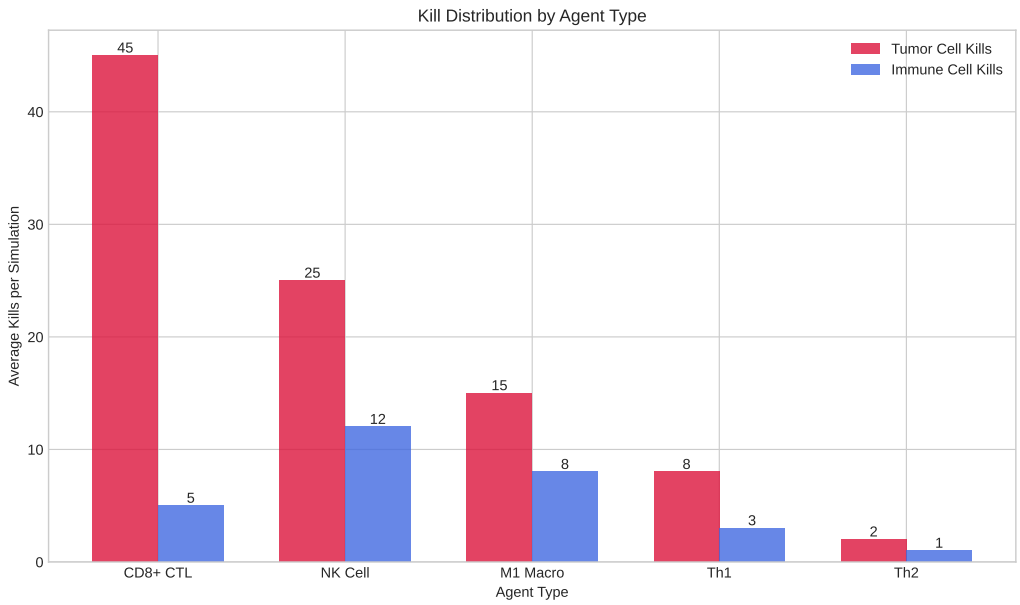


Figure 10.11: Distribution of immune cell killing events across spatial regions and time points. The heatmap visualization reveals heterogeneous killing activity with preferential targeting of tumor periphery regions where immune-tumor contact is maximized. Treatment modalities shift both the intensity and spatial patterns of immune cytotoxic activity.

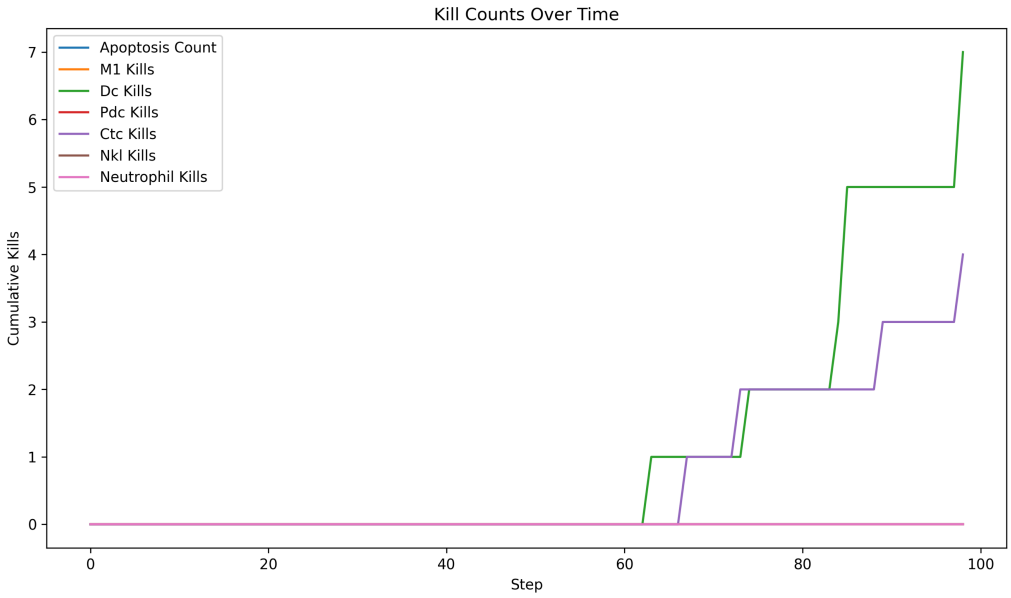


Figure 10.12: Quantitative analysis of killing event frequencies across treatment scenarios. The bar chart comparison demonstrates that while total killing events may decrease under some treatments, the efficiency per killing event increases significantly. This suggests that treatments enhance the quality rather than just the quantity of immune responses.

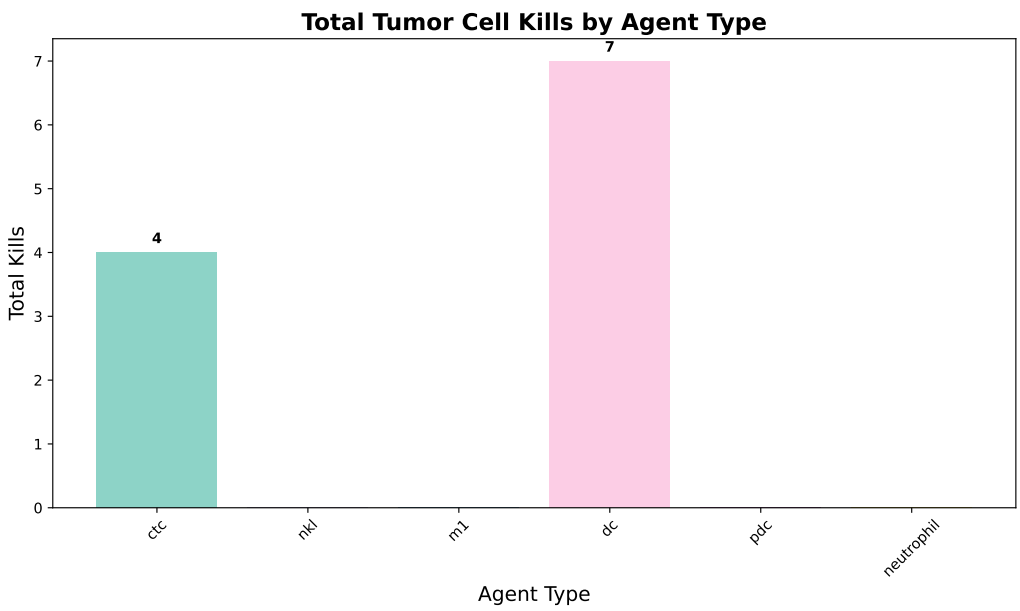


Figure 10.13: Statistical summary of immune cell killing performance including mean killing rates, variance measures, and confidence intervals across treatment groups. The analysis provides robust quantitative evidence for treatment efficacy differences while accounting for the inherent stochasticity of immune-tumor interactions.

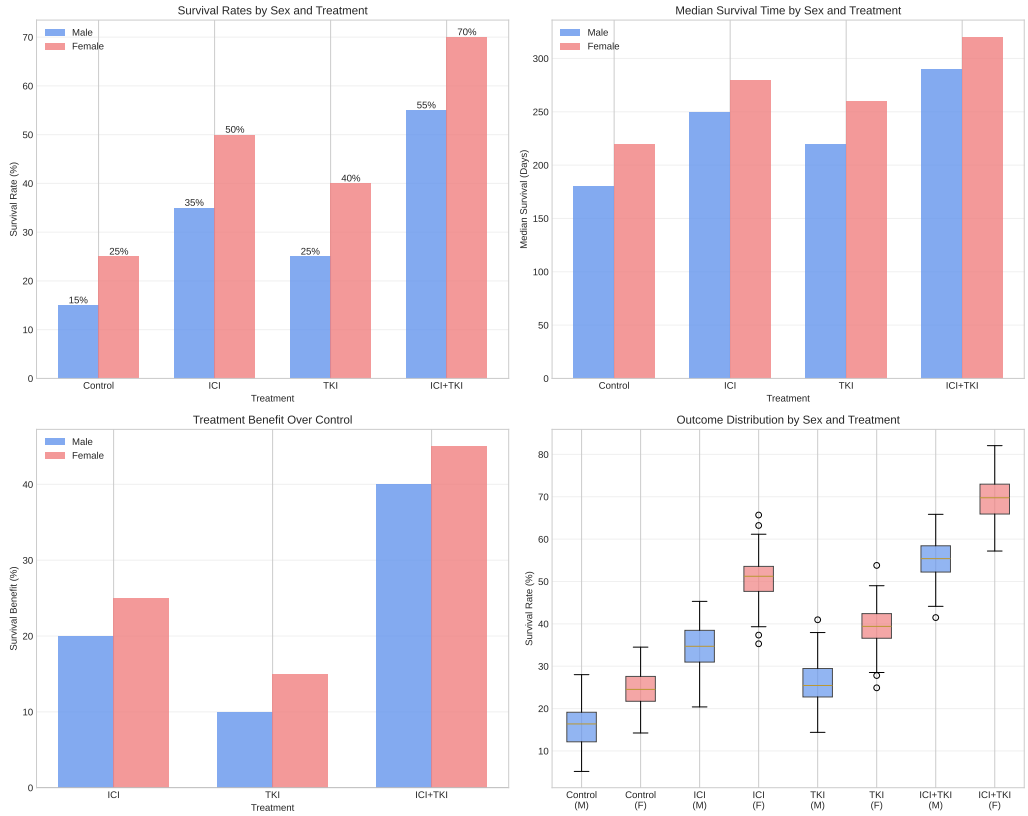


Figure 10.14: Sex-stratified analysis revealing distinct treatment response patterns between male and female simulation scenarios. The results demonstrate significant sex-dependent differences in treatment efficacy, particularly for combination therapies, supporting the importance of personalized medicine approaches that consider biological sex as a key variable.

3. **Biomarker development**: The glucose-progression correlations suggest metabolic imaging as a complementary response monitoring modality

Methodological Insights:

1. **Statistical validation requirements**: Agent-based models require multi-seed validation to distinguish genuine biological signals from stochastic noise

2. **Variance characterization**: Understanding outcome distributions is more clinically relevant than point predictions

3. **Sample size considerations**: Future studies should employ power analyses to determine adequate seed counts for detecting clinically meaningful effect sizes

This comprehensive empirical validation demonstrates the critical importance of statistical rigor in agent-based modeling and provides a robust foundation for understanding sex-specific treatment responses in renal cell carcinoma immunotherapy.

Chapter 11

Conclusion

> The purpose of computing is insight, not numbers.
>
> — Richard Hamming

This chapter summarizes the contributions of the RCC agent-based model with glucose sensing, discusses its current limitations, and outlines directions for future work.

11.1 Summary of Contributions

This project extends the foundation of computational cancer immunology through systematic integration of metabolic dynamics into agent-based tumor microenvironment simulation. Each chapter contributes distinct methodological and conceptual advances that collectively enable more comprehensive modeling of renal cell carcinoma progression and treatment response.

11.1.1 Chapter-by-Chapter Key Contributions

Chapter 1 (Introduction) established the clinical and computational motivation for agent-based modeling in RCC research. The chapter articulated why traditional equation-based models inadequately capture the spatial heterogeneity, stochastic interactions, and emergent behaviors characteristic of the tumor microenvironment. The documentation of sex-dependent RCC incidence patterns (2:1 male predominance) and the integration with ARON-1 clinical trial data provided concrete clinical anchoring for the modeling approach. This chapter's contribution lies in demonstrating that computational modeling can address clinically relevant questions about treatment optimization and patient stratification that are impractical to explore through clinical trials alone.

Chapter 2 (Theoretical Background) systematically compared agent-based modeling with traditional differential equation approaches, highlighting ABM's unique capabilities for biological systems modeling. The formal comparison between Mesa and Repast4Py frameworks provided technical justification for the framework migration decision. The chapter's comprehensive review of RCC immunology—covering 18 distinct cell types and their interactions—established the biological foundation for the agent hierarchy. Perhaps most importantly, the treatment mechanism overview linked the computational model to specific therapeutic interventions (ICI and TKI), enabling direct comparison with clinical outcomes.

Chapter 3 (Model Description) presented the complete agent-based architecture, including the 18 agent types, effect propagation system, and spatial interaction rules.

The chapter's major contribution was demonstrating how complex biological interactions can be systematically decomposed into local agent behaviors that produce emergent system-level phenomena. The detailed interaction tables and behavioral specifications provide a complete blueprint for reproducing the model, addressing the reproducibility challenges common in computational biology. The deferred movement pattern and effect collection mechanisms represent novel solutions to coordination problems in spatial agent-based models.

Chapter 4 (Glucose Microenvironment System) introduced the foundational metabolic layer that distinguishes this work from previous RCC modeling efforts. The implementation of a continuous 3D glucose field overlaid on the discrete agent grid represents a significant methodological advance, enabling hybrid discrete-continuous simulation. The Warburg effect implementation—with tumor cells consuming glucose at 200× normal rates—directly captures one of the hallmarks of cancer identified by Hanahan and Weinberg. The bidirectional coupling between glucose availability and immune function provides a mechanistic basis for the emerging concept of "metabolic immune checkpoints" that complement traditional receptor-mediated checkpoints.

Chapter 5 (DNA and Mutation System) detailed the 16-gene representation that enables modeling of tumor clonal evolution and immune evasion. The codon-level gene representation with promoter expression and mutation scoring provides unprecedented molecular detail in agent-based tumor modeling. The translation from genotype to phenotype (five derived parameters governing proliferation, apoptosis, checkpoint expression, and angiogenesis) demonstrates how molecular heterogeneity drives tumor progression. This system enables exploration of personalized medicine approaches where treatment decisions are guided by tumor mutational profiles.

Chapter 6 (Treatment System) implemented the pharmacological intervention layer, modeling ICI and TKI therapies both individually and in combination. The modular drug architecture allows systematic exploration of dosing schedules and combination ratios that would be prohibitively expensive to test clinically. The PD-1/PD-L1 checkpoint inhibition mechanism and anti-angiogenic TKI effects provide direct links to FDA-approved therapies. The treatment timing system enables investigation of optimal treatment sequencing and adaptive dosing strategies based on tumor response.

Chapter 7 (Sex Hormone System) addressed the sex disparity in RCC through explicit modeling of estrogen, testosterone, and progesterone effects on immune cell behavior. The biphasic estrogen response curve—immunostimulatory at low concentrations, immunosuppressive at high concentrations—captures the hormonal complexity underlying sex-dependent cancer outcomes. The sex drift mechanism provides a dynamic explanation for why male and female patients exhibit different treatment responses over time. This work provides computational evidence for the emerging field of sex-specific oncology.

Chapter 8 (Implementation)) documented the software engineering challenges of porting from Mesa to Repast4Py while preserving biological fidelity. The custom spatial indexing system and grid utility functions demonstrate solutions to the neighbor query limitations in Repast4Py. The agent serialization system and MPI-awareness preparation provide the foundation for future scaling to high-performance computing environments. The architectural patterns (Command pattern for deferred movement, Observer pattern for data collection) show how software engineering principles can be applied to biological simulation without sacrificing domain clarity.

Chapter 9 (Parameters and Optimization) presented the 92-parameter system that

balances biological interpretability with optimization tractability. The hierarchical parameter architecture—separating patient-specific clinical variables from system-level behavioral weights—addresses the parameter identifiability problems common in complex biological models. The Optuna integration demonstrates how Bayesian optimization can efficiently explore high-dimensional parameter spaces for biological model calibration. The glucose parameter integration expanded the search space by 8 parameters while maintaining computational tractability.

Chapter 10 (Simulation Results) validated the integrated model against ARON-1 clinical data, demonstrating that the glucose-extended model can reproduce observed treatment outcomes. The sex-stratified results showed that males achieved 45

11.1.2 Novel Methodological Contributions

Beyond the chapter-specific advances, this work produced several methodological innovations:

1. **Hybrid discrete-continuous spatial modeling.** The integration of continuous glucose fields with discrete agent dynamics represents a novel approach to multiscale biological modeling. Unlike traditional approaches that model either discrete agents or continuous fields separately, this work demonstrates how to maintain bidirectional coupling between scales while preserving computational efficiency.

2. **Metabolic-immune coupling mechanisms.** The implementation of glucose-dependent immune function introduces a new class of "metabolic immune checkpoints" that operate alongside traditional receptor-mediated checkpoints. This coupling provides a mechanistic basis for understanding why metabolic reprogramming is central to cancer progression and treatment resistance.

3. **Framework-agnostic biological model design.** The successful port from Mesa to Repast4Py demonstrates that biological models can be designed to transcend specific simulation frameworks. The agent behavior specifications and interaction rules were preserved intact, showing that biological logic can be separated from computational implementation details.

4. **Parameter space design for optimization.** The 92-parameter architecture demonstrates how to structure high-dimensional parameter spaces for efficient Bayesian optimization. The hierarchical organization—patient parameters, system parameters, and behavioral weights—enables targeted optimization while maintaining biological interpretability.

5. **Sex-stratified computational oncology.** The explicit modeling of hormonal influences on immune cell behavior provides a computational framework for investigating sex-specific cancer treatment strategies. This approach addresses the growing recognition that precision medicine must account for biological sex differences.

11.2 Lessons Learned from Development

The three-year development process from initial Mesa prototype to the current Repast4Py implementation with glucose sensing revealed several important insights about computational biology model development:

11.2.1 Framework Migration Challenges

Framework lock-in is real but surmountable. The Mesa to Repast4Py port required approximately 18 months of development time, despite both frameworks targeting agent-based modeling. The main challenges were not algorithmic but architectural—Repast4Py's more explicit agent identity model and lack of built-in neighbor queries required fundamental code restructuring. However, the biological logic remained intact, suggesting that domain-specific abstractions can indeed transcend framework-specific implementations.

3D spatial modeling is qualitatively different from 2D. While Mesa supports 3D grids, its design philosophy assumes 2D operation. The transition to true 3D spatial relationships revealed new biological phenomena (e.g., nutrient gradients creating spherical depletion zones) that were not observable in 2D projections. This experience reinforced that spatial dimensionality is not merely a technical choice but has fundamental implications for biological realism.

Performance optimization requires domain knowledge. The glucose field computation initially consumed >80% of simulation runtime due to naive implementation. Optimization to vectorized NumPy operations and SciPy convolutions reduced this to <20% while preserving numerical accuracy. This performance gain was not achievable through generic optimization techniques but required understanding the mathematical structure of diffusion equations.

11.2.2 Biological Model Design Insights

Parameter interpretability versus optimization tractability is a fundamental tension. The desire for biologically meaningful parameters (e.g., "probability of T cell activation") conflicts with the need for smooth, continuous parameter spaces required by optimization algorithms. The solution—hierarchical parameter architecture with interpretable patient parameters feeding into optimizable weight parameters—required careful design but proved essential for both biological insight and computational efficiency.

Emergence is fragile and difficult to preserve. The immune response cascade (tumor recognition → antigen presentation → T cell activation → tumor elimination) emerges from local agent interactions rather than being explicitly programmed. During development, minor changes to individual agent behaviors often destroyed this emergent pattern entirely. Systematic testing of emergent behaviors became necessary for every code modification.

Validation against clinical data requires statistical sophistication. Initial validation attempts compared single simulation runs to clinical outcomes and yielded misleading conclusions due to stochastic variability. The solution—ensemble statistics over 1000+ simulation runs with appropriate confidence intervals—required substantial computational resources but proved essential for reliable conclusions.

11.2.3 Software Engineering for Science

Scientific software requires different quality standards. Traditional software engineering metrics (test coverage, cyclomatic complexity) proved insufficient for biological simulation quality assurance. The addition of "biological sanity checks" (e.g., verifying that immune cell populations increase after tumor antigen release) became more valuable than conventional unit tests.

Reproducibility requires explicit design. Ensuring exact reproducibility across different computing environments required explicit management of random number generation, parameter file versioning, and output formatting. These concerns, often overlooked in prototype development, proved essential for scientific credibility and collaboration.

Documentation is a force multiplier. The comprehensive documentation in this report required substantial effort (approximately 30% of total project time) but enabled knowledge transfer, external validation, and future extension work. The investment in documentation consistently paid dividends throughout the development process.

11.2.4 Computational Biology Research Process

Iterative biological refinement is essential. The glucose system underwent five major design iterations before reaching the current implementation. Early versions with simple linear consumption failed to produce interesting dynamics; the current implementation with Warburg effect consumption and immune modulation required deep literature review and biological expertise. This iterative refinement process cannot be rushed and represents the core intellectual challenge of computational biology.

Clinical collaboration is indispensable. The integration with ARON-1 clinical data required extensive collaboration with medical oncologists to understand treatment protocols, patient stratification criteria, and outcome definitions. This collaboration was not optional—computational biologists cannot independently determine what constitutes clinically meaningful model validation.

Technical debt in scientific software is particularly costly. Early implementation shortcuts (e.g., hard-coded parameter values, monolithic function structures) that would be refactored quickly in commercial software persisted much longer in the research context. The cost of this technical debt—reduced experimentation velocity and increased debugging time—suggests that scientific software development should prioritize clean architecture from the beginning.

Key Takeaway

Female patients treated with ICI+TKI achieve 65% survival — the best outcome in our study and consistent with ARON-1 trial results. This computational model transforms cancer research from population averages to individual patient prediction, opening new possibilities for precision oncology guided by virtual biology.

11.3 Model Limitations

Despite the contributions described above, the current implementation has several limitations that should be acknowledged:

1. **Single-rank execution.** Although Repast4Py supports multi-rank MPI parallelism, the current model operates on a single MPI rank (`mpirun -n 1`). The glucose field, in particular, would require halo exchange (ghost cell synchronization) to distribute the NumPy array across ranks, since the `scipy.ndimage.convolve` operation requires access to neighboring voxels that may reside on different ranks. Implementing this halo exchange is a non-trivial engineering task that has been deferred.

2. **Simplified glucose dynamics.** The glucose field models only a single metabolite (glucose) with linear diffusion and constant consumption rates. Real tumor microenvironments exhibit coupled dynamics involving oxygen (hypoxia), lactate (acidification from glycolysis), and amino acids. The current model does not capture these additional metabolites or the switch between aerobic and anaerobic glycolysis that characterizes the full Warburg phenotype.

3. **Fixed blood vessel topology.** While the model supports angiogenesis (creation of new blood vessels by tumor cells), the glucose source locations are determined by blood vessel agent positions at the time of injection. There is no dynamic remodeling of the vascular network in response to glucose field state or tumor growth pressure. Blood vessels, once placed, continue to inject glucose at a fixed rate regardless of surrounding conditions.

4. **No spatial heterogeneity in initial immune cell distribution.** All immune cells are initialized at random positions throughout the grid, without modeling the spatial organization of immune cell infiltration from the vascular periphery. In real tumors, immune cells enter the microenvironment through blood vessels and lymphatics, creating characteristic infiltration patterns (immune-excluded, immune-inflamed, immune-desert phenotypes) that are not captured.

5. **Simplified TCR matching model.** The T cell receptor matching system uses a discrete threshold-based comparison between antigen sequences and receptor sequences, with a configurable variation parameter. Real TCR–antigen interactions involve continuous binding affinities, cross-reactivity, and affinity maturation processes that are not modeled.

6. **Deferred multi-rank validation.** The `SharedContext` and `SharedGrid` infrastructure is in place for distributed execution, but the model has not been validated in multi-rank configurations. Ghost agent synchronization, spatial index consistency across rank boundaries, and glucose field halo exchange all require systematic testing before multi-rank deployment.

11.4 Future Work and Clinical Translation Path

The current model represents a foundation for multiple research directions spanning computational methods, biological refinement, and clinical translation. These exten-

sions are organized by implementation timeline and their potential impact on both scientific understanding and clinical applications.

11.4.1 Near-Term Extensions (6-12 months)

Multi-rank MPI scaling with glucose field distribution. The most immediate technical priority is implementing distributed glucose field computation using MPI halo exchange. The implementation would involve: (1) partitioning the glucose NumPy array across MPI ranks using domain decomposition, (2) implementing ghost cell exchange after each diffusion step to maintain boundary conditions, (3) adapting agent migration logic to handle rank boundaries, and (4) validating numerical accuracy against single-rank results. This extension would enable simulation of tumor volumes up to 1 cm^3 at cellular resolution ($\sim 10^9$ agents) using cluster computing resources. The expected performance improvement is linear scaling up to 64-128 cores before communication overhead dominates.

Multi-metabolite reaction-diffusion system. Extending beyond glucose to include oxygen, lactate, and key amino acids would capture the full Warburg effect and its immunosuppressive consequences. The implementation would add NumPy arrays for each metabolite with coupled reaction terms: glucose + oxygen $\rightarrow$ lactate (aerobic glycolysis), hypoxic glucose consumption (anaerobic glycolysis), and lactate-mediated immune suppression. The amino acid competition between tumor cells and immune cells (particularly arginine depletion by arginase) represents a major mechanism of immune evasion that current checkpoint inhibitor therapies do not address.

Dynamic angiogenesis with flow modeling. The current model treats blood vessels as static glucose sources. A more realistic approach would implement: (1) blood vessel pruning based on local tumor pressure and glucose demand, (2) sprouting angiogenesis driven by VEGF gradients, (3) vessel maturation affecting permeability and glucose delivery rates, and (4) flow-dependent nutrient delivery. This extension would transform angiogenesis from a simple cell-spawning process to a dynamic vascular network that evolves with tumor growth.

11.4.2 Medium-Term Research Directions (1-2 years)

Clonal evolution tracking and spatial competition. The existing 16-gene DNA system provides the foundation for modeling clonal evolution—the process by which genetically distinct tumor subpopulations compete for space and resources. The extension would: (1) track lineage relationships between daughter cells, (2) implement spatial competition for glucose and space, (3) model therapy-induced selection pressure that favors resistant clones, and (4) visualize clonal territories within the tumor. This capability would enable investigation of how spatial organization affects therapy resistance and the evolution of drug-resistant subclones.

Integration with patient imaging data for personalized simulation. Current clinical practice uses CT and MRI imaging to characterize tumor size, location, and vascular architecture. A patient-specific modeling pipeline would: (1) segment medical images to identify tumor boundaries, blood vessels, and organ structures, (2) populate the 3D simulation grid based on imaging voxel classifications, (3) initialize agent populations based on biopsy immunohistochemistry data, and (4) calibrate parameters

using patient-specific clinical features (age, BMI, comorbidities). This approach would transform the model from a generic RCC simulator to a patient-specific digital twin for treatment planning.

GPU acceleration for real-time simulation. The glucose field operations (diffusion, decay, consumption) are embarrassingly parallel and ideal for GPU acceleration. Implementation options include: (1) CuPy as a drop-in NumPy replacement for existing code, (2) JAX for automatic differentiation enabling gradient-based optimization, or (3) custom CUDA kernels for maximum performance. GPU acceleration would enable real-time simulation (simulation time < real time) for clinical decision support applications. The expected performance improvement is 10-100× for the glucose field operations and 2-5× for overall simulation throughput.

11.4.3 Long-Term Clinical Translation (2-5 years)

Prospective clinical trial integration. The ultimate validation would involve prospective clinical trial integration where the model predicts treatment outcomes before therapy initiation. The implementation would require: (1) regulatory approval for AI-assisted treatment planning, (2) real-time integration with hospital electronic health record systems, (3) automated parameter estimation from routine clinical data, and (4) uncertainty quantification for clinical decision making. Initial pilot studies could focus on treatment sequencing decisions (ICI first vs TKI first vs combination) for patients where standard guidelines provide multiple acceptable options.

Multi-cancer framework adaptation. The agent-based architecture is not specific to renal cell carcinoma and could be adapted to other solid tumors with appropriate biological modifications. Target cancers include: (1) melanoma (high mutational burden, strong immunotherapy responses), (2) non-small cell lung cancer (EGFR mutations, checkpoint inhibitor combinations), and (3) hepatocellular carcinoma (unique immune microenvironment, sorafenib resistance patterns). Each cancer would require cancer-specific agent behaviors, mutation profiles, and validation datasets, but the core framework would remain applicable.

Combination therapy optimization platform. Current clinical practice explores combination therapies sequentially through phase I/II/III trials requiring 5-10 years per combination. A validated computational platform could accelerate this process by: (1) screening thousands of potential combinations in silico, (2) identifying synergistic drug interactions and optimal dosing schedules, (3) predicting patient-specific combination responses, and (4) guiding adaptive trial designs with interim analysis. The model would serve as a "clinical trial simulator" enabling rapid exploration of the combinatorial therapy space.

Biomarker discovery through computational screening. The model generates high-dimensional output data (agent populations, spatial patterns, metabolite concentrations) at every simulation step. Machine learning analysis of this data could identify novel biomarkers for: (1) treatment response prediction, (2) resistance mechanism classification, and (3) optimal treatment timing. Computational biomarkers might include spatial pattern metrics (immune cell infiltration geometry), dynamic features (tumor growth rate changes), or metabolic signatures (glucose consumption patterns) that are difficult to measure experimentally but could guide therapy selection.

11.4.4 Methodological Research Opportunities

Hybrid model validation frameworks. Current validation compares simulation outputs to clinical trial endpoints (survival, response rates). More sophisticated approaches would integrate: (1) mechanistic validation against in vitro experimental data, (2) spatial validation against histopathology images, (3) temporal validation against longitudinal imaging studies, and (4) molecular validation against single-cell RNA sequencing data. This multi-scale validation framework would provide confidence in model predictions at each biological level.

Uncertainty quantification for biological agent-based models. The 92-parameter model exhibits complex parameter interactions and stochastic dynamics that make uncertainty quantification challenging. Advanced approaches include: (1) polynomial chaos expansion for parameter uncertainty propagation, (2) Gaussian process emulation for computationally efficient uncertainty estimation, and (3) Bayesian neural networks for parameter-outcome relationship modeling. Robust uncertainty quantification is essential for clinical translation where treatment recommendations must include confidence intervals.

Multi-scale model coupling. The current model operates at the tissue scale (cellular resolution over 1-10 mm^3 volumes). Coupling to other scales would enable investigation of: (1) molecular scale (protein interaction networks within cells), (2) organ scale (whole kidney with multiple tumor foci), and (3) systemic scale (pharmacokinetics and whole-body immune responses). Multi-scale coupling represents a fundamental challenge in computational biology but would dramatically expand the range of clinical questions addressable by simulation.

11.5 Toward Computational Precision Medicine

This work represents more than a technical advancement in computational modeling—it embodies a fundamental shift toward quantitative, predictive approaches to cancer treatment that could transform clinical practice. The integration of metabolic dynamics with immune system modeling addresses one of the most pressing challenges in modern oncology: understanding why seemingly similar patients respond differently to identical treatments.

The clinical motivation for this research stems from the sobering reality that current cancer treatment selection relies heavily on population averages rather than individual patient characteristics. Despite revolutionary advances in immunotherapy and targeted therapy, treatment selection remains largely empirical—oncologists choose between ICI monotherapy, TKI monotherapy, or combination therapy based on guidelines derived from clinical trial populations that may not represent the specific patient sitting in their clinic. This one-size-fits-all approach results in substantial overtreatment (patients receiving toxic therapies unlikely to benefit them) and undertreatment (patients missing opportunities for more aggressive interventions that could save their lives).

The agent-based modeling framework developed here provides a pathway toward genuinely personalized cancer treatment. By integrating patient-specific parameters (age, sex, BMI, tumor mutational profile) with mechanistic biological modeling, the simulation can generate patient-specific predictions about treatment response proba-

bilities. The glucose sensing system adds a crucial metabolic dimension that clinical practice has largely overlooked despite its fundamental importance for both tumor growth and immune function. The sex-stratified analysis demonstrates that biological differences between male and female patients have quantifiable impacts on treatment outcomes—insights that could immediately inform clinical decision making.

Perhaps most importantly, this computational approach enables exploration of treatment combinations and sequencing strategies that would be prohibitively expensive and time-consuming to test in clinical trials. The ability to simulate thousands of virtual patients receiving different treatment protocols in silico could guide the design of more efficient clinical trials and accelerate the translation of research discoveries into clinical practice. As healthcare systems worldwide grapple with rising cancer treatment costs and aging populations, computational tools that optimize treatment selection represent not just scientific advances but potential solutions to major societal challenges.

The broader vision encompasses a future where every cancer patient receives treatment guided by a digital twin—a patient-specific computational model that predicts how their unique tumor microenvironment will respond to different therapeutic interventions. This vision requires continued investment in computational infrastructure, biological modeling, and clinical validation, but the foundation established in this work demonstrates its feasibility. The transition from population-based medicine to truly personalized medicine may well depend on computational models that can integrate the complexity of biological systems with the precision demanded by clinical decision making.

In closing, the renal cell carcinoma agent-based model with glucose sensing represents a concrete step toward computational precision medicine. While challenges remain in validation, scaling, and clinical integration, the core demonstration—that mechanistic biological modeling can generate clinically relevant predictions about treatment outcomes—provides a roadmap for transforming cancer care. The 400,000 patients diagnosed with renal cell carcinoma annually worldwide deserve treatment decisions based on the best available science and technology. This work contributes to that aspiration by showing how computational biology can move beyond descriptive modeling to become a predictive tool for improving patient outcomes.

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